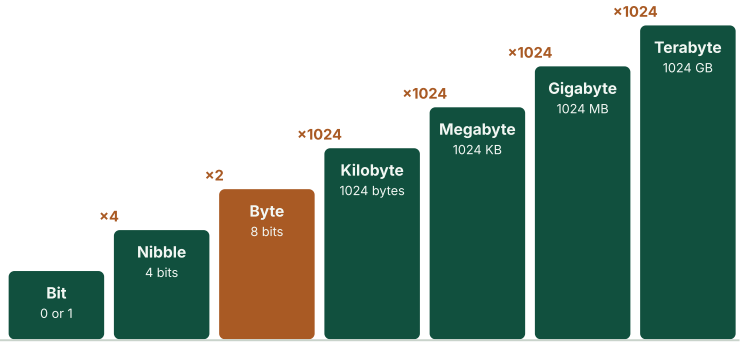
UNIT 1: DIGITAL TECHNOLOGY · DIGITAL DATA

Data, information and units of storage

L01 ANSWERS

I can: Define data and information and explain how processing turns one into the other · Explain why computers store everything in binary and how ASCII represents characters · Order the units of storage from bit to terabyte · Convert between units of storage and work out how many files will fit on a device



Moving to a bigger unit: divide by 1024. Moving to a smaller unit: multiply by 1024.

Use this diagram to help you: Units of storage as a staircase from bit to terabyte, with the multiplier between each step

A Key terms

Write the letter of the matching definition · 6 marks

E Data

A Information

D Bit

B Nibble

F Byte

C Kilobyte (KB)

A Data that has been processed and given context so that it has meaning.

B A group of 4 bits (half a byte). One hexadecimal digit can be stored in a nibble.

C 1024 bytes – enough for a short plain-text file.

D A binary digit: the smallest unit of digital data, either a 0 or a 1 (off or on).

E Raw facts and figures that have no meaning or context on their own.

F A group of 8 bits. One keyboard character (for example a letter, a digit or a punctuation mark) takes up 1 byte.

B Fill the gaps

Use the word bank · 8 marks

byte | information | nibble | data | ASCII | bit | context | 1024

Raw facts and figures with no meaning are called **data**. When data is processed and given **context** it becomes **information**. The smallest unit of storage is a **bit**, which is a single 0 or 1. A **nibble** is 4 bits and a **byte** is 8 bits, enough to store one character using a character set such as **ASCII**. One megabyte is **1024** kilobytes.

C True or false?

Tick one box · 9 marks

	True	False

A byte is 8 bits.	☑	☐
1 GB = 1024 KB. *(1 GB = 1024 MB. 1 MB = 1024 KB, so 1 GB = 1 048 576 KB.)*	☐	☑
'15, 22, 9' on its own is information. *(With no context it has no meaning, so it is data.)*	☐	☑
A terabyte is larger than a gigabyte.	☑	☐
A nibble can store one hexadecimal digit. *(A nibble is 4 bits, which gives 16 values (0–F) – exactly one hex digit.)*	☑	☐
To change bits into bytes you divide by 1024. *(There are 8 bits in a byte, so divide by 8. 1024 links bytes to kilobytes and the larger units.)*	☐	☑
In ASCII, 'A' and 'a' have the same binary code. *(Upper and lower case letters have different codes: A is 01000001 and a is 01100001.)*	☐	☑
One keyboard character is usually stored in 1 byte. *(A character such as a letter or digit is represented by 1 byte (8 bits).)*	☑	☐
An information system takes data as input and produces information as output.	☑	☐

D Exam-style questions

27 marks

1 Using an example of each, describe the difference between data and information. [4]

> Data is raw facts and figures with no meaning / context (1); suitable example, e.g. 18, 22, 19 (1); information is data that has been processed / given context so it has meaning (1); suitable example showing the context, e.g. 'Belfast maximum temperature on Monday was 18°C' (1).

2 Put these units in order from smallest to largest: megabyte, bit, terabyte, byte, nibble, kilobyte, gigabyte. [2]

> bit, nibble, byte, kilobyte, megabyte, gigabyte, terabyte. 2 marks for all in correct order; 1 mark if only one item is out of place.

3 (a) How many bits are in a nibble? (b) How many kilobytes are in a megabyte? (c) How many bytes are needed to store the word 'Fermanagh'? [3]

> (a) 4 (1) (b) 1024 (1) (c) 9 bytes – 9 characters, 1 byte each (1).

4 Place one of the following terms beside the correct definition. (Not all terms will be used.) BYTE, BIT, KILOBYTE. (a) The smallest unit of digital data, represented by either 0 or 1. (b) Describes eight binary digits. [2]

> (a) BIT (1) (b) BYTE (1).

5 (a) A file is 6400 bits in size. How many bytes are required to store the file? (b) A school report is 6144 KB. What is its size in MB? Show your working. [3]

> (a) 6400 ÷ 8 = 800 bytes (1). (b) 6144 ÷ 1024 (1) = 6 MB (1).

6 Convert 5 GB into MB. Show your working. [2]

> 5 × 1024 (1) = 5120 MB (1).

7 Niamh saves a short poem as a plain-text file. The file is 5 KB. Calculate how many bits are needed to store the file. Show your working. [2]

> 5 × 1024 = 5120 bytes (1); 5120 × 8 = 40 960 bits (1).

8 A youth club in Omagh buys a 32 GB USB flash drive to store video clips. Each clip is 512 MB. Calculate how many clips the drive can hold. Show your working. [3]

Convert to the same units: 32 × 1024 = 32 768 MB (1); 32 768 ÷ 512 (1) = 64 clips (1).

9 Cian types a message on his laptop in Coleraine and emails it to a friend, whose tablet shows exactly the same text. Explain how a character set such as ASCII makes this possible. [2]

Computers can only store/process binary, so each character (letter, digit, symbol) needs its own unique binary code (1); both devices use the same character set, so the codes Cian's laptop sends are turned back into the same characters on his friend's tablet (1). Max 2.

10 A GP surgery in Newry uses an information system to manage appointments. Describe how the system turns data into information, using examples from the surgery. [4]

Data is input, e.g. patient name, date of birth, requested time / doctor (1); the system processes the data, e.g. checks free slots, sorts / matches appointments (1); output is information with meaning, e.g. a daily appointment list for each GP / a reminder text to the patient (1); information gives context, e.g. 'Dr Kelly, 10:20, Mrs Byrne' rather than just times and codes (1).

E Quick quiz

Circle one answer · 9 marks

1 Which of these is information rather than data? [1]

A 14, 17, 21 — B Q7X
C Sam scored 17 out of 20 in the maths test ✓ — D 01101

2 A phone has 64 GB of free space. Each photo is 4 MB. About how many photos will fit? [1]

A 16 — B 256
C 16 384 ✓ — D 65 536

3 How many bits make one byte? [1]

A 4 — **B 8 ✓**
C 16 — D 1024

4 Which unit is the largest? [1]

A Gigabyte — B Megabyte
C Kilobyte — **D Terabyte ✓**

5 What is 2 MB in KB? [1]

A 2048 KB ✓ — B 1024 KB
C 16 KB — D 2000 KB

6 A file is 6400 bits. How many bytes is that? [1]

A 51 200 — B 6.25
C 800 ✓ — D 1600

7 Why does a computer use a character set such as ASCII? [1]

A To compress text files — **B To give each character a unique binary code ✓**
C To check spelling — D To translate between languages

8 Roughly how much video would 1 GB hold? [1]

A About 20 minutes of HD video ✓ — B About 1 second of HD video
C About 20 hours of HD video — D About 1 minute of audio

9 How many bytes are needed to store the word 'Strabane'? [1]

A 8 ✓ — B 7
C 64 — D 16

Data types

L02 ANSWERS

I can: Name and describe the five data types in the specification and give an example of each · Explain why choosing the correct data type matters for storage, processing and validation · Choose the most suitable data type for an item of data and justify the choice · Recognise common date and time formats

Field	Example value	Data type
Surname	Maguire	String
Age	15	Integer
Height (m)	1.62	Real
Form initial	B	Character
Date of birth	12/09/2010	Date/time
Phone	07700 900123	String

Phone numbers are strings: the leading 0 must be kept and no maths is done on them.

Use this diagram to help you: A pupil record with each field matched to its data type

A Key terms

Write the letter of the matching definition · 6 marks

C Data type

E Integer

B Real

A Date/time

F Character

D String / text

A Data type that stores dates and/or times so that they can be sorted, calculated with and shown in different formats.

B Data type for numbers that can have a decimal or fractional part, positive or negative. Real numbers also include whole numbers.

C The kind of data an item is (such as integer, real, date/time, character or string), which decides how it is stored and processed.

D Data type for a sequence of characters (letters, digits and symbols) of variable length.

E Data type for whole numbers, positive or negative, with no decimal or fractional part.

F Data type for a single letter, digit or symbol, stored using its code from the character set.

B Fill the gaps

Use the word bank · 7 marks

string · real · data type · character · integer · date/time · type check

Before data is processed it must be given a **data type** . A whole number such as 28 is stored as an **integer** , while a price such as 3.99 is stored as a **real** number. A single letter such as 'Y' is a **character** , and a name or postcode is stored as a **string** . A birthday is stored as **date/time** so that it can be sorted and displayed in different formats. A **type check** rejects data of the wrong kind.

C True or false?

Tick one box · 9 marks

	True	False
A postcode should be stored as a string.	☑	☐
Integers can store numbers with a decimal part. *(Integers are whole numbers only; numbers with a decimal part need the real data type.)*	☐	☑
Choosing the correct data type allows a type check to be used for validation.	☑	☐
A telephone number should be stored as an integer. *(It should be a string: the leading zero would be lost as an integer and phone numbers are not used in calculations.)*	☐	☑
'double' is a numeric data type that can store a decimal number. *(In C#, int stores whole numbers and double stores real numbers with a decimal part.)*	☑	☐
A score of 51 out of 100 is best stored as a string. *(It is a whole number that may be used in calculations, so integer is the most appropriate type.)*	☐	☑
-61 is an integer. *(Integers can be negative as long as they have no decimal part.)*	☑	☐
A string must always contain the same number of characters. *(A string has a variable length, e.g. 'Ann' has 3 characters and 'Londonderry' has 11.)*	☐	☑
12-Sep-2026 is an example of a medium date format.	☑	☐

D Exam-style questions

34 marks

1 State the most suitable data type for: (a) the number of pupils in a class (b) the price of a school lunch (c) a pupil's date of birth (d) a pupil's surname (e) a grade, e.g. A. [5]

> (a) integer (1) (b) real / currency (1) (c) date/time (1) (d) string / text (1) (e) character (1).

2 Explain why a mobile phone number should be stored as a string and not an integer. [2]

> A phone number starts with 0 and the leading zero would be lost if stored as an integer (1); it may contain spaces / a + sign, which integers cannot hold (1); it is never used in calculations (1). Max 2.

3 Describe the difference between the integer and real data types. [2]

> An integer is a whole number with no decimal / fractional part, e.g. 15 (1); a real number can have a decimal / fractional part, e.g. 15.75 (1).

4 Give two reasons why data must be defined as a particular data type before it is processed. [2]

> Any two from: decides what processing can be done, e.g. calculations / sorting (1) / data is stored efficiently / correct amount of storage used (1) / allows validation such as a type check (1) / data is displayed in the correct format (1).

5 A student scores 51 in a Computing module. State the most appropriate data type for storing the student's score. [1]

> Integer (1).

6 Describe each of the following data types and give an example of data that would be stored using it: (a) numeric (integer or real/double) (b) date/time (c) text (character or string). [6]

> For each part, description (1) + suitable example (1). (a) Numeric stores numbers that can be used in calculations: integer is a whole number, e.g. 28 pupils / real has a decimal part, e.g. 1.75 m. (b) Date/time stores a date or a time, e.g. 12/09/2026 or 15:45. (c) Character is a single letter/digit/symbol, e.g. 'Y'; string is a sequence of characters, e.g. 'Enniskillen'.

7 From the list 4, 6.5, -3.2, 9, 15, 42.7: (a) identify all the integers (b) explain why all of the values could be stored as real numbers. [3]

> (a) 4, 9, 15 (1 for all three, no extras) (b) Real numbers can include whole numbers as well as numbers with decimal parts (1); and can be positive or negative (1).

8 State three different ways a date could be formatted when it is displayed by a computer, giving an example of each. [3]

> Any three formats, each with a matching example (1 each): long date, e.g. Saturday, 12 September 2026 / medium date, e.g. 12-Sep-2026 / short date, e.g. 12/09/2026 / long time, e.g. 15:45:30 / medium time, e.g. 03:45 PM / short time, e.g. 15:45.

9 A sports club in Ballymena stores these details for each member: membership number (e.g. 00417), first name, date joined, annual fee (e.g. £85.50), number of matches played and whether they have paid (Y or N). State the most suitable data type for each item and justify your choice for membership number and annual fee. [8]

> Data types (1 each, max 6): membership number – string; first name – string; date joined – date/time; annual fee – real / currency; matches played – integer; paid – character / Yes/No (Boolean). Justifications (1 each): membership number is a string because the leading zeros must be kept and it is not used in calculations; annual fee is real/currency because money needs a decimal part / two decimal places.

10 A Year 9 pupil says 'The postcode BT74 6AA should be stored as an integer because it contains numbers.' Explain why the pupil is wrong. [2]

> The postcode contains letters and a space as well as digits (1); integers can only hold whole numbers / it is not used in calculations, so it must be stored as a string (1).

E Quick quiz

Circle one answer · 8 marks

1 Which data type is best for a postcode such as BT74 6AA? [1]

A Integer
B Real
C String ✓
D Character

2 Why is a mobile phone number not stored as an integer? [1]

A It is too long to store
B Integers cannot be sorted
C It uses less storage
D The leading zero would be lost ✓

3 Which data type should store a height of 1.75 m? [1]

A Real ✓
B Integer
C Character
D Date/time

4 A field stores a single letter for shirt size (S, M or L). Which data type? [1]

A String
B Character ✓
C Integer
D Real

5 Which item should be stored as an integer? [1]

A Price of a ticket
B Number of goals scored ✓
C Phone number
D Team name

6 'int' and 'double' are both examples of which kind of data type? [1]

A Text
B Date/time
C Numeric ✓
D Boolean

7 Which of these is a short date format? [1]

A Saturday, 12 September 2026
B 12-Sep-2026
C 15:45:30
D 12/09/2026 ✓

8 Which is NOT a reason for defining a data type before processing? [1]

A It makes the data more interesting to read ✓

B It decides what processing can be done

C It allows a type check

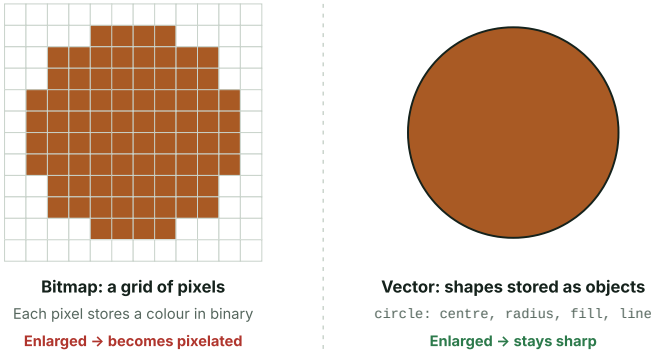
D It helps the data be stored efficiently

UNIT 1: DIGITAL TECHNOLOGY · DIGITAL DATA

Representing images: pixels, resolution, bitmap and vector

L03 ANSWERS

I can: Explain how a bitmap image is made of pixels, each stored as a binary colour code · Explain how resolution and colour depth affect image quality and file size · Calculate the file size of a simple bitmap image · Compare bitmap and vector graphics, giving a suitable use for each



Use this diagram to help you: Bitmap image as a grid of pixels compared with a vector circle stored as an object

A Key terms

Write the letter of the matching definition · 6 marks

B Pixel

E Bitmap graphic

C Vector graphic

D Resolution

F Colour depth

A Pixelation

A The blocky appearance of a bitmap image when it is enlarged too much, because the individual pixels become visible.

B The smallest element of a digital picture that a screen can show or an editor can change. Every pixel holds one colour, recorded as a binary code.

C An image stored as a list of objects (lines, curves and shapes) with properties such as position, size, line colour and fill colour. Can be enlarged without loss of quality. Example file type: SVG.

D The number of pixels in an image (width × height) or per inch. More pixels give a sharper image but a bigger file size.

E An image made of a grid of pixels, each of which can be a different colour. The colour code of every pixel is stored, plus metadata such as the dimensions. Pixelation occurs when it is enlarged. Example file type: JPEG.

F The number of bits used to store the colour of each pixel. n bits give 2^n colours.

B Fill the gaps

Use the word bank · 7 marks

colour depth | file size | pixels | vector | metadata | resolution | pixelated

A bitmap image is made of a grid of **pixels**, and the colour of each one is stored as a binary code. The number of bits used per pixel is the **colour depth**. Increasing the **resolution** gives a

sharper image but increases the **file size**. Data about the image, such as its width and height, is called **metadata**. A **vector** graphic is stored as objects and their properties, so it can be enlarged without loss of quality. Bitmap images become **pixelated** when enlarged too much.

C True or false?

Tick one box · 9 marks

	True	False
Vector graphics lose quality when enlarged. *(Vector graphics are recalculated when resized, so they stay sharp. Bitmaps lose quality.)*	☐	☑
Photographs are best stored as bitmaps.	☑	☐
Higher resolution means a smaller file. *(More pixels must be stored, so higher resolution means a larger file.)*	☐	☑
Vector graphics store shapes with properties such as fill colour.	☑	☐
A 24-bit colour image uses 24 bits to store each pixel. *(24-bit colour uses 8 bits each for red, green and blue.)*	☑	☐
A 2-bit image can use 4 different colours. *(2 bits give $2^2 = 4$ codes: 00, 01, 10, 11.)*	☑	☐
The colour codes alone are enough to rebuild a bitmap image. *(Metadata such as the width and height is also needed – otherwise you would not know where each row ends.)*	☐	☑
An image with 5 colours needs only 2 bits per pixel. *(2 bits only give 4 codes, so 3 bits are needed (8 possible codes).)*	☐	☑
SVG is an example of a vector file type. *(Scalable Vector Graphics stores shapes, not pixels.)*	☑	☐

D Exam-style questions

32 marks

1 Define the following terms: (a) pixel (b) image resolution. [2]

> (a) The smallest unit / element of a digital image that can be shown on screen and edited (1). (b) The number of pixels in an image, given as width × height / pixels per inch (1).

2 Explain how increasing the resolution of an image affects its quality and its file size. [3]

> More pixels are used (1); so the image is sharper / shows more detail / higher quality (1); but more data must be stored so the file size increases (1).

3 Describe how a vector graphic is stored. [2]

> Stored as a list of objects / shapes such as lines, curves and circles (1); with their properties, e.g. position / coordinates, size, line colour, fill colour, line thickness (1).

4 Describe two differences between bitmap and vector-based graphics. [4]

> Two differences, each with both sides (2 each): a bitmap is stored as a grid of pixels whereas a vector is stored as objects and properties / a bitmap pixelates when enlarged whereas a vector can be resized without loss of quality / bitmap files are usually larger whereas vector files are usually smaller / a bitmap is edited pixel by pixel whereas a vector is edited object by object / bitmaps suit photographs whereas vectors suit logos and diagrams. 1 mark only if one side given.

5 Ruairí takes a small photo of his team from the club website and stretches it to fill an A3 poster. The printed photo looks blocky. Explain why this happens. [2]

> The photo is a bitmap made of a fixed number of pixels; enlarging it makes each pixel bigger, with no new detail added (1); the pixels become visible as small square blocks – pixelation – so the image looks blocky / loses

quality (1).

6 Describe what needs to be stored so that a bitmap image can be saved on a storage device and displayed again. [2]

The colour code (binary value) of each pixel (1); metadata, e.g. width and height / resolution / colour depth, so the image can be rebuilt (1).

7 A pupil designs a 10 × 10 pixel image using 4 colours. Calculate the minimum file size of the image data in bytes. Show your working. [3]

4 colours need 2 bits per pixel (1); 10 × 10 × 2 = 200 bits (1); 200 ÷ 8 = 25 bytes (1).

8 A photo for a school newsletter is 800 × 600 pixels and uses 24-bit colour. Calculate the size of the image data in kilobytes. Show your working. [3]

800 × 600 × 24 = 11 520 000 bits (1); ÷ 8 = 1 440 000 bytes (1); ÷ 1024 = 1406.25 KB (1).

9 An image uses 16 different colours. State how many bits are needed to store each pixel and explain why. [2]

4 bits (1); because $2^4 = 16$, so 4 bits give exactly 16 different codes / 3 bits would only give 8 (1).

10 A designer is creating a company logo for a Derry bakery that will be printed on paper bags and on a large shop sign. Recommend a graphic type and justify your choice. [3]

Vector (1); can be resized from a small bag print to a large sign without losing quality / pixelating (1); smaller file size / objects such as text and shapes can be edited separately, e.g. changing colours (1).

11 A Belfast estate agent wants to put house photographs and a map of each area on its website. Discuss the use of bitmap and vector graphics for the website. In your answer, refer to how each is stored, quality and file size. [6]

Indicative content: photographs should be bitmaps because they contain many colour variations; bitmaps store a grid of pixels with a colour code for each; high-resolution photos look sharp but have large file sizes, so pages load slowly unless compressed; enlarging small photos causes pixelation. Maps suit vectors: stored as objects (roads, boundaries, labels) with properties; can be zoomed without losing quality; smaller files load quickly; objects can be updated easily. Levels: 5–6 marks – both types discussed with storage, quality and file size, clearly linked to the website; 3–4 marks – both types discussed but some points not developed or not linked to context; 1–2 marks – a few relevant points, mostly about one type.

E Quick quiz

Circle one answer · 9 marks

1 What happens when a bitmap image is enlarged a lot? [1]

A It stays sharp
B It becomes a vector
C It becomes pixelated ✓
D The file size halves

2 Which image is best stored as a vector graphic? [1]

A A school logo ✓
B A holiday photograph
C A scanned X-ray
D A webcam image

3 What does a bitmap image store for each pixel? [1]

A A shape
B A colour value in binary ✓
C A line thickness
D A list of coordinates for curves

4 Increasing the resolution of an image will... [1]

A reduce the file size
B not change the file size
C turn it into a vector
D increase the file size ✓

5 How many colours can a 2-bit colour image use? [1]

A 2
B 4 ✓
C 8
D 16

6 A 10 × 10 image uses 4 colours. What is the size of the image data? [1]

A 25 bytes ✓
B 100 bytes
C 200 bytes
D 40 bytes

7 Which of these is metadata stored with a bitmap image? [1]

A The colour of pixel 1
B The image's width and height ✓
C The fill colour of a circle
D The printer used

8 Which is a property stored in a vector graphic? [1]

A Fill colour of a shape ✓
B Pixel colour depth
C Number of pixels per inch
D Compression ratio

9 Which file type is a vector graphic? [1]

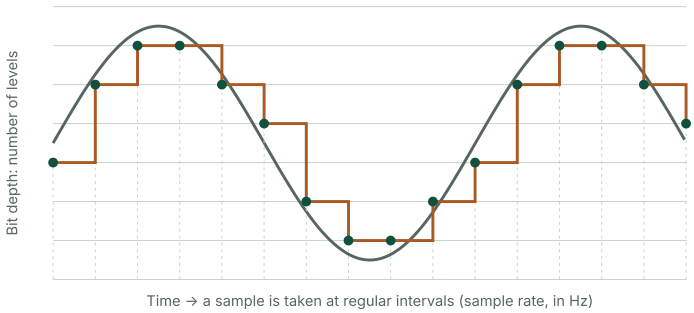
A JPEG
B PNG
C GIF
D SVG ✓

UNIT 1: DIGITAL TECHNOLOGY · DIGITAL DATA

Representing sound; buffering and streaming

L04 ANSWERS

I can: Explain why sound must be converted from analogue to digital and describe what an ADC does · Describe how sample rate, bit depth and bit rate affect quality and file size · Calculate the bit rate and the file size of a simple sound recording · Explain how streaming and buffering let a video play while it downloads



More samples and more levels → closer to the original sound → bigger file

Use this diagram to help you: A sound wave sampled at regular intervals, with sample rate on the time axis and bit depth as the number of levels

A Key terms

Write the letter of the matching definition · 6 marks

B Analogue signal

E Analogue-to-digital converter (ADC)

C Sampling

A Sample rate

F Bit depth

D Bit rate

A The number of samples of a sound taken each second, measured in hertz (Hz). CD audio uses 44 100 Hz (44.1 kHz).

B A signal that varies smoothly and continuously, representing a physical quantity such as a sound wave.

C Measuring the level (amplitude) of an analogue signal at regular time intervals; each reading is rounded to a whole-number level and stored in binary.

D The number of bits of data processed for each second of sound, usually in kilobits per second; for one channel it equals sample rate × bit depth.

E A device that converts an analogue signal, such as sound from a microphone, into digital (binary) data by sampling it at regular intervals.

F How many bits are allocated to every sound sample. More bits give more possible levels, so each sample is recorded more accurately.

B Fill the gaps

Use the word bank · 8 marks

hertz · sample · bit depth · bit rate · ADC · sample rate · analogue · buffer

Sound is an **analogue** signal, so it must be converted to binary by an **ADC** before a computer can store it. Each measurement of the wave is called a **sample**. The number of measurements taken per second is the **sample rate**, measured in **hertz**. The number of bits

allocated to every single measurement is the **bit depth**, and the number of bits processed per second is the **bit rate**. When streaming a video, the next part is stored temporarily in a **buffer** so playback carries on smoothly.

C True or false?

Tick one box · 9 marks

	True	False
A higher sample rate gives a smaller file. *(A higher sample rate means more samples to store, so the file is larger (but the quality is better).)*	☐	☑
Streaming lets media play before the whole file arrives.	☑	☐
Bit rate is measured in bits per second.	☑	☐
A buffer is a permanent storage device. *(A buffer is an area of memory that holds data temporarily until it is needed.)*	☐	☑
Standard CD audio takes 44 100 samples per second.	☑	☐
Each sample is rounded to the nearest whole-number level before it is stored as binary.	☑	☐
For one channel, bit rate = sample rate + bit depth. *(The values are multiplied: bit rate = sample rate × bit depth.)*	☐	☑
A bit depth of 16 bits gives 65 536 possible values for each sample.	☑	☐
Doubling the length of an uncompressed recording halves its file size. *(File size is proportional to length, so doubling the length doubles the file size.)*	☐	☑

D Exam-style questions

32 marks

1 Explain why analogue-to-digital conversion is needed when recording sound on a computer. [2]

> Sound is an analogue signal / continuous wave (1) / computers can only store and process digital (binary) data, so the sound must be converted (1).

2 State the purpose of an analogue-to-digital converter (ADC). [1]

> To convert an analogue signal (e.g. from a microphone) into digital / binary data (by sampling it) (1).

3 What is meant by sample rate? State the unit it is measured in. [2]

> The number of samples taken per second (when recording sound) (1) / measured in hertz (Hz) (1).

4 Describe how increasing the bit depth affects a sound recording. [2]

> More bits are used for each sample, so there are more possible values / levels (1) / each sample is stored more accurately so the sound quality improves (1) / but the file size increases (1). Max 2.

5 Distinguish between bit depth and bit rate. [2]

> Bit depth is how many bits are allocated to each sample (1) / bit rate is the number of bits processed per second of sound (1).

6 Standard CD audio is sampled at 44.1 kHz with a bit depth of 16 bits. Calculate the bit rate for one channel. Show your working. [2]

> Bit rate = sample rate × bit depth (1) / 44 100 × 16 = 705 600 bits per second (705.6 kbps) (1).

7 Orla records a 2-minute mono podcast intro for her school radio club at a sample rate of 22 050 Hz and a bit depth of 16 bits. Calculate the file size in bytes, and then in megabytes to two decimal places. Show your working. [4]

> Formula: sample rate × bit depth × channels × time (1) / time = 2 × 60 = 120 s (1) / 22 050 × 16 × 1 × 120 = 42 336 000 bits; ÷ 8 = 5 292 000 bytes (1) / 5 292 000 ÷ 1024 ÷ 1024 = 5.05 MB (1). Accept 5.29 MB if 1 MB = 1 000 000 bytes is stated.

8 Explain how buffering allows a film to be streamed smoothly on a smart TV. [3]

> A buffer is an area of (temporary) memory (1) / the next part of the film is downloaded into the buffer while the current part is being watched (1) / if the connection slows or is briefly lost, the film keeps playing from the data in the buffer, so there are no interruptions (1).

9 Give one advantage and one disadvantage of streaming a film compared with downloading it. [2]

> Advantage, any one: can start watching almost immediately / does not use much storage space on the device / no need to wait for the whole file (1). Disadvantage, any one: needs a constant internet connection / may pause if the buffer empties / quality drops on slow connections / uses data every time it is watched (1).

10 Explain how sound is recorded and stored on a computer. [6]

> Indicative content: sound is an analogue wave; a microphone converts it to an analogue electrical signal; an ADC samples the signal at regular intervals; each sample's amplitude is rounded to a whole-number level; each level is stored as a binary number; the sample rate is the number of samples per second (Hz); the bit depth is the number of bits per sample; higher sample rate / bit depth give better quality but larger files. Level 3 (5–6): clear, ordered description using correct terms including ADC, sample rate and bit depth. Level 2 (3–4): describes most stages with some correct terms. Level 1 (1–2): limited, mentions recording/converting with few terms.

11 A music teacher in Omagh wants to record pupils' performances for their GCSE Music coursework. Discuss the choice of sample rate and bit depth for these recordings, referring to sound quality, file size and how the files will be stored and shared. [6]

> Indicative content: higher sample rate (e.g. 44.1 kHz or more) captures the wave more accurately; higher bit depth (16 or 24 bits) gives more levels and less noise; music for assessment needs good quality so moderators hear the performance accurately; higher settings give much larger files (e.g. CD-quality stereo ≈ 10 MB per minute); larger files need more storage on the school network or cloud and take longer to upload/share; could record at high quality then save a compressed copy (e.g. MP3) for sharing; lower settings save space but may sound dull or noisy. Level 3 (5–6): balanced discussion of quality and file size with a justified conclusion. Level 2 (3–4): some discussion of both factors. Level 1 (1–2): simple points, little linking.

E Quick quiz

Circle one answer · 8 marks

1 What does an analogue-to-digital converter (ADC) do when sound is recorded? [1]

A Converts digital data into sound for the speakers
B Compresses the sound file to make it smaller
C Samples the analogue signal and turns each sample into a binary number ✓
D Stores the next part of a song in memory

2 What is the bit rate of a mono sound sampled at 8 000 Hz with an 8-bit depth? [1]

A 16 000 bps
B 64 000 bps ✓
C 8 008 bps
D 1 000 bps

3 What is the unit of sample rate? [1]

A Hertz (Hz) ✓
B Bits
C Bytes
D Pixels

4 Which change makes each sample more accurate? [1]

A Lower sample rate
B Fewer channels
C Lower bit rate
D Higher bit depth ✓

5 Why is some detail lost when sound is sampled? [1]

A The microphone is digital

B What the wave does between samples is not recorded ✓

C Binary cannot store numbers

D The buffer is too small

6 A 10-second stereo clip is recorded at 44 100 Hz with 16-bit samples. How many bits is it? [1]

A 705 600

B 1 411 200

C 7 056 000

D 14 112 000 ✓

7 What is a buffer? [1]

A A cable that speeds up downloads

B A compressed video format

C Memory that temporarily holds data before it is used ✓

D A type of microphone

8 What happens when a streamed video stops and shows a loading icon? [1]

A The buffer has run out of data ✓

B The file has finished downloading

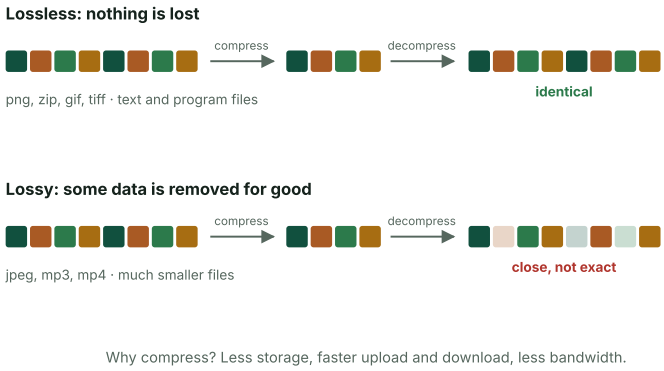
C The bit depth is too high

D The ADC has stopped working

Data portability, file formats and compression

L05 ANSWERS

I can: Define data portability and explain why standard file formats support it · Describe the purpose and typical use of each of the 16 file formats in the specification · Choose a suitable file format for a task and justify the choice · Explain why data is compressed and compare lossy and lossless compression



Use this diagram to help you: Lossless compression restores every block of data; lossy compression permanently removes some

A Key terms

Write the letter of the matching definition · 6 marks

D **Data portability**

A **File format**

E **Compression**

B **Lossy compression**

F **Lossless compression**

C **CSV**

A The standard way data is organised inside a file, shown by its file extension (e.g. .jpg, .csv), which determines which programs can open it.

B Compression that reduces file size by permanently removing some data, so some quality is lost and the original cannot be fully restored, e.g. jpeg, mp3.

C Comma-separated values: a text format in which each line is a record and the fields are separated by commas; used to move table data between spreadsheets and databases.

D Being able to move data between computers, devices or software applications without typing it in again, usually by saving it in a standard file format.

E Reducing the size of a file so it needs less storage and can be uploaded or downloaded more quickly.

F Compression that reduces file size without losing any data, so the file can be restored exactly as it was, e.g. zip, png.

B Fill the gaps

Use the word bank · 7 marks

pdf | lossless | portability | compression | csv | Lossy | midi

Moving data between applications without retyping it is called data **portability** . A **csv** file stores table data with commas between the fields, while a **pdf** keeps its layout the same on any device. Making a file smaller is called **compression**. **Lossy** compression permanently

removes some data, as in a jpeg photo, but **lossless** compression, such as zipping a file, lets the original be rebuilt exactly. The **midi** format stores musical instructions rather than recorded sound.

C True or false?

Tick one box · 9 marks

	True	False
midi stores recorded sound. *(midi stores instructions such as notes, instruments and timing, not recorded sound.)*	☐	☑
jpeg uses lossy compression.	☑	☐
tiff is often used for high-quality printing.	☑	☐
Compressing a file increases the time it takes to upload. *(Compressed files are smaller, so they upload and download faster.)*	☐	☑
Zipping a file with WinZip is an example of lossless compression.	☑	☐
wav is an uncompressed audio format.	☑	☐
wma was developed by Apple for iTunes. *(wma (Windows Media Audio) was developed by Microsoft for Windows Media Player.)*	☐	☑
Lossy compression is suitable for a spreadsheet of pupils' marks. *(Lossy compression would permanently change data; spreadsheets need lossless compression.)*	☐	☑
A pdf keeps the fonts and layout of a document the same on any device.	☑	☐

D Exam-style questions

38 marks

1 What is meant by data portability? [2]

> Being able to move data between computers / devices / software applications (1) / without it needing to be typed in again (1).

2 Name a suitable file format for each: (a) a photo for a website (b) a document that must look the same on every device (c) data exported from a database to a spreadsheet (d) an animated image with few colours. [4]

> (a) jpeg (accept png) (1) / (b) pdf (1) / (c) csv (1) / (d) gif (1).

3 Describe the difference between a midi file and an mp3 file. [2]

> A midi file stores instructions / musical data such as notes, instrument and timing (1) / an mp3 stores recorded sound in compressed form (1). Accept midi files are much smaller.

4 Give two reasons why data is compressed. [2]

> Any two from: reduces storage space needed (1) / faster to upload / download / transfer (1) / less bandwidth needed, smoother streaming (1) / fits within email attachment limits (1).

5 Explain the difference between lossy and lossless compression. [2]

> Lossy compression permanently removes some data, so quality is reduced / cannot be restored (1) / lossless compression reduces file size without losing any data, so the original can be recreated exactly (1).

6 Explain why rtf may be chosen instead of a word processor's own file format. [2]

rtf can be opened by different word processors / on different operating systems (1) / while keeping basic formatting such as fonts and bold (1).

7 Explain why a user would compress a file before emailing it, and state which type of compression should be used for a spreadsheet of exam results. [3]

Smaller file so it uploads / sends faster (1) / may be under the attachment size limit (1) / lossless compression should be used so no data / marks are lost or changed (1).

8 For each of the following file formats, state whether it is normally compressed or uncompressed: jpeg, wav, mp3, txt. [4]

jpeg – compressed (1) / wav – uncompressed (1) / mp3 – compressed (1) / txt – uncompressed (1).

9 A camera shop in Enniskillen takes photos of its stock. Compare jpeg and tiff as formats for (a) photos on the shop's website and (b) photos printed in a large poster for the window. [4]

(a) jpeg: lossy compression gives small files (1) so pages load quickly on the website (1). (b) tiff: keeps high quality / detail with no loss (1) which is needed when the image is enlarged and printed, although files are large (1).

10 Daniel needs to send his CV to several employers, who use different word processors and computers. Name two text or document file formats he could use and describe why each would suit him. [4]

Any two suitable formats, 1 for name and 1 for a description linked to the CV, e.g. pdf (1) fonts and layout look exactly the same on any device and it is hard to alter (1) / rtf (1) keeps basic formatting such as bold and fonts and opens in most word processors on any operating system (1) / txt (1) opens in any program on any computer, although all formatting is lost (1). Do not credit csv (unsuitable for a CV).

11 A 3-minute song is stored as an uncompressed wav file of about 30 MB and as an mp3 of about 2.75 MB. Explain why the mp3 is smaller and describe one drawback of using the mp3. [3]

mp3 uses lossy compression (1) / it removes sounds / data that most listeners would not notice (1) / drawback: some quality is permanently lost / cannot be restored to the original / not ideal for editing (1).

12 A local council in Northern Ireland wants to put a planning document, photographs of the sites and a table of planning decisions on its website for the public. Discuss suitable file formats for each type of data, referring to portability and compression. [6]

Indicative content: planning document as pdf – layout and fonts fixed, opens on any device, cannot easily be altered by the public; photographs as jpeg – lossy compression gives small files that download quickly, slight quality loss acceptable on screen; png if diagrams/plans need sharp lines; table of decisions as csv – can be opened in any spreadsheet/database, so the public or journalists can analyse it; standard formats give portability across Windows/Apple/phones; compression reduces bandwidth and server storage; avoid tiff/wav-style large uncompressed files for web. Level 3 (5–6): formats for all three data types with justified links to portability and compression. Level 2 (3–4): suitable formats for at least two types with some justification. Level 1 (1–2): formats named with little or no justification.

E Quick quiz

Circle one answer · 9 marks

1 What is meant by data portability? [1]

A Carrying a laptop between rooms
B Moving data between computers or applications without re-entering it ✓
C Making a file as small as possible
D Storing data in the cloud

2 Which audio format stores musical instructions rather than recorded sound? [1]

A wav
B mp3
C wma
D midi ✓

3 Which format stores data with each field separated by a comma? [1]

A rtf
B csv ✓
C tiff
D wma

4 Which image format supports simple animation? [1]

A gif ✓
B tiff
C jpeg
D pict

5 Lossy compression... [1]

A keeps every bit of the original
B always increases file size
C permanently removes some data ✓
D is used for program files

6 Which format keeps a document's layout the same on any device? [1]

A txt
B csv
C midi
D pdf ✓

7 Which format contains no formatting and can be opened by any word processor? [1]

A txt ✓
B rtf
C pdf
D pict

8 Which format is best for a logo that needs a transparent background and no loss of quality? [1]

A jpeg
B mp4
C png ✓
D csv

9 Which format was created by Microsoft to play audio and video in sync? [1]

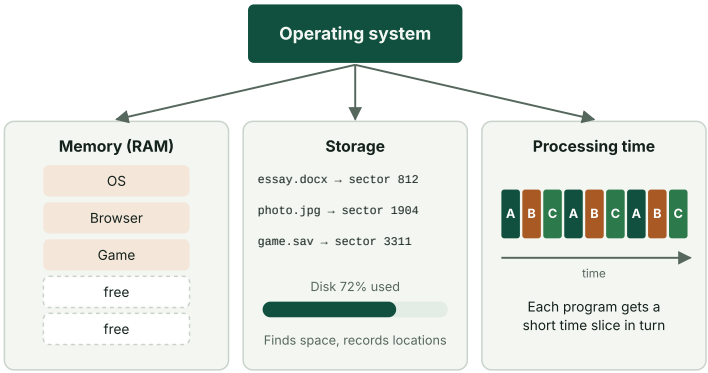
A mp3
B avi ✓
C midi
D wav

UNIT 1: DIGITAL TECHNOLOGY · SOFTWARE

System software and modes of processing

L06 ANSWERS

I can: Distinguish between system software and application software, and between packaged and tailored software · Describe how the operating system allocates memory, storage and processing time · Describe real-time, batch and multi-user processing with examples · Describe how batch processing uses a master file and a transaction file to produce bills



Use this diagram to help you: The operating system allocating memory, storage and processing time

A Key terms

Write the letter of the matching definition · 6 marks

B System software

C Application software

E Operating system

A Boot-up sequence

F Real-time processing

D Batch processing

A The steps carried out when a computer is switched on: instructions in ROM check the hardware and load the operating system from storage into RAM.

B Software that runs and manages the computer hardware and acts as the interface between the hardware and application programs, e.g. the operating system, utilities and device drivers.

C Software designed to let the user carry out a specific task, such as word processing, spreadsheets or web browsing.

D Similar data is collected into batches over a period and processed together later, without human involvement, often overnight, e.g. bills and payroll.

E System software that manages the computer's resources – allocating memory, storage and processing time – and lets the hardware and application software work together.

F Each input is processed the moment it is entered, so the output is produced in time to affect the next input, e.g. ticket booking or air traffic control.

B Fill the gaps

Use the word bank · 8 marks

time slice | real-time | batch | operating system | master file | transaction file | ROM | multi-user

The **operating system** is system software that allocates memory, storage and processing time. It gives each task a **time slice** of the processor in turn. When a computer is switched on, instructions stored

in **ROM** check the hardware and load the operating system. Ticket booking systems use **real-time** processing so that seats are updated immediately. Electricity bills are produced by **batch** processing, which matches a temporary **transaction file** against the permanent **master file**. Many users sharing one processor using the round robin method is a **multi-user** system.

C True or false?

Tick one box · 9 marks

	True	False
Batch processing needs a person to check each record as it is processed. *(Batch processing runs without human involvement once the batch has been collected.)*	☐	☑
Time slicing gives each program a share of the processor's time.	☑	☐
The boot-up instructions are stored in RAM. *(They are stored in ROM, which is non-volatile and keeps its contents when the power is off.)*	☐	☑
Airline bookings use real-time processing.	☑	☐
A device driver is system software.	☑	☐
In batch processing the transaction file is kept permanently after processing. *(The transaction file is temporary; the permanent file is the master file, which is updated from it.)*	☐	☑
In a multi-user system the OS uses a round robin approach to share the processor.	☑	☐
Tailored software is usually cheaper than packaged software. *(Tailored software is built for one organisation, which bears the whole cost, so it is usually more expensive.)*	☐	☑
Higher-priority tasks can be given more time slices.	☑	☐

D Exam-style questions

48 marks

1 Describe the difference between system software and application software. [2]

> System software runs / manages the computer hardware, e.g. the operating system (1) / application software lets the user carry out a specific task, e.g. word processing (1).

2 Classify each software example as either system software (S) or application software (A): Windows 11 operating system; Microsoft Excel spreadsheet; patient record database; disk defragmenter software; device driver; Google Chrome web browser. [6]

> Windows 11 – S (1) / Excel – A (1) / patient record database – A (1) / disk defragmenter – S (1) / device driver – S (1) / Google Chrome – A (1).

3 Describe two differences between packaged software and tailored software. [4]

> Any two differences, 2 marks each (point + contrast): packaged is made for many users, tailored for one organisation's needs (2) / packaged is cheaper because cost is shared, tailored is expensive (2) / packaged is available immediately, tailored takes time to develop (2) / packaged has lots of support / tutorials, tailored relies on the developer (2).

4 Describe how system software manages memory. [2]

> Programs / data are loaded from storage into RAM (main memory) (1) / the OS finds free space and allocates it, keeps programs separate and frees it when the program closes (1).

5 Describe how system software manages storage. [2]

> The OS decides where files are saved on the storage device and keeps a record of their locations (1) / frees space when files are deleted / controls access permissions (1).

6 Describe how system software allocates processing time. [2]

> Processor time is divided into time slices (1) / each task is given slices in turn, with higher-priority tasks given more (1).

7 Describe what happens during the boot-up sequence. [3]

> Instructions stored in ROM run when the computer is switched on (1) / the hardware is checked / tested (1) / the operating system is loaded from storage into RAM (1).

8 Omar runs a small business in Newry manufacturing parts for bikes. His staff have their own computer systems. (a) What do the letters CPU stand for in relation to a computer system? (b) List two resources managed by the operating system in a computer system. [3]

> (a) Central Processing Unit (1). (b) Any two from: memory / RAM (1) / storage (1) / processing time / CPU time (1) / input and output devices (1).

9 The Grand Opera House in Belfast sells theatre tickets online. Describe the main features of the real-time processing its booking system uses. [4]

> Data is processed immediately as it is input (1) / output is produced quickly enough to affect the next input (1) / files are updated at once (1) / features linked to the context, e.g. as soon as a seat is bought it is marked as sold, so a second customer cannot book the same seat (1).

10 A district council in Northern Ireland pays its staff once a month. Describe the main features of batch processing as used for the council's payroll. [4]

> Data is collected in groups / batches over a period of time (1) / processed together at a later / agreed time, e.g. overnight (1) / without human involvement (1) / linked to context, e.g. hours / pay details for all staff are collected during the month and every payslip is produced in one run at month end (1).

11 An electricity company uses batch processing to produce customer bills. Describe how the bills are produced, referring to the master file and the transaction file. [5]

> Customer details are stored in a master file (1) / meter readings and account numbers are entered into a transaction file (1) / records are matched using the account number (1) / bill is calculated as units used × price per unit and all bills are produced together in a batch (1) / the transaction file is used to bring the master file up to date for the next billing period (1).

12 Describe how a multi-user system allows many users to share one processor. [3]

> Many users at different terminals share one powerful processor (1) / the OS gives each user a time slice in turn (round robin) (1) / the processor switches so quickly that each user seems to have sole access (1).

13 Explain the difference between real-time processing and multi-user processing. [2]

> Real-time processing is about processing each input immediately so it affects the next input (1) / multi-user processing is about many users sharing one processor using time slices (1).

14 A leisure centre in Derry~Londonderry wants a computer system for (i) online booking of badminton courts and (ii) paying its staff each month. Discuss which mode of processing should be used for each task, justifying your choices. [6]

> Indicative content: (i) court booking should be real-time – availability must update immediately so the same court is not double-booked; customers get instant confirmation; many customers may book at once so it may also be multi-user. (ii) payroll should be batch – lots of similar records, pay only needed once a month, can run overnight without staff, uses a master file of employee details and transaction file of hours worked; real-time would be unnecessary. Level 3 (5–6): correct mode for both with clear justification linked to the scenario. Level 2 (3–4): correct modes with limited justification, or one well justified. Level 1 (1–2): modes named with little or no justification.

E Quick quiz

Circle one answer · 9 marks

1 An airline booking system must mark a seat as sold the moment it is booked. Which mode of processing is this? [1]

A Batch processing
B Real-time processing ✓
C Lossy processing
D Offline processing

2 Which task is best suited to batch processing? [1]

A Booking a concert seat
B Controlling traffic lights
C Producing monthly electricity bills ✓
D Playing an online game

3 Which of these is system software? [1]

A A spreadsheet
B A web browser
C A photo editor
D An operating system ✓

4 What does the OS do when allocating processing time? [1]

A Gives each task a time slice of the processor ✓
B Defragments the disk
C Encrypts files
D Backs up data

5 Where are the first instructions run at boot-up stored? [1]

A RAM
B Cache
C The cloud
D ROM ✓

6 In batch processing, which file is permanent or semi-permanent? [1]

A Master file ✓
B Transaction file
C Buffer
D Temporary file

7 A school's pupil-records database written just for it is... [1]

A packaged software
B tailored software ✓
C utility software
D freeware

8 In a multi-user system, how is the processor shared? [1]

A Users queue until the first user logs off
B Each user gets a time slice in turn (round robin) ✓
C Each user gets their own processor
D Jobs are saved and run overnight

9 What does CPU stand for? [1]

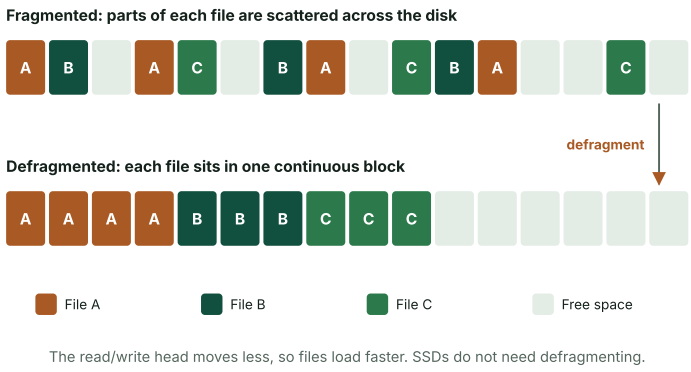
A Computer Personal Unit
B Central Program Utility
C Central Processing Unit ✓
D Control Processing Unit

UNIT 1: DIGITAL TECHNOLOGY · SOFTWARE

Utility applications and antivirus software

L07 ANSWERS

I can: Explain why files become fragmented and how a defragmenter speeds up file access · Describe what a task scheduler does, including round robin sharing of processor time · Describe full and incremental backups, how data is restored and why backups are needed · Explain how antivirus software works and why it must be updated regularly



Use this diagram to help you: Disk blocks before and after defragmentation

A Key terms

Write the letter of the matching definition · 6 marks

B Utility software

D File fragmentation

A Disk defragmenter

E Task scheduler

F Round robin

C Backup

A A utility that rearranges the fragments of files on a disk so each file is stored in continuous blocks and free space is grouped together, speeding up file access.

B A program that carries out one specific task to help the operating system maintain, protect or speed up the computer, e.g. defragmenter, task scheduler, backup, antivirus.

C A copy of data stored separately, preferably off-site, so it can be restored if the original is lost or damaged.

D When a file's data blocks are stored in separate, non-adjacent places on a disk because there is not enough continuous free space.

E A utility that shares processor time between tasks (e.g. round robin) and runs jobs such as backups, virus scans or updates automatically at set times.

F A scheduling method in which each task or user is given a time slice of the processor in turn, returning to the first after the last.

B Fill the gaps

Use the word bank · 7 marks

signatures | task scheduler | incremental backup | quarantine | backup | disk defragmenter | updated

A **disk defragmenter** moves the scattered blocks of each file together so it can be read faster. A **task scheduler** can run jobs such as backups automatically at set times. A copy of data kept separately in case the original is lost is a **backup**. An **incremental backup** only copies data that has changed

since the last backup. Antivirus software compares files with a database of virus **signatures** and moves infected files into **quarantine**. It must be **updated** regularly so that new viruses can be detected.

C True or false?

Tick one box · 9 marks

	True	False
Defragmenting can improve the speed of a hard disk drive.	☑	☐
Antivirus software never needs updating once it is installed. *(It must be updated regularly so its database includes the signatures of new viruses.)*	☐	☑
Restoring means copying data back from a backup.	☑	☐
A task scheduler only runs a job when the user clicks on it. *(A task scheduler runs jobs automatically at a set time or when an event occurs.)*	☐	☑
WinZip is an example of utility software.	☑	☐
Defragmentation puts the parts of each file back into continuous blocks.	☑	☐
A full backup only copies files that have changed. *(That is an incremental backup; a full backup copies all the data.)*	☐	☑
Backups should be stored in the same room as the server for convenience. *(Backups should be stored off-site or in the cloud so a fire, flood or theft cannot destroy both copies.)*	☐	☑
Antivirus software can move an infected file into quarantine so it cannot run.	☑	☐

D Exam-style questions

38 marks

1 The operating system on John's computer includes a utility application. What is the function of the following two tasks carried out by utility software? (i) Disk defragmenting (ii) Task scheduling [4]

> (i) Defragmenting: files become split into fragments stored in different places on the disk (1) / the defragmenter rearranges the fragments so each file is stored in continuous blocks / free space grouped together (1). (ii) Task scheduling: divides processor time between tasks / users, e.g. round robin (1) / or runs jobs such as backups / virus scans automatically at set times (1).

2 Explain why files become fragmented on a hard disk drive. [3]

> The OS breaks files into blocks and stores them in any free space (1) / as files are saved and deleted, free space is left in small gaps spread across the disk (1) / so a new large file has to be split between several non-adjacent gaps (1).

3 Describe how disk defragmenting improves the performance of a computer. [3]

> Moves the blocks of each file so they are stored together / contiguously (1) / groups free space together (1) / the read/write head moves less, so files are read / saved more quickly (1).

4 Explain why a solid-state drive should not be defragmented. [2]

> An SSD has no moving parts, so access time is the same wherever data is stored and there is no speed gain (1) / defragmenting causes extra write operations that wear out the SSD / shorten its life (1).

5 Describe how the round robin method is used to share a processor between several users. [3]

Each user / computer is given a small time slice of processor time (1) / in turn, one after another; after the last it starts again with the first (1) / the processor is so fast each user appears to have uninterrupted access (1).

6 Give two examples of tasks that could be scheduled to run automatically. [2]

Any two from: backup (1) / virus scan (1) / software or OS updates (1) / disk defragmentation (1) / disk clean-up (1).

7 Why do we need to back up our data? [3]

Any three from: hard disk may fail (1) / virus / ransomware may destroy or lock data (1) / user may accidentally delete files (1) / fire, flood or theft may destroy the computer (1) / so the data can be restored and the organisation can continue working (1).

8 Describe the difference between a full and an incremental backup. [2]

A full backup copies all the data (1) / an incremental backup copies only the data that has changed since the last backup (1).

9 State three places where data can be backed up to. [3]

Any three from: external hard drive (1) / cloud storage (1) / another server at a different site (1) / USB flash drive (1) / magnetic tape (1) / NAS on the network (1).

10 Describe how antivirus software detects and deals with a virus. [3]

Scans files / emails / downloads / storage devices (1) / compares them against a database of known virus signatures, or looks for suspicious behaviour (1) / removes, repairs or quarantines infected files (1).

11 Explain why antivirus software must be updated regularly. [2]

New viruses are created all the time (1) / the software can only detect viruses whose signatures are in its database, so without updates new viruses are not detected (1).

12 Explain why it is important to keep software up to date. [2]

Updates / patches fix security weaknesses that viruses and hackers could exploit (1) / and fix bugs or add new features / keep virus definitions current (1).

13 A GP surgery in Ballymena stores all its patient records on one server. Discuss a suitable backup and restore procedure for the surgery, referring to the type of backup, how often it should happen, where the backup should be stored and how utility software can help. [6]

Indicative content: patient records change daily, so daily backups at least; full weekly backup plus nightly incremental to save time and space; scheduled overnight using the task scheduler when the surgery is closed and the network is quiet; copies stored off-site or in secure (encrypted) cloud storage so fire/flood/theft cannot destroy both; more than one copy kept; backups tested by doing trial restores; restore by loading the latest full backup then each incremental in order; antivirus kept updated to reduce risk of ransomware; patient data is sensitive personal data so backups must be kept secure under UK GDPR. Level 3 (5–6): coherent procedure covering type, frequency, location and utility support, linked to the surgery. Level 2 (3–4): several relevant points with some linking. Level 1 (1–2): isolated points.

E Quick quiz

Circle one answer · 8 marks

1 Why does a disk defragmenter make a hard disk drive faster? [1]

A It deletes unwanted files

B It stores each file's blocks together so the read/write head moves less ✓

C It compresses every file

D It scans for viruses

2 Why must antivirus software be updated regularly? [1]

A To make the computer start faster

B To free up disk space

C So it can recognise the signatures of newly created viruses ✓

D So it can defragment the disk

3 An incremental backup copies... [1]

A only files changed since the last backup ✓ B everything on the drive

C only the operating system D only photos

4 Which utility would run a backup automatically every night? [1]

A Defragmenter B Firewall

C Compiler **D Task scheduler ✓**

5 After defragmenting, the blocks A1 B3 A2 A4 B2 A3 B1 B4 become... [1]

A A1 B1 A2 B2 A3 B3 A4 B4 B B1 B2 B3 B4 A1 A2 A3 A4

C A1 A2 A3 A4 B1 B2 B3 B4 ✓ D Unchanged

6 Which is NOT a reason to back up data? [1]

A The hard drive may fail B A virus may destroy the data

C A fire may destroy the building **D It makes the processor faster ✓**

7 In round robin scheduling, what happens after the processor serves the last computer? [1]

A It switches off **B It starts again with the first computer ✓**

C It waits for the user to log off D It serves the busiest computer only

8 What is a virus signature? [1]

A A pattern of code that identifies a particular virus ✓ B The name of the person who wrote the virus

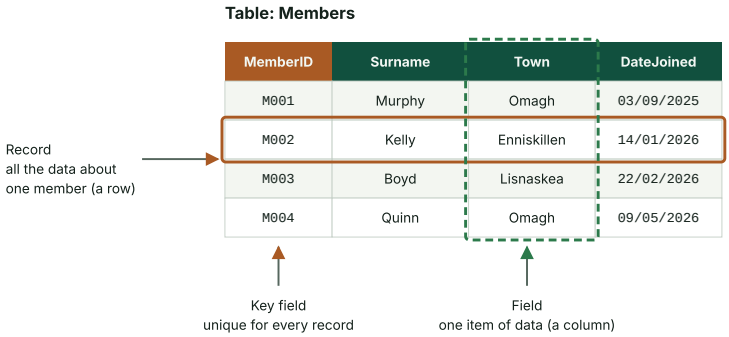
C A password to unlock infected files D A digital certificate on a website

UNIT 1: DIGITAL TECHNOLOGY · DATABASE APPLICATIONS

Database concepts

L08 ANSWERS

I can: Define table, record, field and key field and identify them in a given table · Explain why a key field must be unique and choose a suitable key field · Explain how tables are linked by a relationship using a primary key and a foreign key · Describe what queries, forms, reports and macros are used for · Explain what importing data means and why it is useful



Use this diagram to help you: A database table showing a record, a field and the key field

A Key terms

Write the letter of the matching definition · 6 marks

B Database

C Table

A Flat file

E Record

F Field

D Key field / primary key

A A database that holds all its data in a single two-dimensional table, which often leads to repeated data.

B An organised, structured collection of related data that can be searched, sorted and presented, usually held in one or more tables.

C A database structure that stores data about one type of thing, organised in rows (records) and columns (fields).

D A field whose value is unique for every record, used to identify each record in a table.

E All of the related data about one object or person, forming a complete row in a table. A record is made up of fields.

F A single item of data stored about every object or person, forming a column in a table and identified by its field name.

B Fill the gaps

Use the word bank · 8 marks

field | key field | record | relationship | query | tables | foreign key | report

A database stores data in **tables**. Each row is a **record**, which holds all the data about one item, and each column is a **field**. The **key field** uniquely identifies each record. When the primary key of one table is stored in another table it is called a **foreign key**, and this creates a

relationship . A **query** searches for records that match criteria, and a **report** presents the results in a formatted layout.

C True or false?

Tick one box · 9 marks

	True	False
A field is a row in a table. *(A field is a column. A row is a record.)*	☐	☑
A key field must contain a unique value for every record.	☑	☐
Forms are used to enter, edit and view records.	☑	☐
A relational database stores all its data in one table. *(That is a flat file. A relational database uses several linked tables.)*	☐	☑
A composite key is a primary key made up of two or more fields.	☑	☐
A foreign key must be unique in the table it appears in. *(Many patients can share the same DoctorID. It is the primary key in the Doctor table that must be unique.)*	☐	☑
Enforcing referential integrity stops a record being linked to a parent record that does not exist.	☑	☐
A report is used to search the database for records that match criteria. *(A query searches; a report presents data from a table or query in a formatted layout.)*	☐	☑
Importing data from a csv file saves time and avoids retyping errors.	☑	☐

D Exam-style questions

34 marks

1 Define the terms (a) record and (b) field. [2]

> (a) All the data about one person or item / one row in a table (1). (b) One item of data stored about each record / one column in a table (1).

2 Look at the Erne Valley Swimming Club Members table on the slides. (a) State how many records the table contains. (b) Identify the key field and justify your choice. (c) Explain why Surname would not be a suitable key field. [4]

> (a) 4 (1). (b) MemberID (1) because each member has a different/unique ID (1). (c) Two members can have the same surname, e.g. Maguire (1) so it would not uniquely identify a record (1).

3 Describe the purpose of a form in a database, and give two ways a form can help to keep the data accurate. [3]

> Purpose: an on-screen layout used to enter new records / edit or view existing records (1). Any two from: drop-down lists limit choices (1) / validation rules on fields reject unsuitable data (1) / input masks control format (1) / only the fields needed are shown so fewer mistakes (1) / data is saved to the correct table automatically (1).

4 A leisure centre in Newry wants a printed list of all members whose membership ends this month, grouped by membership type. Explain how a query and a report would each be used to produce this. [4]

> Query: searches the members table using criteria (1) e.g. membership end date between the first and last day of this month (1). Report: uses the query results as its source (1) and presents them in a formatted, printable layout grouped by membership type (1).

5 The receptionist at the leisure centre adds new members and prints a welcome letter many times each day. Explain one benefit of using a macro for this task. [2]

> Any one explained, e.g.: the steps are carried out with one button click (1) so it saves the receptionist time on a task repeated many times a day (1) / the actions are always done in the same order (1) so fewer mistakes are made (1).

6 A swimming club in Lisburn has 200 new members' details in a spreadsheet from its online sign-up form. Explain what importing this data into the club database would involve, and give one benefit of importing rather than typing it in. [2]

> Bringing the data into a database table from the other file, e.g. the spreadsheet or a csv saved from it, so each row becomes a record (1). Benefit: much quicker than retyping 200 members / avoids transcription errors (1).

7 Explain what is meant by a foreign key. Use a Doctor table and a Patient table as an example. [2]

> A foreign key is the primary key of one table stored as a field in another table (1). e.g. DoctorID is the primary key of Doctor and is stored in Patient as a foreign key to link each patient to their doctor (1).

8 One doctor has many patients, but each patient has only one doctor. State the type of relationship between the Doctor and Patient tables and explain one benefit of enforcing referential integrity on it. [3]

> One-to-many (1). Benefit explained, e.g.: a patient cannot be given a DoctorID that does not exist in the Doctor table (1) so the data in the two tables stays consistent / no orphan records (1). Or: a doctor cannot be deleted while patients are still linked to them (1) (1).

9 What is meant by a composite key? Give an example. [2]

> A primary key made up of two or more fields together (1). Example, e.g. MemberID + SessionDate in an attendance table / PatientID + AppointmentDate (1).

10 A youth club in Strabane stores all its data in one flat file. Each time a member books an event, their name, address and phone number are typed again. Explain two advantages of using a relational database instead. [4]

> Any two explained, e.g.: member details are stored once only (1) so less data redundancy / less storage wasted (1) / when details change they are updated in one place (1) so data stays consistent / no conflicting copies (1) / less typing for each booking (1) so fewer input errors (1).

11 A school library in Omagh is designing a database with three tables: Pupils, Books and Loans. Discuss how the database should be designed, referring to tables, key fields, relationships and the database tools the librarian would use. [6]

> Indicative content: Pupils table with key PupilID; Books table with key BookID/ISBN plus copy number; Loans table with LoanID; PupilID and BookID as foreign keys in Loans; one-to-many from Pupils to Loans and from Books to Loans; referential integrity so a loan cannot refer to a pupil or book that does not exist; suitable fields and data types e.g. DateBorrowed, DueDate; form for issuing/returning books with drop-downs; query to find overdue loans (DueDate < today and not returned); report listing overdue books by form class; macro buttons to issue a book or print reminders; import pupil list from the school MIS as csv. Levels: 5–6 marks a well-structured discussion covering tables, keys, relationships and tools with correct terms and context; 3–4 marks several relevant points with some explanation; 1–2 marks isolated points or definitions only.

E Quick quiz

`Circle one answer · 8 marks`

1 In a database table, what is one row called? [1]

A A field
B A record ✓
C A query
D A macro

2 A swimming club stores Surname, Squad, DateOfBirth and MemberID. Which is the best key field? [1]

A MemberID ✓
B Surname
C Squad
D DateOfBirth

3 What is a query used for? [1]

A Entering data one record at a time
B Printing a formatted summary
C Searching for records that match criteria ✓
D Recording keystrokes

4 A set of stored instructions that runs a database task automatically when a button is clicked is a... [1]

A form
B relationship
C field
D macro ✓

5 When the primary key of one table appears as a field in a second table, it is called a... [1]

A composite key
B foreign key ✓
C lookup
D macro

6 One doctor has many patients; each patient has one doctor. Where does the ∞ or crow's foot go? [1]

A At the Doctor end
B At both ends
C At the Patient end ✓
D Nowhere – it is one-to-one

7 What is the main problem with storing everything in a flat file? [1]

A Data is repeated, wasting space and causing inconsistencies ✓
B It cannot be printed
C It cannot be sorted
D It cannot have a key field

8 Which tool would present club members grouped by squad, ready to print? [1]

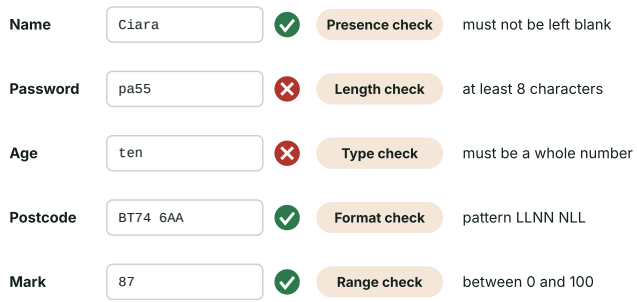
A Form
B Macro
C Relationship
D Report ✓

UNIT 1: DIGITAL TECHNOLOGY · DATABASE APPLICATIONS

Database data types and validation

L09 ANSWERS

I can: Choose and justify a suitable data type for each field in a table · Explain why validation is needed and what it cannot do · Describe presence, length, type, format and range checks with examples · Write validation rules and input masks for database fields · Explain the difference between validation and verification



Use this diagram to help you: Five validation checks applied to form fields, with passes and failures

A Key terms

Write the letter of the matching definition · 6 marks

C Data type

D Validation

B Presence check

A Length check

F Type check

E Format check

A A check that makes sure the data entered has an exact number of characters, or is no shorter or longer than set limits.

B A check that makes sure data has been entered in a required field, so the field cannot be left blank.

C The kind of data a field can hold, e.g. Short Text, Number, Currency or Date/Time. It controls what can be entered and what can be done with the data.

D An automatic check made by the computer as data is input, to make sure the data is sensible, reasonable and inside allowed limits. Data that breaks the rules is rejected; validation cannot confirm the data is correct.

E A check that makes sure the data entered matches a set pattern, e.g. a postcode or an email address containing @.

F A check that makes sure the data entered is of the correct data type, e.g. numeric, text or date.

B Fill the gaps

Use the word bank · 8 marks

format | presence | Date/Time | data type | range | length | Validation | correct

Each field in a table is given a **data type**, such as Currency for money or **Date/Time** for a booking date. **Validation** is an automatic check that data is sensible. A **presence** check makes sure a field is not left blank. A **length** check counts the number of characters. A **format** check

makes sure data matches a pattern such as a postcode. A **range** check makes sure a value is between two limits. Validation cannot tell whether data is **correct**.

C True or false?

Tick one box · 9 marks

	True	False
Validation proves that data is accurate. *(Validation only checks data is sensible and allowed. A valid value can still be wrong.)*	☐	☑
A range check suits a field that stores an exam mark from 0 to 100.	☑	☐
A format check suits a postcode.	☑	☐
A type check counts the number of characters in a field. *(That is a length check. A type check makes sure the data is the correct data type.)*	☐	☑
A telephone number should be stored with the Number data type. *(It should be Short Text: it is never used in calculations and may start with 0, which a Number field would remove.)*	☐	☑
The Currency data type stores money values with two decimal places.	☑	☐
The Lookup Wizard creates a list of values that can be picked from a drop-down list.	☑	☐
In Access, Len([Ability])>7 is an example of a range check. *(Len counts characters, so it is a length check.)*	☐	☑
Typing an email address twice so the computer can compare them is a form of verification.	☑	☐

D Exam-style questions

43 marks

1 Why is validation used when data is entered into a database? [2]

> Any two from: to check data is sensible/reasonable/within acceptable limits (1) / to reduce input errors (1) / so that invalid data is rejected at the input stage (1) / so the database is more accurate/reliable (1).

2 Explain why validation cannot guarantee that data is correct. Use an example. [2]

> Validation only checks that data is allowed/follows the rules, not that it is the true value (1). Example, e.g. a date of birth typed as 12/05/1984 instead of 21/05/1984 is still a valid date and is accepted (1).

3 Describe a range check and give an example of a field where one would be used. [2]

> Checks that a value lies between a lower and an upper limit (1). Example, e.g. an exam mark must be from 0 to 100 / age 11 to 19 (1).

4 Name and describe a suitable validation check for each field: (a) Postcode (b) Surname (c) Number of tickets (maximum 10). [6]

> (a) Format check (1): postcode must match a pattern such as BT followed by digits, a space, a digit and two letters (1). (b) Presence check (1): surname cannot be left blank (1). Accept type check: must contain letters only (1)(1). (c) Range check (1): number of tickets must be from 1 to 10 / <=10 (1). Accept type check: must be a whole number (1)(1).

5 Describe the difference between a length check and a format check. [2]

> Length check: checks the number of characters entered is within limits or an exact number, e.g. password at least 8 characters (1). Format check: checks the data matches a set pattern of characters, e.g. postcode BT74 7AB (1).

6 Describe how a presence check can be set up for a field in a database table. [2]

In Design View select the field and set the Required property to Yes (1) / enter the validation rule Is Not Null (1) and add validation text explaining that the field must be filled in (1). Max 2.

7 A Film Characters table has a field called Ability which must be at least 8 characters long. Write a validation rule for this field and name the type of check. [2]

Len([Ability])>=8 (accept Len([Ability])>7) (1). Length check (1).

8 A group must have a minimum of 5 and a maximum of 20 members. Write a validation rule for the GroupSize field and name the type of check. [2]

>=5 And <=20 (accept Between 5 And 20 / >4 And <21) (1). Range check (1).

9 State the most suitable data type for each field: (a) a URL for a character's fan page (b) a photo stored with each record (c) the price of a cinema ticket (d) a unique number generated automatically for each new record. [4]

(a) Hyperlink (1). (b) Attachment (accept OLE Object) (1). (c) Currency (1). (d) AutoNumber (1).

10 A youth football club in Dungannon stores PlayerPhone, DateOfBirth, ShirtNumber and SubsPaid for each player. State the most suitable data type for each field and justify each choice. [8]

PlayerPhone: Short Text (1) – starts with 0/contains spaces/not used in calculations (1). DateOfBirth: Date/Time (1) – allows age to be calculated / players sorted by age group (1). ShirtNumber: Number (integer) (1) – whole number; allows range check / sorting (accept Short Text with justification that it is not used in calculations) (1). SubsPaid: Yes/No (1) – only two possible values (1). (Accept Currency for SubsPaid if justified as amount paid.)

11 A school canteen in Coleraine records how much money each pupil tops up, from £1 to £50. Write a validation rule for the TopUp field and suitable validation text. [2]

Rule: >=1 And <=50 (accept Between 1 And 50) (1). Validation text, e.g. 'Top-up must be from £1 to £50' (1).

12 Explain the difference between validation and verification. Describe one method of verification. [3]

Validation: an automatic check by the computer that data is sensible/follows rules (1). Verification: checks that data has been entered/copied accurately, matching the original (1). Method, e.g. double entry – data typed twice and the computer compares them / visual check of the screen against the source document (1).

13 A GP surgery in Ballymena is setting up a database to book patient appointments. Discuss how data types and validation checks can be used to make the appointment data as accurate as possible, and explain the limits of validation. [6]

Indicative content: data types – PatientID AutoNumber or Short Text key, health and care number Short Text, AppointmentDate Date/Time, AppointmentTime Date/Time, Reason Long Text, NewPatient Yes/No, DoctorName via Lookup Wizard drop-down so names are consistent; type checks come from data types; presence check on PatientID and date; range check that the appointment date is not in the past (>=Date()) and time within surgery hours; length check on the 10-digit health and care number; format check/input mask on phone number; validation text to explain errors; limits – validation cannot detect valid but wrong data, e.g. wrong date of birth or wrong but valid phone number; verification such as reading details back to the patient or double entry. Levels: 5–6 marks well-developed discussion of both data types and several checks applied to the context, with clear explanation of the limits of validation; 3–4 marks some relevant data types and checks with some context; 1–2 marks basic list of checks or types with little context.

E Quick quiz

Circle one answer · 8 marks

1 Which validation check makes sure a field is not left empty? [1]

A Range check
B Presence check ✓
C Format check
D Type check

2 Which data type is best for a monthly club fee of £24.99? [1]

A Currency ✓
B Number (integer)

C Short Text

D Date/Time

3 Checking that an email address contains an @ symbol is a... [1]

A type check

B range check

C format check ✓

D presence check

4 Which statement about validation is true? [1]

A It guarantees data is correct

B It is carried out by a person reading the data

C It encrypts the data

D It cannot tell if data is correct, only if it is sensible ✓

5 Which validation rule accepts a group of 5 to 20 people, including 5 and 20? [1]

A >5 And <20

B <5 Or >20

C >=5 And <=20 ✓

D Len([GroupSize])>4

6 Which data type stores a file made in other software, such as a spreadsheet, using object linking and embedding? [1]

A Hyperlink

B Calculated

C Lookup Wizard

D OLE Object ✓

7 Which data type automatically creates a new sequential number for each record? [1]

A AutoNumber ✓

B Number

C Yes/No

D Short Text

8 In an Access input mask, what does the character L mean? [1]

A A digit is required

B A letter is required ✓

C A lower-case letter is optional

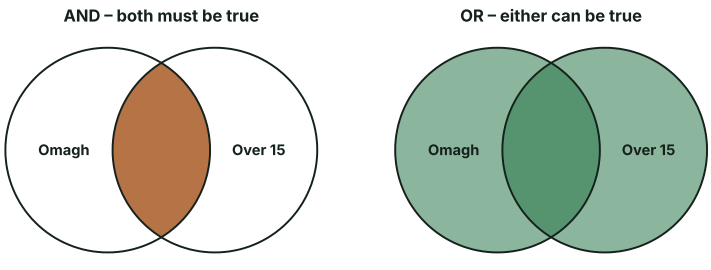
D Any character is allowed

UNIT 1: DIGITAL TECHNOLOGY · DATABASE APPLICATIONS

Queries, logical operators and big data

L10 ANSWERS

I can: Write query criteria using comparison operators and AND, OR and BETWEEN · Set up a query in a query design grid and predict which records it returns · Explain big data using volume, velocity and variety · Tell structured and unstructured data apart · Explain why data analytics is needed to make use of big data



`Town = "Omagh" AND Age > 15` narrows the results

`Town = "Omagh" OR Age > 15` widens the results

BETWEEN 10 AND 20 includes both 10 and 20.

Use this diagram to help you: Venn diagrams: AND returns records in both sets, OR returns records in either

A Key terms

`Write the letter of the matching definition · 6 marks`

B Query

A Criteria

D Comparison operator

E Logical operator

F BETWEEN

C Group by

A The conditions a record must meet to be included in the results of a query, e.g. Age >= 18.

B A way of searching a database to extract the records that meet set criteria. It checks each record and lists the data that satisfies the criteria.

C Organises query or report results by the values of a field, e.g. grouping members by squad.

D A symbol used to compare a value with other data, giving true or false: < less than, <= less than or equal to, > greater than, >= greater than or equal to, = equal to, <> not equal to.

E An operator such as AND, OR or BETWEEN that combines or tests conditions and gives a true or false (Boolean) result.

F A logical operator that selects values within a range, including both end values.

B Fill the gaps

`Use the word bank · 8 marks`

variety | OR | BETWEEN | query | AND | velocity | criteria | analytics

A **query** extracts records that meet set **criteria**. The **AND** operator needs both conditions to be true, while **OR** needs at least one to be true. **BETWEEN** finds values within a range, including both ends. Big data is described by volume, **velocity** and **variety**. Using software to find patterns and trends in big data is called data **analytics**.

C True or false?

Tick one box · 9 marks

	True	False
OR returns records where at least one of the conditions is true.	☑	☐
Velocity refers to the amount of data stored. *(Velocity is the speed at which data is created and must be processed. The amount is volume.)*	☐	☑
BETWEEN 10 AND 20 includes the value 20.	☑	☐
Big data is easy to analyse by hand. *(It is too large, fast and varied; it needs data analytics software and powerful computers.)*	☐	☑
The comparison operator <> means 'not equal to'.	☑	☐
The criteria Goals > 8 returns a player who has scored exactly 8 goals. *(> means greater than, so 8 is not included. Use >= 8 to include it.)*	☐	☑
In a query design grid, criteria typed on the same row are joined with AND.	☑	☐
Social media posts and essays are examples of structured data. *(They are unstructured: they have no fixed format of fields and records.)*	☐	☑
Data mining software is used to find hidden patterns in very large data sets.	☑	☐

D Exam-style questions

34 marks

1 Write the criteria to find all members who are aged 18 or over. [1]

> >=18 (in the Age field) (1). Accept Age >= 18 or >17.

2 Write the criteria to find all bookings made between 1 March 2026 and 31 March 2026. [2]

> Between #01/03/2026# And #31/03/2026# (1 for BETWEEN or >= and <= used, 1 for both correct dates). Accept >=#01/03/2026# And <=#31/03/2026#.

3 Explain the difference between the AND and OR operators in a query. [2]

> AND: both/all conditions must be true for a record to be returned (1). OR: only one/at least one condition needs to be true (1).

4 Describe 'big data' by making reference to volume, velocity and variety. [6]

> Two marks for each V: Volume – the very large amount of data (1) e.g. terabytes of posts stored every day (1). Velocity – the speed at which data is generated and must be processed (1) often in real time, e.g. card payments checked instantly (1). Variety – many different types/formats of data (1) e.g. text, images, video, sensor data; structured and unstructured (1).

5 Explain why data analytics is needed to interpret big data. [2]

> Big data is too large/complex/fast to analyse by normal means or by hand (1). Analytics finds patterns/trends/relationships (1) that help the organisation make decisions (1). Max 2.

6 Describe the difference between structured and unstructured data. Give an example of each. [4]

> Structured: organised in a fixed format/tables of fields and records so it is easy to search and analyse (1) e.g. dates of birth, addresses, card numbers (1). Unstructured: no fixed format/predefined structure so it cannot easily be processed with normal database tools (1) e.g. social media posts, emails, videos (1).

7 Using the Fermanagh Junior Football League Players table on the slides, write the criteria to find every player who is not in the Enniskillen club, and state the PlayerIDs it returns. [2]

> Club <> "Enniskillen" (accept <>"Enniskillen" in the Club column) (1) / returns P01, P03, P04 (1).

8 Using the Fermanagh Junior Football League Players table on the slides, state the PlayerIDs returned by each query: (a) Goals < 9 (b) Club = "Enniskillen" AND Age = 14 (c) Age = 13 OR Goals >= 12. [3]

> (a) P02, P03 (1). (b) P05 (1). (c) P01, P03, P05 (1).

9 A garden centre in Lisburn stores products in a table with the fields ProductName, Category, Price and InStock. Write criteria to find all products in the category "Plants" that cost less than £20. [2]

> Category = "Plants" (1) AND Price < 20 (1). In a grid: "Plants" under Category and <20 under Price on the same Criteria row.

10 The garden centre manager wants a list of products costing from £10 to £25 inclusive, sorted by price, showing only ProductName and Price. Describe how the query should be set up in a query design grid. [4]

> Any four from: add ProductName and Price to the Field row (1) / criteria under Price: Between 10 And 25 or >=10 And <=25 (1) / Sort row under Price set to Ascending (or Descending) (1) / Show ticked for ProductName and Price only; untick any other fields used (1) / run the query to see the results (1).

11 A supermarket chain in Northern Ireland collects data from loyalty cards, self-service tills, its app and social media. Discuss how this is an example of big data and how data analytics could help the business. [6]

> Indicative content: volume – millions of transactions each week from tills and app, stored over years; velocity – data arrives continuously and in real time, e.g. online orders and stock levels updated instantly; variety – structured till and loyalty-card data, unstructured social media posts, reviews and images, app location data; normal database tools cannot cope so specialist analytics needed; analytics finds buying patterns, e.g. which products are bought together; personalised offers to loyalty-card holders; stock planning to reduce waste and avoid empty shelves; spotting trends from social media; deciding store layouts or opening hours; may mention cost of software, cloud computing and skilled staff, and the need to follow UK GDPR when using customer data. Levels: 5–6 marks clear discussion of all three Vs in context and at least two well-explained benefits of analytics; 3–4 marks some Vs explained and at least one benefit; 1–2 marks basic definitions or list of points with little context.

E Quick quiz

Circle one answer · 8 marks

1 Which operator finds values from 10 to 20, including 10 and 20? [1]

A OR
B <
C BETWEEN ✓
D <>

2 Which of the three Vs of big data describes the speed at which data is created and processed? [1]

A Velocity ✓
B Volume
C Variety
D Validation

3 Town = "Omagh" AND Age > 15 returns records where... [1]

A either condition is true
B neither condition is true
C only Town is checked
D both conditions are true ✓

4 Social media posts, videos and GPS data collected together are an example of... [1]

A volume
B variety ✓
C velocity
D validation

5 Does Score >= 50 return a record with a score of exactly 50? [1]

A Yes ✓
B No
C Only with OR
D Only with BETWEEN

6 Which criteria finds every player except those in the Senior squad? [1]

A Squad = "Senior"

B Squad > "Senior"

C Squad <> "Senior" ✓

D Squad BETWEEN "Senior"

7 Which of these is unstructured data? [1]

A A table of dates of birth

B Customer addresses in a database

C A list of exam marks

D Video clips and social media posts ✓

8 What is data analytics? [1]

A Typing data into a form

B Backing up a database

C Using software to find patterns, trends and relationships in data ✓

D Checking data as it is entered

UNIT 1: DIGITAL TECHNOLOGY · SPREADSHEET APPLICATIONS

Spreadsheets 1: structure, formatting, formulae and cell referencing

L11 ANSWERS

I can: Identify cells, rows, columns, ranges and worksheets, and give a cell reference correctly · Choose a suitable data format for a column and describe formatting that improves presentation · Write formulae and simple functions using the correct symbols and range syntax · Explain and use relative and absolute cell references when replicating formulae



Use this diagram to help you: Spreadsheet formulas copied down: relative references change, the absolute reference stays fixed

A Key terms

Write the letter of the matching definition · 6 marks

B Spreadsheet

D Cell range

C Label

E Value

F Formula

A Spreadsheet function

A A built-in formula that does a specific job, e.g. SUM, AVERAGE, MAX, MIN, COUNT, IF and VLOOKUP.

B Application software that organises data in a grid of rows and columns so it can be calculated, analysed and presented. A spreadsheet file (workbook) contains one or more worksheets.

C Text used to identify the contents of a row or column, e.g. Name, Age.

D A block of adjacent cells referred to together, written first cell:last cell, e.g. B5:B12.

E A number entered into a spreadsheet so that calculations can be carried out, e.g. 5, 23.

F An instruction in a cell to carry out a calculation using operators and cell references, e.g. =A1+B1. It always starts with = and the cell shows the result, which updates when the referenced cells change.

B Fill the gaps

Use the word bank · 8 marks

cell · MIN · range · SUM · worksheets · formula · replication · absolute

A spreadsheet can contain several **worksheets**, each made of rows and columns. A **cell** is identified by its column letter and row number, such as C4, and a block of cells such as B2:B10 is called a **range**. Every **formula** starts with an equals sign. The **SUM** function adds the

values in a range and **MIN** returns the smallest value. Copying a formula down a column is called **replication** . A reference written with dollar signs, such as $F$1, is an **absolute** reference because it does not change when copied.

C True or false?

Tick one box · 9 marks

	True	False
$A$1 is an absolute reference.	☑	☐
Formulae use x for multiply. *(Spreadsheets use * for multiply and / for divide.)*	☐	☑
A spreadsheet can contain more than one worksheet.	☑	☐
A label is a number entered so that calculations can be carried out. *(That is a value; a label is text that identifies a row or column, e.g. Name.)*	☐	☑
=SUM(A1:A10) adds A1 and A10 only. *(The colon means a range: it adds every cell from A1 through to A10.)*	☐	☑
When a formula is replicated, relative references change to match the row being used.	☑	☐
The fill handle is the small square at the bottom right of the selected cell.	☑	☐
COUNT counts every cell in a range that is not empty. *(COUNT counts only cells containing numbers; COUNTA counts every non-empty cell.)*	☐	☑
Merge and centre combines selected cells into one cell and centres the contents.	☑	☐

D Exam-style questions

48 marks

1 Describe the difference between a row and a column in a spreadsheet. [2]

> A row runs horizontally / across the sheet and is identified by a number (1) / A column runs vertically / down the sheet and is identified by a letter (1)

2 A sports club records match fees in cells B2 to B12. Write a function to add the values in B2 to B12. [1]

> =SUM(B2:B12) (1) – equals sign, function and correct range all needed

3 Write a function to find the highest value in cells C2 to C20. [1]

> =MAX(C2:C20) (1)

4 Write a function to find the lowest value in cells C2 to C20. [1]

> =MIN(C2:C20) (1)

5 Write a function to calculate the arithmetic mean (average) of the values in A1 to A10. [1]

> =AVERAGE(A1:A10) (1)

6 Write a formula for cell B7 that adds the cells B1 to B6 together. [1]

> =SUM(B1:B6) (1) – accept =B1+B2+B3+B4+B5+B6

7 State the 'golden rule' for entering a formula and write a formula to divide the contents of A3 by the contents of A4. [2]

> A formula must begin with an equals sign = (1) / =A3/A4 (1)

8 A youth club budget sheet holds the money available in I1 and the money spent so far in I2. Write a formula for I3 to show how much money is left. If I1 is £640.00 and I2 is £415.50, state the value shown in I3. [2]

> =I1-I2 (1) / £224.50 (1)

9 A Bangor sailing club records course bookings in a sheet with the columns Booked By, Course, Places, Price Per Place, Member Discount and Amount Due. State the most suitable data format for (a) Booked By, (b) Price Per Place, (c) Member Discount. [3]

> (a) Text (1) / (b) Currency (1) / (c) Percentage (1) – accept Currency for (c) if the discount is a money amount off

10 Cell E2 contains =C2*D2. State the formula in E3 after E2 is replicated into E3. [1]

> =C3*D3 (1)

11 Cell C1 contains =A1*B1. The formula is replicated down to C16 using the fill handle. (a) State the formula in C9. (b) Name the type of cell referencing used. [2]

> (a) =A9*B9 (1) / (b) Relative (cell referencing) (1)

12 Describe what is meant by replicating formulae and give one benefit. [2]

> Copying a formula into other cells, e.g. down a column using the fill handle, so the references adjust to each row (1) / Benefit: saves time / reduces errors from retyping / consistent formulae (1)

13 Explain why an absolute cell reference would be used for a VAT rate stored in cell H1. [2]

> Every formula must use the same VAT rate in H1 (1) / so the reference must not change when the formula is replicated / an absolute reference $H$1 stays fixed (1) – accept: the rate can be changed in one place and all results update

14 In a bakery's order sheet, cell E2 contains =D2*(1-$H$1) and is replicated down to E6. Cell H1 holds the discount rate. Explain why $H$1 has been used. [2]

> $H$1 is an absolute reference that does not change when the formula is copied down to E6 (1) / so every order uses the same discount rate in H1; a relative H1 would become H2, H3... which are blank, giving wrong results (1)

15 Explain the difference between relative and absolute cell referencing. Use an example of each. [4]

> Relative: the reference changes when replicated to match the new row/column (1); example: =B2*C2 becomes =B3*C3 (1) / Absolute: the reference stays the same when replicated, shown with $ signs (1); example: $F$1 holding a VAT rate in =G2*(1+$F$1) (1)

16 A school tuck shop sheet has takings of £27.00, £24.00, £16.50, £39.60 and £13.20 in cells D2 to D6. State the result of (a) =SUM(D2:D6), (b) =MAX(D2:D6), (c) =MIN(D2:D6). [3]

> (a) £120.30 (1) / (b) £39.60 (1) / (c) £13.20 (1)

17 A teacher has entered test scores in D2 to D30, but some pupils were absent and their cells are blank. Write a function to count how many pupils have a score. [1]

> =COUNT(D2:D30) (1) – COUNT ignores blank cells

18 Column C of a camp sign-up sheet (C2 to C40) shows "Yes" if the medical form has been returned. Write a function to count how many forms have been returned. [2]

> =COUNTIF(C2:C40,"Yes") (2) – 1 mark for COUNTIF with the correct range, 1 for the condition "Yes" in quotes

19 Explain why a formula such as =B2*C2 should be used in a cell instead of typing in the answer. [2]

> If a value in B2 or C2 changes, the formula recalculates automatically (1) / so the answer is always up to date and accurate; a typed answer would be wrong after the change / avoids calculation errors (1)

20 Give one reason for selecting a print area. [1]

> Only the needed part of the sheet is printed / saves paper and ink / hides working cells or data that should not

be printed (1)

21 Describe what merge and centre does and give an example of when it would be used. [2]

Combines two or more adjacent cells into one cell and centres the content across them (1) / e.g. a title in row 1 spread across the width of the table (1)

22 Describe two features that could be used to improve the presentation of a spreadsheet that will be printed. [4]

Any two features, each identified (1) and described (1), e.g. headers/footers (1) add a title, date or page number to every page (1) / borders or shading (1) separate headings from data so the table is easier to read (1) / bold or larger font for headings (1) / merge and centre the title (1) / currency format (1) / set the print area or landscape orientation (1) so the table fits on the page (1)

23 A Fermanagh football club records its members' details, subscriptions and payments on paper. Discuss the benefits and drawbacks of moving these records to a spreadsheet. In your answer refer to formulae, formatting and printing. [6]

Indicative content – Benefits: formulae (e.g. SUM of subscriptions, amount owed =Due-Paid) calculate totals automatically and recalculate when a payment is added; fewer arithmetic errors; COUNTIF can count unpaid members; absolute reference for the yearly fee so it can be changed once; currency/date formats make records clear; conditional formatting/sorting to find who owes money; print area and headers/footers to print tidy lists for committee meetings; easier to edit, back up and share. Drawbacks: time and cost of setting up and training volunteers; wrong formulae or missing $ signs give wrong totals without warning; data entry errors still possible; personal data must be kept secure under UK GDPR (password, backups); need access to a computer. Levels: 5–6 a balanced discussion with benefits and drawbacks linked to the club and all three areas; 3–4 several valid points, some explanation, may be one-sided; 1–2 a few simple points with little explanation.

E Quick quiz

Circle one answer · 9 marks

1 Which cell reference identifies column D, row 7? [1]

A 7D
B D7 ✓
C D:7
D 7D

2 In a tuck shop sheet, which function finds the smallest value in D2 to D6? [1]

A =SMALL(D2:D6)
B =MIN(D2:D6) ✓
C =LOW(D2:D6)
D =MIN(D2,D6)

3 Every formula must start with... [1]

A +
B #
C = ✓
D $

4 Which is an absolute reference? [1]

A B2
B B$
C #B2
D $B$2 ✓

5 =B4/C4 replicated to row 5 becomes... [1]

A =B5/C5 ✓
B =B4/C4
C =C4/D4
D =$B$5/$C$5

6 A cell formatted as Percentage holds 0.2. What does it show? [1]

A 0.2
B 0.2%
C £0.20
D 20% ✓

7 =D1-C1 does what? [1]

A Subtracts D1 from C1
B Adds C1 and D1
C Subtracts C1 from D1 ✓
D Finds the difference between columns C and D

8 E2 contains =D2*(1-H1) and is filled down. What is in E3? [1]

A =D3*(1-$H$1)

B =D2*(1-H2)

C =D3*(1-H1)

D =D3*(1-H2) ✔

9 Cells D2:D6 hold 27, 24, 16.5, 39.6 and 13.2. What does =COUNTIF(D2:D6,">20") return? [1]

A 3 ✔

B 2

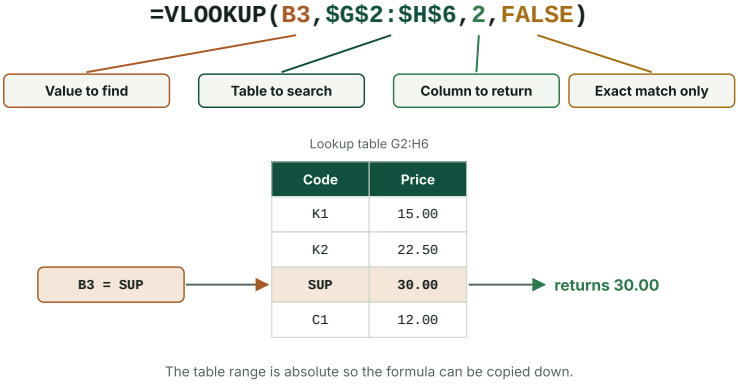
C 5

D 90.6

Spreadsheets 2: IF, VLOOKUP, validation, templates, modelling, charts and macros

L12 ANSWERS

I can: Write IF, nested IF and VLOOKUP functions and explain what each part does · Use conditional formatting, data validation and templates, and explain why they help users · Use a spreadsheet model to answer what-if questions · Choose, create and label a suitable chart, and describe what a macro is and why it is useful



Use this diagram to help you: The four parts of a VLOOKUP formula and the row it matches in a lookup table

A Key terms

Write the letter of the matching definition · 6 marks

B IF function

C Nested IF

A VLOOKUP

D Lookup table

F Conditional formatting

E Data validation

A Vertical lookup: a function that searches for a value in the first column of a table and returns a value from a specified column in the same row. It updates automatically if the source data changes.

B A function that tests a condition and returns one value if it is true and another if it is false, e.g. =IF(A1>60,"PASS","FAIL").

C An IF function placed inside another IF function so that more than two outcomes are possible, e.g. =IF(B2>=80,"Distinction",IF(B2>=65,"Merit","Fail")).

D The range of cells searched by VLOOKUP; the values to be matched are in its first column.

E An automatic check when data is entered that it is sensible and within acceptable limits, e.g. a range check of 1 to 16. It cannot check the data is correct.

F Formatting (e.g. fill or font colour) that is applied automatically to cells only when a specified condition is met, e.g. hours worked over 40 highlighted.

B Fill the gaps

Use the word bank · 9 marks

IF · Conditional formatting · error alert · VLOOKUP · template · what-if · nested IF · validation · macro

The **IF** function returns one value when a condition is true and another when it is false; placing one inside another makes a **nested IF**. **VLOOKUP** searches the first column of a lookup

table and returns a value from the same row. **Conditional formatting** changes a cell's appearance automatically when a rule is met. Data **validation** limits what can be entered, and an **error alert** asks the user to try again. A **template** gives a ready-made layout for others to use. Changing inputs in a model to see the effect is called **what-if** analysis, and a **macro** automates a repeated task.

C True or false?

Tick one box · 9 marks

	True	False
VLOOKUP searches the last column of the table. *(It searches the first column of the lookup table.)*	☐	☑
What-if analysis is used in data modelling.	☑	☐
A chart should have axis titles.	☑	☐
In an IF function, the second part is what happens when the condition is false. *(The second part is the value if true; the third part is the value if false.)*	☐	☑
An IF statement placed inside another IF statement is said to be nested.	☑	☐
VLOOKUP can only return a value from the first column of the table. *(It searches the first column and returns from the column number you specify.)*	☐	☑
Data validation can check that the data entered is correct. *(It can only check that data is sensible and within acceptable limits.)*	☐	☑
With FALSE as the last part, VLOOKUP shows #N/A if the value is not in the table.	☑	☐
A macro can be created by recording the tasks to be carried out.	☑	☐

D Exam-style questions

72 marks

1 Cell D2 contains a test mark. Write an IF function that displays 'Merit' if the mark is 70 or more, otherwise 'Pass'. [2]

> =IF(D2>=70,"Merit","Pass") – correct condition D2>=70 (1) / both results in the right order with quotation marks (1)

2 Explain what the function =IF(A1>60,"PASS","FAIL") does. State the result if A1 contains 60. [3]

> Checks whether the value in A1 is greater than 60 (1) / if true displays PASS, otherwise displays FAIL (1) / A1 = 60 gives FAIL because 60 is not greater than 60 (1)

3 Cell B2 contains a student's mark. Write a nested IF function for C2 that displays "Distinction" for a mark of 80 or more, "Merit" for a mark of 65 or more and "Fail" otherwise. [3]

> =IF(B2>=80,"Distinction",IF(B2>=65,"Merit","Fail")) – first condition and Distinction (1) / nested IF with B2>=65 and Merit (1) / Fail as final false value with correct brackets and quotes (1)

4 An online shop charges £4.95 delivery, stored in K1, unless the order total in J2 is £50 or more, when delivery is free. Write an IF function for K2 that can be replicated down column K. State the result in K2 when J2 is £35.50. [3]

> =IF(J2>=50,0,$K$1) – condition J2>=50 (1) / 0 if true and $K$1 (absolute) if false (1) / J2 = £35.50 gives £4.95 (1). Accept =IF(J2<50,$K$1,0).

5 A Year 12 formal committee keeps a budget sheet. Cell C8 holds the amount allowed for the DJ and D8 the amount actually spent. Write an IF function for F8 that displays "Overspent" if the amount spent is more than the amount allowed, otherwise "OK". [2]

> =IF(D8>C8,"Overspent","OK") – correct condition (1) / results in the right order with quotes (1). Accept =IF(C8<D8,"Overspent","OK") or =IF(D8<=C8,"OK","Overspent").

6 A travel club converts pounds to euros. H1 holds the base rate and H2 the bonus rate. Groups paying less than £100 (in column H, starting at H5) get the base rate; everyone else gets the bonus rate. Write the formula for I5 so that it can be replicated down column I, and explain why absolute references are needed. [4]

> =IF(H5<100,$H$1,$H$2) – condition (1) / both results correct (1) / H1 and H2 hold the fixed rates, so they must not change when the formula is copied down (1) / a relative H1 would become H2, H3… and pick up the wrong values (1)

7 Explain what =VLOOKUP(B3,$F$2:$G$10,2,FALSE) does. [3]

> Looks up the value in B3 (1) / in the first column of the table F2:F10 (1) / and returns the matching value from column 2 of the table (column G) in the same row; FALSE means only an exact match is accepted (1)

8 A table of activities and costs is stored in A1:B5: Activity/Cost, Kayaking £12, Rock Climbing £10, Archery £8.50, Zip Line £15. Explain what =VLOOKUP("Rock Climbing",A1:B5,2,FALSE) does and state the value returned. [4]

> Searches the first column of A1:B5 for the text Rock Climbing (1) / exact match because of FALSE (1) / returns the value from column 2 (Cost) of the same row (1) / value returned: £10 (1)

9 Explain why the lookup table in a VLOOKUP formula is often written with absolute references, e.g. $A$2:$B$5. [2]

> So that the table range stays the same when the formula is replicated down a column (1) / otherwise the range moves down, e.g. A3:B6, and some lookups miss rows or return #N/A / wrong results (1)

10 A leisure centre table in G20:I24 has Code in column G, Activity in column H and Price in column I. A booking code is entered in B20. (a) Write a VLOOKUP for C20 to show the price. (b) State what is displayed if a code is entered that is not in the table. [3]

> (a) =VLOOKUP(B20,$G$20:$I$24,3,FALSE) – B20 and correct table range (1) / column 3 and FALSE (1) / (b) #N/A (an error) is displayed (1)

11 Describe what is meant by conditional formatting and give an example of its use. [2]

> The format of a cell (e.g. fill or font colour) changes automatically when a set condition is met (1) / e.g. a test mark below 40 shown in red / stock below 5 highlighted (1)

12 Explain why conditional formatting is better than changing the colour of cells by hand. [2]

> It updates automatically when data changes, so the colour is always correct (1) / it is quicker and no cells are missed or coloured wrongly by mistake (1)

13 Values entered in B1:B6 must be between -£50 and +£100. Describe how data validation could be used and state one limitation of validation. [3]

> Select B1:B6 and choose Data Validation (1) / allow Decimal between -50 and 100, with an error alert if data is outside the range (1) / limitation: it only checks data is within limits, not that it is correct, e.g. 20 typed instead of 25 is accepted (1)

14 In a Noughts and Crosses spreadsheet, state a suitable validation rule for the nine board cells. [2]

> A list check / drop-down list (1) allowing only X or O (1)

15 Describe one benefit of using a template. [2]

Saves time because the layout, formatting and formulae are already set up (1) / gives consistency: every copy looks the same / users with little skill can use it and are less likely to make errors (1)

16 Describe how data could be imported into a spreadsheet from a database, and give one benefit of importing rather than typing the data. [3]

Export the database table as a CSV/text file or connect to it (1) / use the import wizard (Get External Data), answering each step's dialogue box, e.g. choosing the delimiter (1) / benefit: no retyping so fewer typing errors / saves time (1)

17 Explain what is meant by data modelling. Give an example of how a spreadsheet model could be used. [3]

Using a spreadsheet to represent a real situation with values and formulae (rules) (1) / changing inputs to predict what would happen / what-if analysis (1) / example, e.g. changing ticket prices to see the effect on a club's profit (1)

18 A burger van owner has a spreadsheet model of income, costs and profit. Explain how the model could help the owner decide what price to charge. [4]

The owner can enter different prices (1) / formulae recalculate income, costs and profit automatically (1) / can ask what-if questions, e.g. fewer sales at a higher price (1) / compare options without the risk of trying them for real, and choose the price giving the best profit (1)

19 A sports camp model has fixed costs of £1,500 and costs of £18 per child. (a) Calculate the profit if 120 children book at £35 each. (b) A what-if test raises the price to £38 and 105 children book. Calculate the new profit. [4]

(a) Income 35 × 120 = £4,200; costs 1,500 + 18 × 120 = £3,660 (1) / profit £540 (1) / (b) Income 38 × 105 = £3,990; costs 1,500 + 18 × 105 = £3,390 (1) / profit £600 (1)

20 State three features a chart must have to be clearly labelled. [3]

Any three: chart title (1) / axis titles (1) / units on axes (1) / legend (where there is more than one series) (1) / data labels (1)

21 State three types of chart and say what each is used to show. [3]

Any three, each with its use: column/bar – compare categories (1) / line – show trends over time (1) / pie – show parts of a whole / percentages (1) / scatter – relationship between two variables (1)

22 A youth club has its monthly attendance figures for January to June in cells B2 to B7, with the month names in A2 to A7. State the cell range that should be selected to create a chart, and name and justify the most suitable type of chart. [3]

A2:B7 (1) / line graph (1) / because it shows the trend in attendance over time (1). Accept column chart if justified as comparing months.

23 What is a macro? Describe one benefit of using a macro in a spreadsheet. [3]

A small program that records a series of actions so they can be replayed with one click / shortcut (1) / benefit identified (1) and described (1), e.g. saves time on repeated tasks such as printing a report, because the steps are done automatically / reduces errors as steps are done the same way each time

24 A Newry café uses a spreadsheet to take orders. The manager wants to make the sheet easier and more accurate for part-time staff to use. Evaluate how templates, data validation, VLOOKUP and macros could help. [6]

Indicative content – Template: order sheet with layout, prices and formulae ready; every order looks the same; protected formula cells cannot be overwritten by staff. Validation: drop-down lists of menu items and quantities 1–20 stop typing errors; error alerts guide staff; but it cannot check the right item was chosen. VLOOKUP: staff choose an item and the price is found from one price table, so prices are always correct and can be updated in one place; #N/A if an item is spelt wrongly. Macros: a button to clear the order or print the receipt saves time and avoids missed steps; needs .xlsm and macros must be set up and tested. Overall evaluation: together these

> make the sheet quicker and more accurate for untrained staff, but setting them up takes skill and time. Levels: 5–6 all four features explained in the café context with a judgement; 3–4 most features with some explanation; 1–2 simple statements.

E Quick quiz

Circle one answer · 9 marks

1 =IF(B2>10,"Yes","No") shows what if B2 is 10? [1]

A Yes
B No ✓
C 10
D Error

2 In =VLOOKUP(B2,$G$2:$H$6,2,FALSE), what does the 2 tell the spreadsheet? [1]

A Which column of the table to search
B How many rows to check
C Which column of the table to return a value from ✓
D Where to put the answer

3 Turning a cell red when stock is below 5 is... [1]

A conditional formatting ✓
B a macro
C a VLOOKUP
D a template

4 Best chart to show temperature changes over a month? [1]

A Pie chart
B Scatter graph
C Line graph ✓
D No chart

5 Peter scores 70. What does =IF(B2>=80,"Distinction",IF(B2>=65,"Merit","Fail")) show? [1]

A Distinction
B Merit ✓
C Fail
D 70

6 Which Data Validation tab sets the message shown when invalid data is entered? [1]

A Settings
B Input Message
C Conditional Formatting
D Error Alert ✓

7 Which file type must you use to save a workbook containing a macro? [1]

A .xlsm ✓
B .xlsx
C .csv
D .xltx

8 Changing the price in a model to see the effect on profit is called... [1]

A validation
B importing
C conditional formatting
D what-if analysis ✓

9 J2 holds £50.00. What does =IF(J2>=50,0,4.95) return? [1]

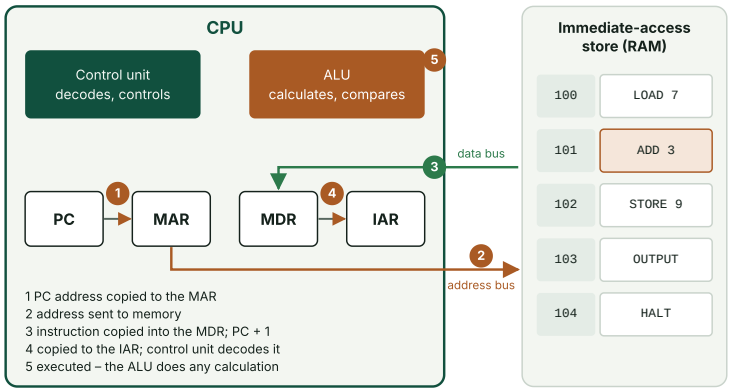
A 4.95
B 0 ✓
C 50
D FALSE

UNIT 1: DIGITAL TECHNOLOGY · COMPUTER HARDWARE

The CPU and the fetch–execute cycle

L13 ANSWERS

I can: Explain the purpose of the CPU using processing language, not just 'the brain' · Describe what the control unit, ALU and immediate-access store each do · Describe the job of the PC, MAR, MDR, IAR and accumulator · Describe each stage of the fetch–decode–execute cycle in the correct order



Use this diagram to help you: Inside the CPU: the fetch–execute cycle using the PC, MAR, MDR and IAR with main memory

A Key terms

Write the letter of the matching definition · 6 marks

B CPU

C ALU

D Control unit

A Immediate-access store

F Register

E Program counter

A Main memory (RAM) holding the programs and data currently in use; each location has a unique address.

B Central processing unit. A microprocessor whose purpose is to process data and carry out the instructions of programs, managing the functions of the computer.

C Arithmetic logic unit: the part of the CPU that carries out arithmetic calculations and logical operations and comparisons.

D The part of the CPU that fetches and decodes instructions and sends control signals to co-ordinate the other components.

E Register holding the address of the next instruction to be fetched; incremented after each fetch.

F A very small, very fast memory location inside the CPU used for a specific purpose.

B Fill the gaps

Use the word bank · 8 marks

MAR | microprocessor | accumulator | MDR | ALU | program counter | control unit | decode

The CPU is a **microprocessor** that processes data and carries out program instructions. The **control unit** decodes each instruction and sends control signals. The **ALU** carries out calculations and logical comparisons, and stores each result in the **accumulator**. The **program counter** holds the address of the next instruction. The **MAR** holds the address being accessed, while

the **MDR** holds the data or instruction fetched from memory. The cycle is fetch, **decode**, execute.

C True or false?

Tick one box · 9 marks

	True	False
The ALU decodes instructions. *(The control unit decodes instructions. The ALU does calculations and logical comparisons.)*	☐	☑
The program counter is increased by one after each instruction is fetched.	☑	☐
The immediate-access store holds the programs and data currently in use.	☑	☐
The MAR holds the data that has just been fetched from memory. *(The MAR holds an address. The data or instruction fetched from memory goes into the MDR.)*	☐	☑
The CPU is a microprocessor.	☑	☐
Writing 'the CPU is the brain of the computer' is enough to earn the marks for its purpose. *(You must explain its purpose: it processes data and carries out the instructions of programs.)*	☐	☑
The result of an ALU calculation is first stored in the accumulator.	☑	☐
Every location in main memory has its own unique address.	☑	☐
The stages of the cycle happen in the order fetch, execute, decode. *(The order is fetch, decode, execute.)*	☐	☑

D Exam-style questions

33 marks

1 Explain the purpose of the central processing unit (CPU). [2]

> It processes data (1) / it carries out (executes) the instructions of programs (1). Also accept: manages/controls the functions of the computer by sending control signals (1). 'Brain of the computer' alone = 0.

2 State the three main components of the CPU. [3]

> Control unit (1) / arithmetic logic unit (ALU) (1) / immediate-access store (IAS) or main memory (1).

3 Describe the role of the arithmetic logic unit (ALU). Give an example of an operation it carries out. [3]

> Carries out arithmetic calculations (1) / carries out logical operations or comparisons, e.g. AND, OR, NOT, greater than (1) / valid example, e.g. adding two numbers or checking if a balance is below a limit (1). Accept: results held in the accumulator (1).

4 Describe the role of the control unit. [3]

> Any three from: decides which instruction to carry out next (1) / fetches the instruction from memory (1) / decodes the instruction (1) / sends control signals to other components so the instruction is executed (1) / co-ordinates/manages the fetch–execute cycle (1).

5 What is the immediate-access store? Explain why it is needed for the fetch–execute cycle. [2]

> Main memory that holds the programs and data currently in use (1) / instructions are fetched from it by address, so without it there is nothing to fetch (1).

6 Expand the acronyms MAR and MDR, and explain the difference between them. [4]

> MAR = memory address register (1) / MDR = memory data register (1) / the MAR holds the address of the memory location being accessed (1) / the MDR holds the data or instruction being read from or written to that

location (1).

7 State what is held in the program counter, and describe what happens to it after an instruction is fetched. [2]

> The address of the next instruction to be fetched (1) / it is incremented (increased by one) so it points to the next instruction (1).

8 Describe the fetch stage of the fetch–execute cycle, referring to the program counter, MAR, MDR and IAR. [4]

> The address in the PC is copied to the MAR (1) / the instruction at that address is copied from memory into the MDR (1) / the instruction is copied to the IAR (1) / the PC is incremented (1).

9 A café in Enniskillen uses a till that adds up the price of each order. Describe what happens in the decode and execute stages when the till's CPU runs an instruction to add the price of a drink to the running total. [4]

> Decode: the control unit works out what the instruction means / that it is an ADD (1) / and which data is needed (1). Execute: the address of the drink's price is placed in the MAR and the value is fetched through the MDR (1) / the ALU adds it to the running total held in the accumulator (1).

10 Discuss how the components of the CPU and its registers work together to carry out a program. Your answer should refer to the control unit, the ALU, the immediate-access store and the fetch–execute cycle. [6]

> Indicative content: control unit fetches, decodes, sends control signals; PC holds the next address and is incremented; MAR holds the address being accessed; MDR holds data/instructions to and from memory; IAR holds the instruction being executed; IAS holds the program and data, each location addressed; ALU carries out arithmetic/logic during execute; accumulator holds results; the cycle repeats. Levels: 5–6 marks = detailed, accurate, well-ordered description of how the parts interact, correct terms throughout; 3–4 = several correct roles with some linking to the cycle; 1–2 = a few roles named or described with little linking.

E Quick quiz

Circle one answer · 8 marks

1 Which part of the CPU carries out calculations and logical comparisons? [1]

A Control unit
B Arithmetic logic unit (ALU) ✓
C Memory address register
D Program counter

2 Which register holds the address of the next instruction to be fetched? [1]

A Program counter ✓
B Memory data register
C Accumulator
D Instruction address register

3 What does the memory data register (MDR) hold? [1]

A The address of the location being accessed
B The address of the next instruction
C The instruction or data just fetched from, or about to be written to, memory ✓
D The clock speed of the CPU

4 Which component decodes instructions and sends control signals? [1]

A ALU
B Accumulator
C Immediate-access store
D Control unit ✓

5 What is the correct order of the cycle? [1]

A Execute, fetch, decode
B Fetch, decode, execute ✓
C Decode, fetch, execute
D Fetch, execute, decode

6 Which best explains the purpose of the CPU? [1]

A It stores files permanently
B It displays output to the user
C It processes data and carries out the instructions of programs ✓
D It connects the computer to a network

7 In the first step of the fetch stage, the contents of the program counter are copied into... [1]

A the MAR ✓

B the MDR

C the accumulator

D the ALU

8 Where is the result of an ALU calculation first held? [1]

A Program counter

B Memory address register

C Hard disc

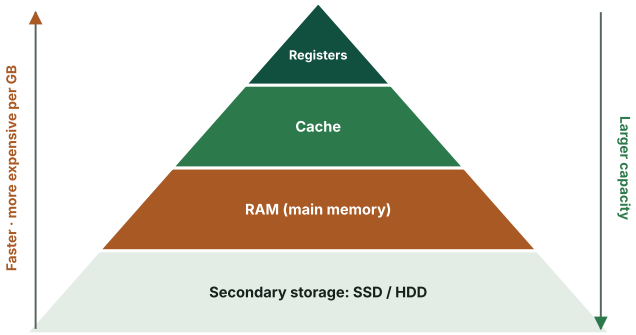
D Accumulator ✓

UNIT 1: DIGITAL TECHNOLOGY · COMPUTER HARDWARE

CPU performance and memory: RAM, ROM and cache

L14 ANSWERS

I can: Explain how clock speed, cache size and number of cores each affect CPU performance · Explain the purpose of RAM, ROM and cache · Compare RAM, ROM and cache using the terms volatile and non-volatile · Recommend and justify a processor and memory specification for a given user



ROM is separate: small, non-volatile, holds the boot-up instructions.

Use this diagram to help you: Memory hierarchy pyramid from registers to secondary storage

A Key terms

Write the letter of the matching definition · 6 marks

E Clock speed

C Core

A Cache

B RAM

F ROM

D Volatile

A Small, very fast memory on or near the CPU that holds frequently used instructions and data, so they need not be fetched from slower RAM.

B Random access memory: volatile read/write main memory that holds the operating system, programs and data currently in use.

C An independent processing unit inside a CPU; each core has a control unit, an ALU and registers, so it can process instructions at the same time as other cores.

D Describes memory that loses its contents when the power is switched off, such as RAM and cache.

E The number of fetch–execute cycles a CPU can carry out per second, measured in hertz (e.g. 3 GHz = 3 billion cycles per second).

F Read only memory: non-volatile memory whose contents cannot normally be changed, used to store the boot-up instructions (BIOS).

B Fill the gaps

Use the word bank · 8 marks

RAM · boot · volatile · cores · hertz · Cache · ROM · virtual memory

Clock speed is measured in **hertz**, and a 3 GHz processor carries out three billion cycles per second. A CPU with more **cores** can process several instructions at the same time.

Cache is a small amount of very fast memory that holds frequently used instructions and data. **RAM** holds the programs and data currently in use and is **volatile**, so its contents are lost

when the power is off. **ROM** is non-volatile and stores the **boot** program. When RAM is full, the operating system uses **virtual memory** on the storage drive.

C True or false?

Tick one box · 9 marks

	True	False
ROM is volatile. *(ROM is non-volatile: it keeps its contents when the power is off. RAM and cache are volatile.)*	☐	☑
A larger cache usually improves performance.	☑	☐
RAM loses its contents when the power is switched off.	☑	☐
Clock speed is the number of cores in a processor. *(Clock speed is the number of fetch–execute cycles per second, measured in hertz.)*	☐	☑
A 2 GHz processor can carry out two billion cycles per second.	☑	☐
Doubling the number of cores always doubles the speed of every program. *(Only software written to run in parallel benefits fully; cores also share memory and other resources.)*	☐	☑
ROM holds the instructions needed to start up the computer.	☑	☐
Cache is slower than RAM but has a larger capacity. *(Cache is faster than RAM but has a much smaller capacity.)*	☐	☑
Using virtual memory slows a computer down because storage is slower than RAM.	☑	☐

D Exam-style questions

28 marks

1 State the unit used to measure clock speed. [1]

> Hertz / Hz / gigahertz (GHz) (1).

2 Describe the impact of increasing the clock speed on the performance of a CPU. [2]

> Higher clock speed means more fetch–execute cycles per second (1) / so more instructions are processed per second and performance is faster (1).

3 A computer in the school library has a 3.2 GHz processor. State how many cycles per second this is, written in full. [1]

> 3,200,000,000 (three thousand two hundred million / 3.2 billion) cycles per second (1).

4 Explain how having more cores can improve performance, and give one reason why doubling the number of cores might not double the speed. [3]

> Each core can fetch, decode and execute instructions independently (1) / so more instructions can be processed at the same time / in parallel (1). Any one reason: software may not be written to use several cores / some tasks cannot be split up / cores share memory and must co-ordinate (1).

5 Two otherwise identical laptops in a Newry shop have 4 MB and 12 MB of cache. Explain what cache memory is for and why the 12 MB laptop is likely to perform better. [3]

> Cache stores frequently used instructions and data (1) / it is faster to access than RAM (1) / a larger cache holds more of this data, so the CPU needs fewer slower RAM accesses, so processing is faster (1).

6 Explain the purpose of RAM. [2]

> Holds the operating system, programs and data currently in use (1) / so the CPU can access them quickly / directly (1).

7 Explain the purpose of ROM. Include its role when the computer is switched on. [3]

> Stores the instructions needed to start up (boot) the computer / BIOS (1) / these run when the computer is switched on, checking hardware (1) / and loading the operating system from storage into RAM (1). Accept: content is permanent / non-volatile so it is available at start-up (1).

8 Describe two differences between RAM and ROM. [4]

> Any two matched differences, two marks each: RAM is volatile (1) whereas ROM is non-volatile (1) / RAM can be read from and written to (1) whereas ROM can only be read (1) / RAM holds programs and data currently in use (1) whereas ROM holds the boot program (1) / RAM is usually much larger (1) than ROM (1).

9 A pupil's laptop slows down when many programs are open. Explain why adding more RAM could help. [3]

> When RAM is full, the OS uses virtual memory on the storage drive (1) / storage is much slower than RAM, so swapping data slows the computer (1) / more RAM means more programs and data can be held in RAM at once, so less virtual memory is used and the laptop runs faster (1).

10 A sports club in Omagh wants a new computer to edit match videos for its website. Compare two possible specifications: Computer A (2.6 GHz, 4 cores, 6 MB cache, 8 GB RAM) and Computer B (3.4 GHz, 8 cores, 16 MB cache, 16 GB RAM). Discuss which is more suitable, referring to clock speed, cores, cache and RAM. [6]

> Indicative content: B has a higher clock speed (3.4 vs 2.6 GHz), so more cycles per second and faster processing; B has 8 cores, so video rendering, which can be split up, is processed in parallel; B's larger cache (16 MB) means fewer slow RAM accesses; B's 16 GB RAM can hold large video files and editing software without relying on virtual memory; A may be cheaper and adequate for simple tasks but would be slow for video; conclusion: B is more suitable, justified by the demands of video editing. Levels: 5–6 = all four factors compared and linked to video editing, with a justified conclusion; 3–4 = most factors compared with some linking; 1–2 = basic statements, e.g. 'B is faster', with little explanation.

E Quick quiz

Circle one answer · 8 marks

1 Clock speed is measured in... [1]

A bytes
B hertz ✓
C pixels
D bits per second

2 Why does a larger cache usually improve performance? [1]

A It increases the clock speed
B It adds more cores
C More frequently used data is kept close to the CPU, so there are fewer slow trips to RAM ✓
D It stores the operating system permanently

3 Which type of memory is volatile and holds the programs currently in use? [1]

A ROM
B SSD
C Blu-ray
D RAM ✓

4 What is stored in ROM? [1]

A The start-up (boot) instructions ✓
B Your saved documents
C The web pages you are viewing
D Frequently used data copied from RAM

5 Extra cores help most when... [1]

A the screen resolution is higher
B software can split its work to run in parallel ✓
C the computer is switched off
D the ROM is larger

6 A processor runs at 4 GHz. How many cycles can it carry out per second? [1]

A 4,000
B 4,000,000
C 4,000,000,000 ✓
D 400,000,000

7 Which statement about cache is correct? [1]

A It is faster than RAM and has a smaller capacity ✔

B It is non-volatile

C It is part of the hard disc

D It is read only

8 What is virtual memory? [1]

A Memory inside the CPU

B Another name for ROM

C Memory on a cloud server

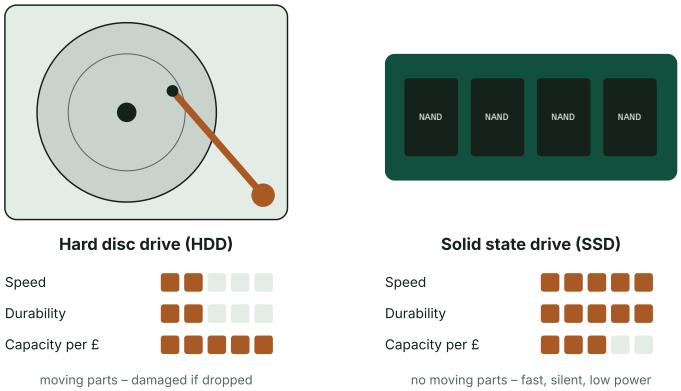
D Part of the storage drive used as extra RAM when RAM is full ✔

UNIT 1: DIGITAL TECHNOLOGY · COMPUTER HARDWARE

Input, output and storage devices

L15 ANSWERS

I can: Describe how each device in the specification works and what it is used for · Give advantages and disadvantages of each device, linked to a context · Compare HDD, SSD and HD optical storage using capacity, speed, durability, portability and cost · Recommend devices for a scenario and justify the choice



Use this diagram to help you: Hard disc drive with spinning platter compared with a solid state drive made of memory chips

A Key terms

Write the letter of the matching definition · 6 marks

E Input device

C Output device

D Graphics digitiser

A Touch screen

F Laser printer

B 3D printer

A A display with a touch-sensitive surface that works as both an input and an output device.

B An output device that builds a solid object layer by layer from a computer-aided design (CAD) model.

C Hardware used to present information from a computer, such as speakers or a printer.

D An input device made of a flat pad and a stylus, used to capture freehand drawings or signatures.

E Hardware used to enter data into a computer, such as a microphone, mouse or graphics digitiser.

F An output device that uses a laser to form the page on a charged drum, which attracts toner that is fused onto paper; fast and high quality.

B Fill the gaps

Use the word bank · 7 marks

touch screen | 3D printer | SSD | HDD | graphics digitiser | laser printer | Blu-ray

A **graphics digitiser** is used with a stylus to draw freehand. A **touch screen** is both an input and an output device. A **laser printer** uses toner and a drum to produce high-quality pages quickly. A **3D printer** builds solid objects layer by layer from a CAD design. An **SSD** uses flash memory and has no moving parts, while an **HDD** stores data magnetically and has a lower cost per gigabyte. **Blu-ray** discs use a blue-violet laser to store high-definition films.

C True or false?

Tick one box · 9 marks

	True	False
SSDs are more durable than HDDs because they have no moving parts.	☑	☐
A laser printer is an input device. *(A laser printer is an output device: it produces printed pages from the computer's data.)*	☐	☑
Blu-ray is a type of high-definition optical storage media.	☑	☐
3D printers are fast and economical for making large numbers of large objects. *(3D printers are slow for large objects and not yet economical for mass production; they suit prototypes and one-off items.)*	☐	☑
Voice recognition accuracy can drop when there is background noise.	☑	☐
An HDD usually has a lower cost per gigabyte than an SSD.	☑	☐
A graphics digitiser is better than a mouse for selecting items from menus. *(A graphics digitiser is designed for freehand drawing; it is not well suited to selecting menus.)*	☐	☑
Laser printers can print onto multipart carbon-copy forms. *(Laser printers are non-impact printers, so they cannot produce carbon copies on multipart stationery.)*	☐	☑
A Blu-ray disc can store more than a DVD because its laser makes smaller pits that are packed more closely.	☑	☐

D Exam-style questions

31 marks

1 State whether each of the following is an input device, an output device or both: microphone, speakers, touch screen. [3]

> Microphone – input (1) / speakers – output (1) / touch screen – both (1).

2 Describe one advantage and one disadvantage of using a touch screen to take orders in a fast-food restaurant. [2]

> Advantage, one from: customers or staff need little training / quick to select items / no separate keyboard or mouse (1). Disadvantage, one from: screen gets dirty or greasy, a hygiene issue (1) / limited number of options on screen at once / can be damaged with heavy use.

3 Describe what a graphics digitiser is and give a typical use. [2]

> A flat pad/tablet used with a stylus to capture freehand drawing (1). Use: graphic design / artwork / animation / capturing signatures (1).

4 Aoife, a GCSE pupil in a busy family home in Lurgan, plans to dictate her homework into a microphone using voice recognition instead of typing it. Describe two problems she might have. [4]

> Any two, two marks each: background noise in the house (1) can cause words to be recognised incorrectly, so errors must be corrected (1) / the system may need training to recognise the pupil's voice or accent (1) which takes time (1) / punctuation and formatting are awkward to dictate (1) so editing is still needed (1). Accept: sound files need more storage than text (1+1 if explained).

5 Give two advantages of a laser printer for a school office that prints large numbers of letters to parents. [2]

> Any two from: fast for large volumes / bulk printing (1) / high-quality text and graphics (1) / quiet (1) / low cost per page for large runs (1) / can be shared on the school network (1).

6 The Royal Victoria Hospital in Belfast is thinking of buying a 3D printer. Explain how a 3D printer makes an item and suggest one way the hospital could use it. [3]

> Builds a solid object (1) / layer by layer (1) / from a design or model created in CAD software (1). Max 2 for features. Use: e.g. prosthetic limbs / custom splints / models of organs for planning surgery (1).

7 Explain why an SSD is more suitable than an HDD for a laptop used by a building-site inspector. [2]

> An SSD has no moving parts (1) / so it is less likely to be damaged if the laptop is dropped or knocked on site (1). Accept: lighter/lower power, so better battery life (1).

8 Give one advantage of an HDD over an SSD. [1]

> Lower cost per gigabyte / larger capacity for the same price (1).

9 Apart from an SSD, flash memory is found in other everyday devices. Identify two of these, describe each one and give a situation in which a teenager might use it. [4]

> Any two, two marks each: USB flash drive (1) – small, portable plug-and-play storage, e.g. carrying coursework between home and school (1) / memory card such as SD or microSD (1) – extra storage for photos in a camera or phone (1) / smart card (1) – chip in a bank card or door entry card storing a small amount of data, e.g. chip and PIN payments (1).

10 Explain why a Blu-ray disc can store more data than a DVD of the same size. [2]

> Blu-ray uses a blue-violet laser with a shorter wavelength than a DVD's red laser (1) / so the pits are smaller and packed more densely, giving more capacity on the same size disc (1).

11 A council leisure centre in Derry~Londonderry is setting up a reception desk. Staff will book classes, print membership cards and store records of 5,000 members. Discuss suitable input, output and storage devices for the desk, justifying each choice. [6]

> Indicative content: input – mouse and keyboard for booking (familiar, accurate), or touch screen for quick selection; card reader or barcode scanner accepted if justified. Output – monitor; card/laser printer for membership cards and letters (fast, high quality). Storage – HDD for capacity at low cost, or SSD for speed and reliability; 5,000 records is small so capacity is not an issue, back-up should be considered. Each choice must be justified against the reception context (busy, staff not all IT experts, quiet environment). Levels: 5–6 = input, output and storage all covered with justified, contextual choices; 3–4 = at least two categories with some justification; 1–2 = devices listed with little or no justification.

E Quick quiz

Circle one answer · 8 marks

1 Which storage device has no moving parts? [1]

A Hard disc drive
B Solid state drive ✓
C Blu-ray drive
D DVD drive

2 How does a 3D printer create an object? [1]

A It prints toner onto paper using a drum
B It scans a real object with a laser
C It builds the object one layer at a time ✓
D It carves the object from a solid block

3 Which device is both an input and an output device? [1]

A Touch screen ✓
B Microphone
C Speakers
D Laser printer

4 Blu-ray is an example of... [1]

A magnetic storage
B solid state storage
C RAM
D high-definition optical storage ✓

5 What is the main advantage of an HDD over an SSD? [1]

A Faster access speed
B More durable
C Lower cost per gigabyte ✓
D Silent operation

6 Which device would best suit a designer who needs to draw detailed artwork by hand? [1]

A Graphics digitiser ✓
B Microphone

C Mouse

D Speakers

7 Which is a disadvantage of using a microphone for voice input? [1]

A It is slower than typing

B Background noise can cause words to be recognised wrongly ✓

C It needs a flat surface

D It cannot be used hands-free

8 Why is a laser printer a good choice for a busy school office? [1]

A It can print 3D objects

B It is cheap to buy in colour

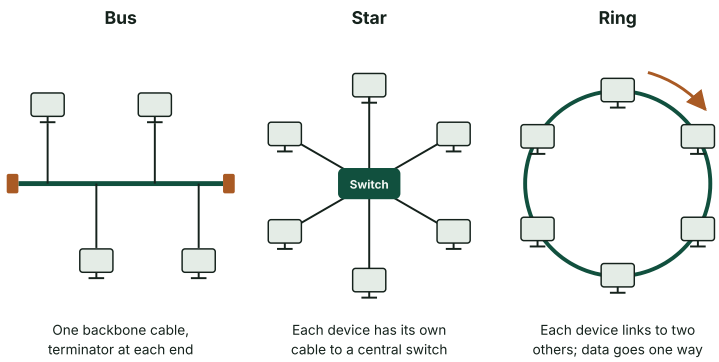
C It can print carbon-copy forms

D It prints large volumes quickly at high quality ✓

Networks: LAN, WAN, internet, hardware and topologies

L16 ANSWERS

I can: Compare a LAN and a WAN and give examples of each · Weigh up the advantages and disadvantages of a network for an organisation · Explain how the internet, the World Wide Web, an intranet and the Internet of Things differ · Describe what a NIC, network cables, a switch and a router do · Describe and compare the bus, star and ring topologies



Use this diagram to help you: Bus, star and ring network topologies

A Key terms

Write the letter of the matching definition · 6 marks

E LAN

D WAN

B Standalone computer

A Internet

F World Wide Web

C Intranet

A The global network of networks: the hardware and connections that carry data between computers worldwide.

B A computer that is not connected to a network, so its files and devices cannot be shared.

C A private network owned by an organisation that uses internet technology but can only be accessed by authorised users who log on.

D Wide area network: a network covering a large geographical area that uses telecommunications links such as fibre optic cables or satellites.

E Local area network: a network covering a small geographical area such as one building or site, usually owned by the organisation.

F A collection of linked web pages stored on web servers and accessed over the internet using a browser.

B Fill the gaps

Use the word bank · 8 marks

switch | NIC | star | terminator | WAN | router | LAN | intranet

A **LAN** covers a small area such as one school, while a **WAN** covers a large geographical area using telecommunications links. Every computer needs a **NIC** to connect to the network. Within the LAN a **switch** forwards data to the intended computer, and a

router connects the LAN to the internet using IP addresses. In a bus network a **terminator** at each end of the backbone stops signals bouncing back. In a **star** network every computer has its own cable to a central device. A private network that only authorised staff can log on to is an **intranet**.

C True or false?

Tick one box · 8 marks

	True	False
The World Wide Web and the internet are the same thing. *(The internet is the global network of networks (hardware and connections); the WWW is a collection of web pages accessed over it.)*	☐	☑
IoT stands for Internet of Things.	☑	☐
In a ring network, one broken cable can stop the whole network.	☑	☐
A switch connects a LAN to the internet. *(A router connects networks such as a LAN and the internet; a switch connects computers within a LAN.)*	☐	☑
A NIC lets a computer connect to a network.	☑	☐
In a bus network, if the backbone cable fails only one computer is affected. *(If the backbone fails the whole bus network stops; only a fault in a single computer's drop cable affects just that computer.)*	☐	☑
Anyone with an internet connection can use a company's intranet. *(An intranet is private; only authorised users who log on can access it.)*	☐	☑
A WAN usually uses telecommunications links such as fibre optic cables or satellites.	☑	☐

D Exam-style questions

36 marks

1 Southern Regional College links the computers in each of its campuses, and also links its Newry, Armagh and Banbridge campuses together. Describe two differences between the network inside one campus (a LAN) and the network joining the campuses (a WAN). [4]

> Any two differences, each fully described (2 marks each): LAN covers a small geographical area such as one building/site (1) whereas a WAN covers a large geographical area such as countries/the world (1) / LAN hardware is usually owned by the organisation (1) whereas a WAN uses telecommunications links rented from other companies (1) / LAN uses network cables or Wi-Fi (1) whereas a WAN uses links such as fibre optic cables, satellites or telephone lines (1)

2 Explain the difference between the internet and the World Wide Web. [2]

> The internet is a global network of networks / the hardware and connections that carry data (1) / the WWW is a collection of web pages/websites that are accessed over the internet using a browser (1)

3 A grammar school in Coleraine has an intranet for its pupils and staff. Explain how the school's intranet differs from the internet. [4]

> Internet: public / accessible to anyone with a communications link (1) / global network of networks (1) / Intranet: private network owned by an organisation (1) / only staff and pupils who log on with their own username and password can use it (1) / uses the same technology/protocols as the internet (1) / used to share internal information such as school resources and notices (1) – max 2 for each side, comparison needed for full marks

4 Expand the abbreviation IoT and describe two ways a teenager might use IoT devices at home. [5]

> Internet of Things (1) / any two uses described, e.g. smart speaker to play music or answer questions by voice (1) and it connects over Wi-Fi to the internet (1) / fitness watch collects heart rate/sleep data (1) and sends it to a phone app for analysis (1) / smart thermostat lets heating be switched on remotely (1) using an app over the internet (1) / video doorbell shows who is at the door on a phone (1) wherever the user is (1)

5 Describe the function of a network interface card (NIC). [2]

> A circuit board installed in a computer (1) that allows it to connect to a network (1) / provides a port for the network cable (1) / wireless NIC allows wireless connection (1) – max 2

6 A GP surgery in Banbridge has a LAN connected to the internet. Describe the function of the switch and the router in this network. [4]

> Switch: central connection point for computers on the LAN (1) / checks the destination of data and forwards it only to the intended computer (1) / Router: connects the surgery LAN to another network such as the internet (1) / uses IP addresses to forward data along the best route to its destination (1) / may filter traffic with a firewall (1) – max 2 each

7 A small office in Cookstown has four PCs connected in a bus network. Describe how this network is laid out and explain why it needs terminators. [4]

> Each PC is attached to one shared main cable, the backbone (1) / signals can move either way along the backbone (1) / a terminator is placed at each end of the backbone (1) / to stop signals bouncing back onto the cable (1) / new computers are added by attaching to the backbone (1) / if the backbone fails the whole network fails (1) – max 4

8 Describe a star topology and give one advantage and one disadvantage of it. [3]

> Each computer has its own cable to a central switch/server (1) / Advantage: if one cable fails only that computer is affected / easy to add computers / fast because data goes only to the intended device (1) / Disadvantage: if the central switch/server fails the whole network fails / more cable needed so more expensive (1)

9 Explain why adding a new computer to a ring network is more difficult than adding one to a star network. [2]

> In a ring the cable between two existing computers must be broken (1) and each end connected to the new computer, which disrupts the network while this is done (1) / in a star the new computer just needs its own cable plugged into the switch (1) – max 2

10 A garden centre in Lisburn has six standalone computers and two printers. The owner is thinking about replacing them with a network. Discuss the advantages and disadvantages of doing this. [6]

> Indicative content – Advantages: shared printers save buying one per PC; files on a server can be opened from any computer; data shared between staff (e.g. stock and orders); central backup more efficient; access rights protect customer data; software installed/updated once. Disadvantages: cost of server, NICs, switch, router, cabling; need someone to maintain the network (staff cost); viruses can spread to all computers; slowdowns if bandwidth is limited; server failure stops access to files. Levels: 5–6 marks = balanced discussion of both sides applied to the garden centre, with a justified conclusion and good use of terms; 3–4 = several points with some development, may be one-sided; 1–2 = a few simple points, little development.

E Quick quiz

`Circle one answer · 9 marks`

1 Which device connects a school's LAN to the internet and uses IP addresses to send data along the best route? [1]

A Switch
B Router ✓
C Network interface card
D Terminator

2 In which topology is every computer connected by its own cable to a central switch? [1]

A Bus
B Ring
C Star ✓
D Backbone

3 What is the job of a terminator in a bus network? [1]

A To connect the bus to the internet

B To stop signals bouncing back along the backbone ✓

C To speed up the backbone

D To give each computer an IP address

4 What does IoT stand for? [1]

A Internet of Technology

B Intranet of Things

C Internet of Things ✓

D Input of Things

5 Which best describes the World Wide Web? [1]

A The cables and routers linking computers worldwide

B A collection of linked web pages accessed over the internet ✓

C A private network for one company

D A type of LAN

6 Which is a disadvantage of a network compared with standalone computers? [1]

A Files can be backed up centrally

B Printers can be shared

C A virus can spread quickly to all computers ✓

D Software can be installed once for everyone

7 Which statement about a ring network is true? [1]

A Data travels in one direction around the loop ✓

B It needs a terminator at each end

C Each computer connects to a central switch

D A fault in one cable affects only one computer

8 A WAN differs from a LAN because a WAN... [1]

A is always wireless

B covers a large geographical area using telecommunications links ✓

C cannot connect to the internet

D only links two computers

9 Which device inside a computer provides the port for a network cable? [1]

A Router

B Switch

C File server

D Network interface card ✓

Network communication technologies

L17 ANSWERS

I can: Describe how Wi-Fi, Bluetooth, optical fibre, 4G and 5G transmit data · Give advantages, disadvantages and typical uses of each technology · Compare wireless and optical fibre for cost, speed and reliability · Choose and justify the best technology for a scenario



Use this diagram to help you: Typical range of Bluetooth, Wi-Fi, 4G, 5G and optical fibre

A Key terms

Write the letter of the matching definition · 6 marks

C **Wi-Fi**

E **Bluetooth**

B **Optical fibre**

D **4G**

F **5G**

A **Bandwidth**

A The amount of data that can be transmitted in a given time, measured in bits per second.

B Cable made of thin glass strands that carries data as pulses of light, giving very high bandwidth and immunity to electrical interference.

C Wireless technology that uses radio waves and a wireless router or access point to connect devices to a network without cables.

D Fourth-generation mobile technology giving high-speed wireless data through a network of masts.

E Short-range, low-power wireless technology that connects two nearby devices directly after they have been paired.

F Fifth-generation mobile technology with faster speeds, much lower latency and capacity for more devices than 4G, but needing more masts.

B Fill the gaps

Use the word bank · 8 marks

interference | repeater | latency | Bluetooth | Wi-Fi | Optical fibre | 5G | bandwidth

Wi-Fi uses radio waves and a wireless router so devices can join a network without cables, but the signal weakens with distance. **Bluetooth** connects two nearby devices directly once they have been paired. **Optical fibre** sends data as pulses of light along glass strands, so it is not affected by electrical **interference**. Over very long distances a **repeater** boosts the signal. **5G** has

much lower **latency** than 4G, meaning less delay. The amount of data that can be sent in a given time is called **bandwidth** .

C True or false?

Tick one box · 8 marks

	True	False
Optical fibre is affected by electrical interference. *(Optical fibre carries light, so it is not affected by electrical interference.)*	☐	☑
Bluetooth is suitable for connecting wireless earbuds to a phone.	☑	☐
Bluetooth needs a router before two devices can communicate. *(Bluetooth devices connect directly to each other after pairing; no router is needed.)*	☐	☑
The further a device is from a wireless router, the weaker the Wi-Fi signal.	☑	☐
Wi-Fi is always more secure than a cabled connection. *(Wi-Fi signals travel through the air and can be intercepted, so they must be protected with a password and encryption.)*	☐	☑
5G has lower latency than 4G.	☑	☐
Each high-speed 5G mast covers a larger area than a 4G mast. *(High-frequency 5G signals travel shorter distances, so more masts are needed.)*	☐	☑
Optical fibre is expensive to install.	☑	☐

D Exam-style questions

40 marks

1 Describe Bluetooth technology and give one typical use. [3]

> Short-range wireless technology (1) / uses radio signals to connect two devices directly (1) / devices must be paired/synchronised (1) / low power (1) / no router needed (1) – max 2 / use, e.g. wireless headphones, hands-free car kit, wireless mouse, smartwatch to phone (1)

2 A family in Larne connect their phones, tablets and smart TV to the internet using Wi-Fi. State two features of the wireless technology they are using. [2]

> Any two: uses radio waves (1) / needs a wireless router or access point (1) / no network cables required (1) / devices have a built-in wireless adapter (1) / signal strength depends on distance from the router (1) / should be protected by a password/encryption (1) / speed depends on the Wi-Fi standard (1)

3 Give two advantages and one disadvantage of optical fibre as a communications medium. [3]

> Advantages (any two): very high bandwidth/speed (1) / not affected by electrical interference (1) / very hard to tap so data is secure (1) / carries data long distances with little loss (1) / Disadvantage (any one): expensive to install (1) / fragile – can break if bent (1) / needs repeaters over very long distances (1) / specialist skills needed to repair (1)

4 Explain one advantage of 5G over 4G. [2]

> One advantage explained, e.g. lower latency (1) so less delay for live video/gaming/video calls (1) / faster download speeds (1) so large files and HD video load quickly (1) / more devices can connect in one area (1) so it works better in crowds/for IoT (1)

5 A café in Portrush wants to offer free internet access to customers. Explain why Wi-Fi is suitable and describe one security measure the owner should take. [4]

> Suitable: customers can connect their own phones/laptops without cables (1) / many devices can connect at once (1) / cheap and easy to set up with a wireless router (1) – max 2 / Security measure: password-protect the

network (1) and use WPA2/WPA3 encryption so data cannot be read if intercepted (1) / separate guest network so customers cannot reach the café's own computers (1) – max 2

6 Explain why optical fibre is a better choice than Wi-Fi for linking the two main buildings of a hospital in Derry~Londonderry. [4]

Any four: fibre has much higher bandwidth (1) so large files such as scans transfer quickly (1) / fibre is not affected by electrical interference from hospital equipment (1) / fibre is very hard to tap, protecting confidential patient data (1) / Wi-Fi signal weakens with distance and walls between buildings (1) / Wi-Fi is less reliable and can be intercepted (1)

7 A farmer near Enniskillen wants to monitor animals in remote fields using IoT sensors. Explain why 4G or 5G may be more suitable than Wi-Fi, and describe one problem the farmer could face. [4]

Suitability: mobile data works over a wide area / outdoors (1) / Wi-Fi only reaches tens of metres from a router (1) / no cables or router needed in the fields (1) – max 2 / Problem: poor or no mobile coverage in rural areas (1) because masts are concentrated where more people live (1) / ongoing data costs (1) / 5G may not be available yet (1) – max 2

8 A secondary school in Armagh is choosing between Wi-Fi and optical fibre cabling to connect its new sixth-form centre to the main building. Compare the two options, referring to cost, speed and reliability. [6]

Cost: Wi-Fi is cheaper to set up as no cables are laid between the buildings (1) / fibre is expensive to install, e.g. digging (1) / Speed: fibre has much higher bandwidth/faster (1) / wireless speed falls with distance and number of users (1) / Reliability: fibre is not affected by interference (1) / wireless is affected by walls, distance and other devices (1) – 2 marks per criterion, comparison needed

9 Evaluate the use of Wi-Fi in a secondary school. [4]

Advantages: no cabling so cheap and quick to install (1) / pupils and staff can move between rooms with laptops/tablets (1) / many devices connect at once (1) / Disadvantages: signal weaker far from access points / thick walls (1) / slows when a whole class connects (1) / must be secured to stop outsiders joining or intercepting data (1) – max 4, at least one of each for full marks

10 A Belfast events company streams live concerts and needs staff to share video and data at different venues. Evaluate the effectiveness of Wi-Fi, optical fibre and 5G for the company. [8]

Indicative content – Wi-Fi: cheap and quick to set up at each venue, staff can move around, many devices; but limited range, slows with many users, interference in crowded venues, must be encrypted. Optical fibre: very high bandwidth for live HD video, reliable, secure, no interference; but expensive and slow to install, fixed so not suitable for temporary venues unless already present. 5G: high speed and low latency for live streaming, works anywhere with coverage, copes with many devices in a crowd; but coverage patchy, data costs, needs 5G devices. Justified conclusion, e.g. fibre at the main base, 5G and Wi-Fi at venues. Levels: 7–8 marks = all three technologies evaluated with advantages and disadvantages applied to the scenario, clear justified conclusion, accurate terms; 4–6 = good points on at least two technologies, some application, limited conclusion; 1–3 = simple descriptions with little evaluation.

E Quick quiz

Circle one answer · 9 marks

1 A hospital needs a very fast link between two buildings that cannot be affected by electrical interference from medical equipment. Which technology is best? [1]

A Bluetooth
B Optical fibre ✓
C Wi-Fi
D 4G

2 A pupil wants to connect wireless earbuds to their phone. Which technology will they use? [1]

A Bluetooth ✓
B Optical fibre
C 5G
D A network cable

3 What is a key benefit of 5G compared with 4G? [1]

A Each mast covers a larger area
B Much lower latency ✓
C No masts are needed
D It works without a SIM or network

4 What is latency? [1]

A The amount of data sent per second
B The range of a signal
C The delay before data arrives ✓
D The number of users on a network

5 Which is a disadvantage of Wi-Fi? [1]

A It needs a cable to every device
B It cannot connect phones
C It only works for headphones
D Signal is weakened by walls and distance ✓

6 Why is optical fibre considered secure? [1]

A It is very hard to tap into the cable ✓
B It is password protected
C It only works over 10 metres
D It uses radio waves

7 Before two Bluetooth devices can share data they must be... [1]

A connected to a router
B plugged into a switch
C paired ✓
D given a URL

8 Which unit is used to measure Wi-Fi speed? [1]

A Megabytes of storage (MB)
B Gigahertz (GHz)
C Pixels per inch
D Megabits per second (Mbps) ✓

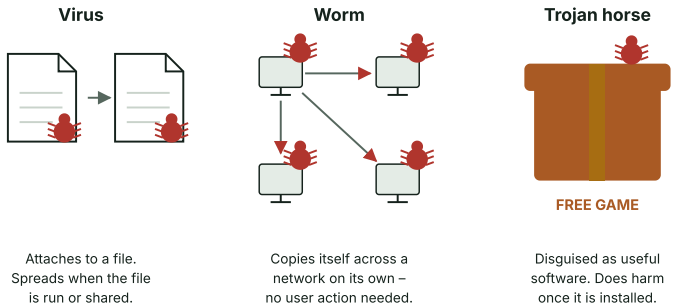
9 A farmer in a remote field with no Wi-Fi needs internet access for sensors. What is the most likely choice? [1]

A Bluetooth
B Mobile data (4G/5G) ✓
C Optical fibre to each sensor
D A bus network

Cybercrime and malware

L18 ANSWERS

I can: Define cybercrime and describe each of the nine threats to cybersecurity in the specification · Recognise the threat being described in a scenario and explain its effect on a person or organisation · Define malware and describe how it gets onto a computer · Distinguish between a virus, worm, Trojan horse, key logger and spyware · Explain how anti-virus software protects a computer



Also: key loggers record keystrokes · spyware secretly sends your data to others.

Use this diagram to help you: How a virus, a worm and a Trojan horse spread

A Key terms

Write the letter of the matching definition · 6 marks

C **Cybercrime**

E **Hacking**

D **Phishing**

B **Denial of service**

F **Cyber stalking**

A **Cyber defamation**

A Posting untrue statements online that harm the good name of a person or an organisation.

B An attack that floods a server or website with requests so that it cannot respond to genuine users.

C Any crime committed using a computer, a network or the internet.

D Emails, texts or websites that pretend to come from a trusted organisation in order to trick people into giving personal details such as passwords or bank details.

E Gaining unauthorised access to a computer system or its data, for example to steal, change or delete data.

F Using technology to repeatedly harass, track or frighten a person.

B Fill the gaps

Use the word bank · 8 marks

worm | virus | Trojan horse | hacking | denial of service | cybercrime | phishing | key logger

Using a computer or the internet to commit a crime is called **cybercrime**. Gaining unauthorised access to a system is **hacking**. Fake emails pretending to be from a bank, asking you to confirm your details, are an example of **phishing**. Flooding a server with requests so genuine users cannot get in is a **denial of service** attack. A **worm** copies itself and spreads across a network without needing a host file, while a **virus** attaches itself to a file. A **Trojan horse** is disguised as useful software, and a **key logger** secretly records every key pressed.

C True or false?

Tick one box · 8 marks

	True	False
A virus needs a host file to spread.	☑	☐
A worm needs the user to open an infected file before it can spread. *(A worm copies itself and spreads across a network on its own, without a host file or user action.)*	☐	☑
Cyber defamation means publishing false, damaging statements about someone online.	☑	☐
A denial of service attack always steals data from the server. *(A DoS attack stops genuine users accessing a service by flooding it with requests; it does not have to steal anything.)*	☐	☑
A Trojan horse copies itself to other computers automatically. *(A Trojan horse does not self-replicate; it relies on tricking the user into installing it.)*	☐	☑
Spyware can slow a computer down because it uses memory and processing power.	☑	☐
Spam is bulk unwanted email, often adverts or scams.	☑	☐
Anti-virus software only needs to be installed once and never updated. *(New malware appears constantly, so anti-virus software must be updated regularly to recognise it.)*	☐	☑

D Exam-style questions

29 marks

1 Define the term cybercrime. [1]

> A crime committed using a computer, a network or the internet (1).

2 Describe phishing. [2]

> An email/text/website that pretends to be from a trusted/genuine organisation, e.g. a bank (1) / that tricks the user into giving personal details such as passwords or card numbers, e.g. by clicking a link (1).

3 A Newry sports shop sells kit through its website. Describe how a denial of service attack would affect the shop. [2]

> The website/server is flooded with requests so it cannot respond / goes offline (1) / so genuine customers cannot access the site, so the shop loses sales/reputation (1).

4 Define malware. [2]

> Malicious software (0 on its own) / that is designed to damage or disrupt a computer system (1) / or to gain unauthorised access to the system or its data, e.g. to steal data (1).

5 Describe the difference between a virus and a worm. [2]

> A virus needs a host file/program and spreads when that file is run or shared (1) / a worm does not need a host and copies itself across a network by itself (1). Must compare both for 2 marks.

6 Explain how a key logger could be used to steal money from a bank account. [2]

> The key logger secretly records the keys pressed (1) / so the criminal can read the bank login username/password/card details and use them to log in and transfer money (1).

7 Describe two ways that malware could get onto a pupil's laptop. [2]

> Any two from: opening an infected email attachment (1) / clicking a link in a message or website (1) / downloading software or apps from an untrusted website (1) / plugging in an infected USB stick (1) / installing a fake program disguised as useful software (1).

8 Describe the difference between a Trojan horse and spyware. [4]

> Trojan horse: disguised as useful/legitimate software (1) / the user is tricked into installing it; it then deletes files or opens a back door for a hacker (1). Spyware: installed secretly, often bundled with free downloads (1) /

monitors user activity such as websites visited or details typed and sends it to a third party (1).

9 A youth club in Derry~Londonderry has a social media page and stores members' details on a laptop. Name and describe two threats to cybersecurity, other than malware, that the club could face. [4]

Any two threats, 1 mark for name and 1 for a description in context, e.g. Hacking (1): someone gains unauthorised access to the laptop and steals members' details (1) / Data theft (1): members' personal details copied and used for identity fraud (1) / Cyber defamation (1): false damaging posts about the club on social media (1) / Cyber stalking (1): a member is harassed through the club page (1) / Spamming (1): page or inbox flooded with junk messages (1) / Phishing (1): fake emails trick the leader into giving the page password (1).

10 Explain why anti-virus software must be updated regularly. [2]

New malware/viruses are created all the time (1) / the software can only detect threats it knows about (in its definitions/signatures), so without updates new malware would not be recognised and could infect the computer (1).

11 A family-run gift shop in Enniskillen has recently started selling online. Discuss the threats that phishing and denial of service attacks pose to the shop and its customers. [6]

Indicative content. Phishing: fake emails/texts that appear to come from a trusted source, e.g. the shop's bank, payment provider or a supplier; trick staff into revealing logins or bank details; criminals could take money or access the website/customer data; customers may get fake messages that appear to be from the shop, so they lose money and trust the shop less; data protection consequences if customer data is leaked. Denial of service: site flooded with requests and goes offline; genuine customers cannot order; lost sales, especially at busy times; damaged reputation; customers may move to competitors; cost of fixing. Levels: 5–6 both threats clearly described with several impacts on shop and customers, good use of context, accurate terms; 3–4 both threats described with some impact or one in depth; 1–2 basic points, limited description, little context.

E Quick quiz

Circle one answer · 8 marks

1 Malware that is disguised as a useful program, so the user installs it themselves, is a... [1]

A worm
B Trojan horse ✓
C spam
D key logger

2 A text claiming to be from your bank asks you to click a link and 'verify' your PIN. What is this? [1]

A Spamming
B Hacking
C Phishing ✓
D Cyber defamation

3 Which malware spreads across a network by itself without a host file? [1]

A Worm ✓
B Virus
C Trojan horse
D Spyware

4 Flooding a website with requests so genuine customers cannot use it is... [1]

A digital forgery
B cyber stalking
C data theft
D denial of service ✓

5 Publishing false, damaging claims about a person or business online is... [1]

A cyber defamation ✓
B spamming
C phishing
D hacking

6 Which malware records every key a user presses? [1]

A Worm
B Virus
C Key logger ✓
D Firewall

7 Stolen customer details used to apply for a credit card in someone else's name is... [1]

A identity theft ✓
B spamming
C a worm
D digital forgery

8 Editing a scanned passport photo to make a fake document is... [1]

A cyber stalking

B digital forgery ✓

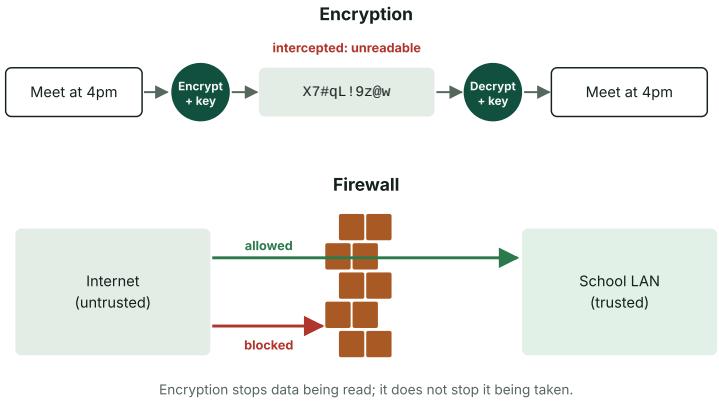
C denial of service

D spyware

Protecting networks and data; protocols

L19 ANSWERS

I can: Explain how encryption protects data that is intercepted or stolen · Describe how usernames, strong passwords and levels of access stop unauthorised use of a network · Explain how backups and firewalls protect a network and its data · Describe what a protocol is and why devices need one · Describe the purpose of FTP, HTTP and HTTPS and compare HTTP with HTTPS



Use this diagram to help you: Encryption makes intercepted data unreadable; a firewall filters network traffic

A Key terms

Write the letter of the matching definition · 6 marks

D Encryption

A Decryption

E Firewall

B Access rights / levels of access

F Read only

C Strong password

A Turning encrypted data back into readable form using the correct key.

B Permissions linked to a user ID that control what each user can see and do, e.g. no access, read only, read and write.

C A long password that mixes upper and lower case letters, digits and symbols and is not a dictionary word or personal detail.

D Scrambling data using an algorithm and a key so that it cannot be understood by anyone who does not have the key.

E Hardware or software that monitors and filters data entering or leaving a network, blocking traffic that does not meet security rules.

F An access level where a user can view data but cannot change it.

B Fill the gaps

Use the word bank · 10 marks

protocol | firewall | read only | FTP | Levels of access | strong password | two-factor authentication | backup | Encryption | HTTPS

Encryption scrambles data so that it cannot be understood without the key. Each user logs on with a unique user ID and a **strong password**, and **two-factor authentication** adds a second check such as a code sent to a phone. **Levels of access** control what each user can do, for example a pupil may have

read only access to a file. A **backup** is a copy of data that can be restored if the original is lost. A **firewall** filters data entering or leaving a network. A **protocol** is an agreed set of rules for transmitting data. **FTP** is used to transfer files, and **HTTPS** encrypts web traffic.

C True or false?

Tick one box · 9 marks

	True	False
HTTPS is less secure than HTTP. *(HTTPS is more secure because it encrypts the data sent between the browser and the web server.)*	☐	☑
Levels of access limit what each user can do with the data.	☑	☐
Encryption stops data being intercepted. *(Encryption does not stop interception; it makes intercepted data unreadable without the key.)*	☐	☑
Two-factor authentication needs two different forms of verification.	☑	☐
A firewall scans files on the hard drive and removes viruses. *(That is anti-virus software. A firewall monitors and filters network traffic against security rules.)*	☐	☑
A backup should be stored in a different location from the original data.	☑	☐
FTP is used to transfer files between computers over a network.	☑	☐
A protocol is a type of network cable. *(A protocol is an agreed set of rules for sending and receiving data.)*	☐	☑
A user with read-only access can view a file but cannot change it.	☑	☐

D Exam-style questions

34 marks

1 Explain how encryption protects data. [2]

> Data is scrambled/encoded using a key/algorithm (1) / so if it is intercepted or stolen it cannot be understood; only someone with the key can decrypt it (1).

2 Describe the purpose of a firewall. [2]

> Monitors/filters data entering and leaving a network (1) / blocks traffic that does not meet the security rules, e.g. stops hackers gaining access from the internet (1).

3 Explain how levels of access protect patient data in a hospital. [2]

> Staff are given access only to the data they need for their job, e.g. receptionists see appointments but not medical notes (1) / so confidential patient data is not seen or changed by unauthorised staff; read-only access prevents records being altered (1).

4 Describe the role of a protocol in data transfer. [2]

> A protocol is an agreed set of rules for sending/receiving data (1) / it allows different devices to communicate, e.g. sets data format, addressing and error checking (1).

5 Expand FTP and describe its purpose. [2]

> File Transfer Protocol (1) / used to send/upload and receive/download files between computers or to/from a file server over a network or the internet (1).

6 Describe the difference between HTTP and HTTPS. [2]

> HTTP sends data as plain text / unencrypted, so it can be read if intercepted (1) / HTTPS encrypts the data, so it

is secure, e.g. for banking/payment (1). Must compare both.

7 State three features of a strong password. [3]

Any three from: at least 8 characters (1) / mix of upper and lower case letters (1) / includes digits (1) / includes symbols/special characters (1) / not a dictionary word or personal information such as names or dates of birth (1).

8 Identify and describe three different levels of access a school network could give to its users. [6]

For each of three levels, 1 mark for the name and 1 mark for the description: No access (1): cannot open/see the files, e.g. pupils cannot see the staff area (1) / Read only (1): can view but not change the data (1) / Read and copy (1): can view and copy the file but not change the original (1) / Read and write (1): can view, edit, add and delete data, e.g. teachers editing marks (1) / Full control (1): can change permissions and install software, e.g. the network manager (1).

9 A dental surgery in Omagh stores patient records on its network server. Describe how backups should be carried out so that the records can be recovered if the server fails. [3]

Any three from: back up regularly / automatically on a schedule, e.g. every night (1) / store the backup separately from the server, off-site or in the cloud (1) / keep more than one backup copy/generation (1) / use secure/encrypted storage because data is confidential (1) / test that the backup can be restored (1) / use restore software to copy data back if the server fails (1).

10 Describe two measures, other than passwords, that a network logon system can use to reduce the risk of unauthorised access. [2]

Any two from: password shown as asterisks so others cannot see it (1) / account locked after a set number of failed attempts (1) / users forced to change passwords regularly (1) / two-factor authentication, e.g. a code sent to a phone (1) / a log of who logged on, when and where (1).

11 Explain why an online shop should use HTTPS on its checkout page. [2]

Customers enter personal and card details at checkout (1) / HTTPS encrypts this data so if it is intercepted it cannot be read and used for fraud, and the padlock reassures customers (1).

12 A leisure centre run by a local council keeps members' personal and payment details on its network and lets members book classes online. Discuss how the council could use encryption, passwords, levels of access, backup and firewalls to protect this data. [6]

Indicative content. Encryption: personal and payment data encrypted when stored and sent; online booking uses HTTPS; intercepted or stolen data unreadable without the key. Passwords: staff and members have unique IDs and strong passwords; two-factor authentication; lockout after failed attempts. Levels of access: reception staff can view bookings but not payment details; only managers can edit records; limits damage if one account is hacked. Backup: regular automatic backups stored off-site/cloud; restore after server failure or ransomware. Firewall: filters traffic between the network and the internet; blocks hackers; logs attempts. Legal duty under UK GDPR / Data Protection Act 2018 to keep data secure. Levels: 5–6 at least four methods clearly explained with how each protects the data, applied to the leisure centre, accurate terms; 3–4 three or more methods described with some explanation; 1–2 methods named or briefly described, little explanation or context.

E Quick quiz

Circle one answer · 8 marks

1 What does a firewall do? [1]

A Removes viruses from infected files

B Monitors and filters network traffic, blocking anything that breaks the security rules ✓

C Makes a copy of data each night

D Encrypts all passwords

2 How does HTTPS differ from HTTP? [1]

A HTTPS encrypts the data sent ✓

B HTTPS is only used for files

C HTTPS needs no web browser

D HTTPS works without the internet

3 Which protocol is used to upload a website's files to a web server? [1]

A HTTP
B HTTPS
C FTP ✓
D TCP

4 A protocol is... [1]

A a type of malware
B a network cable
C a strong password
D an agreed set of rules for transmitting data ✓

5 Which is the strongest password? [1]

A password123
B Liam2010
C t7#Rq!9vLm2p ✓
D fermanagh

6 A pupil can open the class notes but cannot change them. What access level is this? [1]

A Read only ✓
B Read and write
C Full control
D No access

7 Why should a backup be stored off-site or in the cloud? [1]

A It makes the backup run faster
B A fire or theft at the building will not destroy both copies ✓
C It encrypts the data
D It stops hackers logging in

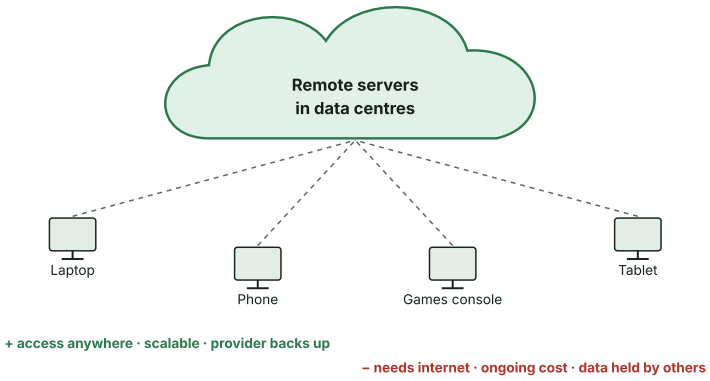
8 What happens to encrypted data if it is intercepted? [1]

A It is deleted automatically
B It is sent back to the sender
C It cannot be understood without the key ✓
D It becomes a virus

Cloud technology

L20 ANSWERS

I can: Define cloud computing and give examples of cloud services · Describe the advantages and disadvantages of the cloud for an organisation · Evaluate the impact of the cloud on gaming for players and game producers · Explain how cloud storage and collaborative tools change the way people store and share files



Use this diagram to help you: Cloud computing: devices reach remote servers over the internet

A Key terms

Write the letter of the matching definition · 6 marks

B Cloud computing

E Cloud storage

A Data centre

C Scalability

F Pay as you go

D Downtime

A A building containing large numbers of servers that store data and run cloud services.

B Storing data and running software on remote servers that are accessed on demand over the internet, instead of on a local computer.

C The ability to increase or decrease storage, bandwidth or users quickly as an organisation's needs change.

D A period when a service is unavailable, for example because of an internet failure or provider maintenance.

E Keeping files on a provider's remote servers so they can be accessed from any device with an internet connection.

F A payment model where an organisation pays only for the cloud resources it actually uses.

B Fill the gaps

Use the word bank · 8 marks

downtime | internet | collaboration | Synchronisation | scalable | cloud gaming | version history | Cloud computing

Cloud computing means using storage and software on remote servers accessed over the internet. Because storage can be increased or reduced as needed, the cloud is **scalable**. Its main weakness is that it needs an **internet** connection; a period when the service cannot be used is called **downtime**. In **cloud gaming**, the game runs on the provider's servers and is streamed to the player's

device. **Synchronisation** makes sure the latest version of a file appears on every device, and **version history** lets users restore an earlier copy. Several users editing a document at the same time is an example of **collaboration**.

C True or false?

Tick one box · 8 marks

	True	False
Cloud services can be used without any internet connection. *(Cloud services are accessed over the internet; without a connection users cannot reach their cloud files or software (apart from any copies already synced to the device).)*	☐	☑
Cloud gaming lets high-quality games run on low-powered devices.	☑	☐
Cloud providers usually take responsibility for backing up data.	☑	☐
Downtime is a risk of using cloud services.	☑	☐
Data stored in the cloud is guaranteed to be completely secure. *(No provider can guarantee total security; cloud data can still be hacked, so strong passwords and two-factor authentication matter.)*	☐	☑
Synchronisation means a change made on one device appears on the user's other devices.	☑	☐
An organisation using the cloud no longer has to follow data protection law. *(The organisation is still responsible for personal data under the UK GDPR and the Data Protection Act 2018.)*	☐	☑
Selling games through the cloud can reduce piracy compared with selling discs.	☑	☐

D Exam-style questions

39 marks

1 Define the term cloud computing. [2]

> Storing data and/or running software/applications (1) on remote servers accessed over the internet (1) / resources delivered on demand (1) – max 2

2 A small accountancy firm in Newry keeps all its files on one ageing server in the office. State three benefits the firm would gain by switching to cloud computing. [3]

> Any three: cost effective / pay as you go / no servers to buy (1) / scalable – storage can be increased or decreased (1) / very large storage capacity (1) / automatic software updates (1) / automatic backup and recovery by the provider (1) / access from anywhere on any device (1) / staff can collaborate on the same files (1) / reduced carbon footprint (1)

3 A solicitor's office in Armagh is moving its files to the cloud. Describe two disadvantages of cloud computing for the office. [4]

> Two disadvantages, each identified (1) and described in context (1): relies on an internet connection (1) so if the broadband fails staff cannot open client files (1) / client data held by a third party (1) could be hacked, and the office must still meet UK GDPR (1) / provider outages or maintenance (1) mean files are unavailable and the office cannot fix it (1) / ongoing subscription costs (1) that continue every month (1) / bandwidth limits (1) with extra charges if exceeded (1) / slow customer support (1)

4 Explain what is meant by 'downtime' and why it is a risk for a business that relies on the cloud. [2]

> Downtime is a period when a service is unavailable (1) / the business cannot access its data or software, so work stops / sales are lost (1)

5 Sinéad saves her GCSE coursework to OneDrive instead of a USB stick. Describe how her files are stored in the cloud and explain why she is less likely to lose her work. [4]

> Any four: files are stored on high-capacity remote servers in data centres, not on Sinéad's own device (1) / accessed over the internet (1) / work is saved automatically every few seconds (1) / files stay online until the user deletes them (1) / the provider keeps backup copies on more than one server (1) / the provider can recover files if a server crashes (1) / files are synced across devices (1)

6 Describe one impact of cloud computing on file sharing. [2]

> One impact described, e.g. users share a link instead of emailing large attachments (1) and permissions control who can view or edit (1) / several users can edit the same file at the same time (1) wherever they are (1)

7 Explain how synchronisation and version history help a group of pupils working on a shared presentation. [4]

> Synchronisation: a change made by one pupil appears in everyone's copy (1) so they always work on the latest version / no conflicting copies (1) / Version history: records who changed what and when (1) so the group can see contributions / restore an earlier version if something is deleted or broken (1)

8 Explain why cloud gaming may not be suitable for someone living in a rural area. [2]

> Rural areas may have slow broadband or poor mobile coverage (1) / cloud gaming needs a fast, low-latency connection, so the game would lag / be low quality / disconnect (1)

9 Describe two ways in which cloud computing benefits game producers. [4]

> Any two described: no need to make discs or pay shops (1) so costs fall and lower prices can be offered (1) / harder to pirate (1) so fewer lost sales (1) / access to a global market (1) so more sales/profit (1) / updates and patches released immediately via servers (1) keeping customers happy (1) / demos released online (1) to attract customers (1)

10 Conor, a Year 11 pupil in Strabane, is deciding whether to sign up to a cloud gaming service instead of buying a new console. Evaluate the impact cloud gaming would have on him as a player. [6]

> Indicative content – Positive: play high-quality games on cheap, low-powered devices; no need to keep upgrading hardware; less local storage needed; instant downloads on release day; saves synced across devices; updates and patches applied automatically. Negative: needs fast broadband with low latency or lag ruins play; heavy data use; ongoing subscription costs; games may be removed or service closed; poor rural connections. Levels: 5–6 marks = balanced evaluation of benefits and drawbacks applied to Conor, well developed, with a conclusion, good spelling, punctuation and terms (QWC); 3–4 = several points, some development, may be one-sided; 1–2 = a few simple points.

11 A charity has volunteers in Belfast, Ballymena and Enniskillen working on a fundraising project. Discuss the benefits and drawbacks of using cloud-based collaborative tools for the project. [6]

> Indicative content – Benefits: volunteers work on the same documents at the same time from three towns; no need to travel; sync means everyone sees the latest version; version history shows who changed what and allows restoring; comments and chat; files backed up by the provider; accessible on any device; often free or low cost for charities. Drawbacks: every volunteer needs reliable internet; some may lack digital skills; permissions must be managed to protect donor data (UK GDPR); risk of accidental edits or deletions; account security (strong passwords, two-factor authentication); provider outages. Levels: 5–6 = balanced discussion applied to the charity with a justified conclusion; 3–4 = several relevant points, partly developed; 1–2 = limited, generic points.

E Quick quiz

Circle one answer · 9 marks

1 Which of these best defines cloud computing? [1]

A Storing data on a USB stick

B Using storage and software on remote servers accessed over the internet ✓

C Connecting devices with Bluetooth

D Working on a standalone computer offline

2 A school's broadband fails for a morning. Why is this a problem if its files are in the cloud? [1]

A Cloud files are deleted when the internet goes down

B Staff cannot reach the files or cloud software until the connection returns ✓

C The cloud provider charges a fine

D The files become corrupted

3 In cloud gaming, where is the game actually processed? [1]

A On the player's device

B On a game disc

C On the provider's remote servers ✓

D On the router

4 Several pupils editing the same Google Doc at the same time is an example of... [1]

A a collaborative tool ✓

B batch processing

C a standalone system

D a bus network

5 What does 'scalable' mean when describing cloud storage? [1]

A It is very secure

B It uses optical fibre

C It cannot be changed

D Capacity can easily be increased or decreased ✓

6 Which is a disadvantage of cloud computing for an organisation? [1]

A Staff can work from home

B Software is updated automatically

C Data is held by a third party and could be hacked ✓

D Data is backed up regularly

7 How does selling games through the cloud help game producers? [1]

A Games can no longer be updated

B It removes the cost of discs and shops and reaches a global market ✓

C Players need more powerful consoles

D It makes broadband unnecessary

8 What does version history allow a user to do? [1]

A See who changed a file and restore an earlier version ✓

B Speed up the internet

C Make files smaller

D Stop anyone else opening the file

9 Why does cloud gaming need a connection with low latency? [1]

A To store more games

B To reduce the price of the game

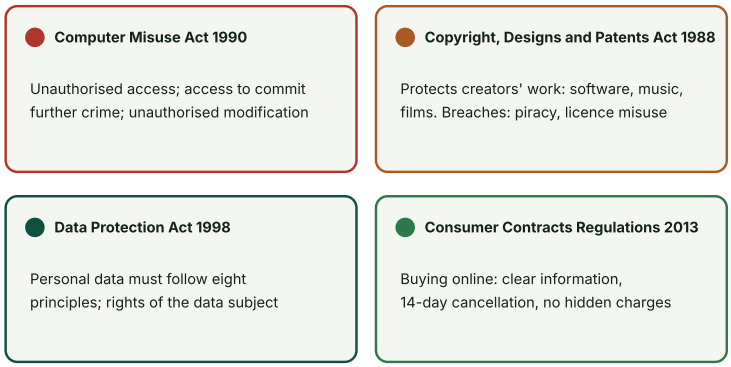
C To make the screen brighter

D So the game responds quickly without lag ✓

Legislation 1: Consumer Contracts, Copyright and Computer Misuse

L21 ANSWERS

I can: Give the full, exact title of each law and say what it protects · Describe the rights online shoppers have under the Consumer Contracts Regulations 2013, including the exceptions · Explain copyright and software licences and identify breaches such as software piracy · Describe the offences in the Computer Misuse Act 1990 and link hackers, viruses and spyware to them · Match a scenario to the correct law and explain why



Learn the full titles – examiners only credit the exact name.

Use this diagram to help you: Four laws examined in Unit 1 and what each covers

A Key terms

Write the letter of the matching definition · 6 marks

B Consumer Contracts Regulations 2013

E Cooling-off period

A Copyright, Designs and Patents Act 1988

D Intellectual property

F Software licence

C Software piracy

A Protects the intellectual property of creators, including software, music, images and text, so it cannot be copied, shared, sold or adapted without the owner's permission.

B The Consumer Contracts (Information, Cancellation and Additional Charges) Regulations 2013: protect people who buy online, by phone or by mail order by requiring clear information, giving a 14-day right to cancel and banning hidden extra charges.

C Illegally copying, downloading, distributing or selling software without the copyright owner's permission.

D Original creative work, such as software, music, designs or writing, which belongs to the person or organisation that created it.

E The 14 days in which a consumer can cancel a distance purchase without giving a reason and get a refund; for goods it starts the day after delivery.

F A legal agreement that states how a piece of software may be used, including how many users or computers may run it.

B Fill the gaps

Use the word bank · 8 marks

copyright | hacker | licence | cooling-off period | unauthorised access | pre-ticked box | piracy | Consumer Contracts Regulations 2013

People who shop online are protected by the **Consumer Contracts Regulations 2013**. For most goods they have a 14-day **cooling-off period** to cancel. Extra payments cannot be added using a **pre-ticked box**. Original work such as software is protected by **copyright** under the Copyright, Designs and Patents Act 1988. Illegally copying and selling software is called software **piracy**, and installing a program on more computers than allowed breaks the software **licence**. A **hacker** who gains **unauthorised access** to a computer commits an offence under the Computer Misuse Act 1990.

C True or false?

Tick one box · 8 marks

	True	False
The Computer Misuse Act 1990 makes unauthorised access to computer material an offence, even if nothing is changed.	☑	☐
Pre-ticked boxes that add extra charges at checkout are allowed if the price is shown. *(Pre-ticked boxes for extra payments are banned by the Consumer Contracts Regulations 2013; the customer must actively opt in.)*	☐	☑
Copying a paid game and selling it to friends breaches the Copyright, Designs and Patents Act 1988.	☑	☐
The cooling-off period for goods bought online is 30 days. *(It is 14 days, starting the day after the goods are delivered.)*	☐	☑
You must register your work to get copyright protection in the UK. *(Copyright is automatic in the UK as soon as an original work is created; there is no register.)*	☐	☑
You usually lose the right to cancel a digital download once it has started, if you agreed to this.	☑	☐
Freeware is not protected by copyright because it is free. *(Freeware is free to use but the owner still holds the copyright, so it cannot be copied, changed or sold without permission.)*	☐	☑
Planting a virus on a school network is an offence under section 3 of the Computer Misuse Act 1990.	☑	☐

D Exam-style questions

34 marks

1 Give the full title of the legislation that gives people buying goods online a 14-day period in which to cancel their order. [1]

> The Consumer Contracts (Information, Cancellation and Additional Charges) Regulations 2013 (1). Accept 'Consumer Contracts Regulations 2013'; the year is needed.

2 Describe two rights given to consumers by the Consumer Contracts Regulations 2013. [4]

> Any two rights, each with description (2 + 2): right to clear information before buying (1) / e.g. trader identity, total price including delivery, how to cancel (1); right to cancel within 14 days (1) / without giving a reason, starting the day after delivery (1); right to a refund (1) / within 14 days of returning the goods, including standard delivery (1); protection from hidden or extra charges (1) / pre-ticked boxes banned, must opt in (1); helplines charged at no more than basic rate (1) / so customers are not overcharged for support (1).

3 Ciara buys a game download from an online store and starts playing it. Two days later she asks for a refund. Explain why the store does not have to give her one. [2]

Ciara agreed to lose her right to cancel when the download started (1) / digital content cannot be returned once downloading or streaming has begun, so the right to cancel no longer applies (1).

4 State three pieces of information an online trader must give a customer before the customer pays. [3]

Any three (1 each): trader's name/identity / trader's address or contact details / description or main features of the product / total price including VAT / delivery charges / payment methods / how to cancel and return items / compatibility of digital content with devices.

5 Identify two typical breaches of the Copyright, Designs and Patents Act 1988. [2]

Any two (1 each): software piracy – copying/selling/distributing software without permission / installing software on more computers than the licence allows / sharing a licence key / using software after the licence or subscription ends / using a home or student licence for business / downloading music or films from illegal sites / using images or music without permission.

6 A primary school in Omagh has bought a single-user licence for a graphics package. Explain why installing it on all 15 classroom laptops breaks the law, and describe a type of licence the school should buy instead. [3]

A single-user licence allows the software on only one computer (1) / installing it on 15 laptops is a breach of the licence and the Copyright, Designs and Patents Act 1988 (1) / the school should buy a multi-user/network licence for 15 users or a site licence covering all computers on the site (1).

7 Describe three offences under the Computer Misuse Act 1990. [6]

Three offences, each named (1) and described/exemplified (1): section 1 unauthorised access to computer material (1) / e.g. using someone's password to view their files (1); section 2 unauthorised access with intent to commit or facilitate further offences (1) / e.g. accessing a bank account to steal money (1); section 3 unauthorised acts intending to impair a computer / unauthorised modification of data (1) / e.g. planting a virus, deleting files, DoS attack (1). Also accept 3ZA (serious damage) or 3A (making/supplying tools) with description.

8 A hacker gets into a Belfast sports shop's website and changes the prices of products. Explain which offences under the Computer Misuse Act 1990 have been committed. [4]

Gaining entry to the website without permission is unauthorised access (1) / section 1 of the Computer Misuse Act 1990 (1); changing the prices is unauthorised modification of data / an unauthorised act impairing the system (1) / section 3 (1). Accept section 2 if the candidate explains intent to commit fraud, e.g. buying goods cheaply (max 4).

9 Describe what is meant by spyware and explain how installing it relates to the Computer Misuse Act 1990. [3]

Spyware is malicious software that secretly monitors a user's activity, e.g. keystrokes or browsing (1) / and sends the information to a third party without the user's knowledge (1); installing it without permission is an unauthorised act/modification under the Computer Misuse Act 1990 / using captured passwords to log in is unauthorised access (1).

10 A youth club in Armagh runs a website and sells tickets online. Discuss how the Consumer Contracts Regulations 2013, the Copyright, Designs and Patents Act 1988 and the Computer Misuse Act 1990 affect the club and its members. [6]

Indicative content: Consumer Contracts Regulations 2013 – club must give clear information before payment (identity, total price, terms), no pre-ticked extra charges, helpline at basic rate; note that tickets for leisure events on a specific date are an exception to the 14-day cancellation right. CDPA 1988 – club must not use photos, music, logos or fonts on its site without permission; must hold valid licences for website/design software; members must not copy content. CMA 1990 – protects the club if its site or ticket database is hacked (unauthorised access s1, fraud s2, defacement or DoS s3); members must not access the admin area without permission; club should keep systems secure. Levels: 5–6 marks: all three laws accurately explained and applied to the club with good use of terms and well-organised writing; 3–4: two laws explained or three described with some application; 1–2: limited, mostly naming laws with little application.

E Quick quiz

Circle one answer · 8 marks

1 A secondary school installs a single-user licence for a design package on 30 PCs in its ICT suite. Which law has it broken? [1]

A Computer Misuse Act 1990 — **B Copyright, Designs and Patents Act 1988 ✓**

C Consumer Contracts Regulations 2013 — D Data Protection Act 2018

2 How long is the cooling-off period for most goods bought online under the Consumer Contracts Regulations 2013? [1]

A 7 days — B 28 days

C 14 days ✓ — D 30 days

3 Guessing a friend's password and reading their messages, without changing anything, is... [1]

A unauthorised access under the Computer Misuse Act 1990 ✓ — B legal because nothing was changed

C a breach of copyright — D a consumer rights issue

4 Which practice is banned by the Consumer Contracts Regulations 2013? [1]

A Charging for delivery — B Selling digital games

C Asking for a delivery address — **D Pre-ticked boxes for extra charges ✓**

5 Which of these can usually be returned within the cooling-off period? [1]

A A game download that has been started — **B Unworn trainers in their box ✓**

C A hoodie printed with your name — D An unsealed Blu-ray

6 Software that secretly records keystrokes and sends them to a criminal is called... [1]

A a firewall — B freeware

C spyware ✓ — D a licence key

7 A hacker gets into an online bank account in order to transfer money. Which section of the Computer Misuse Act 1990 best fits? [1]

A Section 2 – unauthorised access with intent to commit further offences ✓ — B Section 1 – unauthorised access only

C Section 3A – supplying hacking tools — D It is not covered by the Act

8 Under UK law, copyright in an original piece of software... [1]

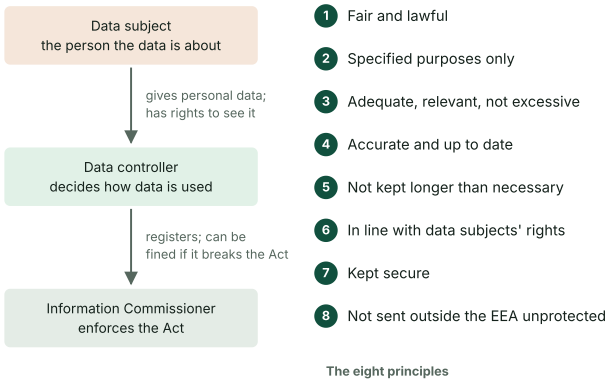
A must be registered with the government — B only applies if the © symbol is used

C lasts for exactly 10 years — **D applies automatically when it is created ✓**

Legislation 2: Data protection (UK GDPR and the Data Protection Act 2018)

L22 ANSWERS

I can: Explain what personal data and special category data are, with examples · State and apply the data protection principles to a scenario · Describe the rights of a data subject, including a subject access request · Describe the roles of the data controller, Data Protection Officer and the regulator



Use this diagram to help you: Data Protection Act roles and its eight principles

A Key terms

Write the letter of the matching definition · 6 marks

C **UK GDPR**	**A** UK law that works alongside the UK GDPR to control how personal data is collected, stored and used; it replaced the Data Protection Act 1998.
A **Data Protection Act 2018**	**B** Personal data that needs extra protection because its misuse could cause discrimination or harm, e.g. health, religion or ethnic origin.
E **Personal data**	**C** The UK version of the General Data Protection Regulation, which, with the Data Protection Act 2018, sets the rules for processing personal data in the UK.
B **Special category data**	**D** The person or organisation that decides why and how personal data is processed and is responsible for following data protection law.
F **Data subject**	**E** Any information about a living person who can be identified from it, such as their name, address, photo or email address.
D **Data controller**	**F** The living individual whom the personal data is about.

B Fill the gaps

Use the word bank · 7 marks

secure | subject access request | accurate | special category data | Information Commission | data controller | data subject

The person the data is about is the **data subject**. The organisation that decides how and why the data is used is the **data controller**. Data about a person's health or religion is **special category data** and needs extra protection. A person can see the data held about them by making a

subject access request. Data must be kept **secure**, must be **accurate** and must not be kept for longer than necessary. The law is enforced by the **Information Commission**, which replaced the Information Commissioner in 2026.

C True or false?

Tick one box · 8 marks

	True	False
The UK GDPR and the Data Protection Act 2018 replaced the Data Protection Act 1998.	☑	☐
Data controllers can keep personal data for as long as they like. *(Storage limitation: personal data must not be kept for longer than is necessary for its purpose.)*	☐	☑
Data subjects can ask an organisation for a copy of the personal data it holds about them.	☑	☐
The data subject is the organisation that decides how data is used. *(That is the data controller; the data subject is the living person the data is about.)*	☐	☑
A person's religion and health are examples of special category data.	☑	☐
An organisation can always charge £10 for a subject access request. *(Subject access requests are normally free; a fee can only be charged if a request is clearly unfounded or excessive.)*	☐	☑
Personal data must be accurate and kept up to date.	☑	☐
Data protection law only applies to data stored on computers. *(It also covers paper records kept in an organised filing system.)*	☐	☑

D Exam-style questions

31 marks

1 What is meant by the term 'data subject'? [1]

> The living individual whom the personal data is about (1).

2 State four principles of data protection law. [4]

> Any four (1 each), current or 1998 wording: processed lawfully, fairly and transparently / collected for a specified purpose and not used for other purposes / adequate, relevant and not excessive (limited to what is necessary) / accurate and kept up to date / not kept longer than necessary / kept secure (integrity and confidentiality) / processed in line with the rights of the data subject (1998) / not transferred outside the UK/EEA without adequate protection (1998) / accountability.

3 A shop in Ballymena stores customers' card details on an unencrypted laptop, which is stolen. State which data protection principle has been broken and describe one measure that would have prevented the breach. [3]

> Data must be kept secure / integrity and confidentiality principle (1); any one measure described (2): encrypt the laptop's storage (1) so data cannot be read if it is stolen (1) / do not store card details on portable devices (1), keep them on a secure server or use a payment provider (1) / password-protect the laptop with strong passwords (1) so a thief cannot log in (1).

4 A leisure centre asks new members for their name, address, date of birth, religion and blood group. Explain which principle this breaks. [2]

> Data minimisation / data must be adequate, relevant and not excessive (1); religion and blood group are not needed to run a membership (1).

5 Describe two rights of the data subject. [4]

> Two rights, each identified (1) and described (1): right of access (1) / can make a subject access request to see a copy of their data, normally free within one month (1); right to rectification (1) / have inaccurate data corrected (1); right to erasure (1) / have data deleted when no longer needed (1); right to object (1) / stop data being used for direct marketing (1); right to be informed (1) / told what data is held and why (1); right to complain/compensation (1) / complain to the Information Commission (accept ICO / Information Commissioner) or claim compensation for harm (1).

6 Describe the role of the data controller in an organisation. [2]

> Decides why and how personal data is processed (1) / is responsible for making sure the organisation follows data protection law, e.g. keeps data secure, answers subject access requests, reports breaches (1).

7 Describe two responsibilities of the Information Commission (formerly the ICO). [4]

> Any two, each identified (1) and expanded (1): enforces the law (1) by investigating complaints from data subjects (1); can take action (1) such as fines or enforcement notices against organisations that break the law (1); promotes good practice (1) by giving advice and guidance to organisations (1); makes the public aware of their rights (1); keeps a register of organisations that pay the data protection fee (1). Accept Information Commissioner / ICO throughout.

8 Explain why a school must have a Data Protection Officer and describe one task they carry out. [2]

> Schools are public bodies, which must have a DPO by law (1); any one task (1): advises staff on data protection / monitors that the school follows the law / organises staff training / is the contact point for the regulator and for data subjects.

9 Give two examples of special category data and explain why it needs extra protection. [3]

> Any two examples (1 each): racial or ethnic origin / religious beliefs / political opinions / trade union membership / health / sexual orientation / genetic or biometric data; misuse could lead to discrimination or serious harm, so the law requires extra protection (1).

10 A hospital trust in Northern Ireland is moving its patient records to a new online system. Discuss how the trust should make sure it follows data protection law. [6]

> Indicative content: identify the trust as data controller and patients as data subjects; health is special category data needing extra protection; collect only necessary data and use it only for patient care (purpose limitation, minimisation); check accuracy when migrating records; secure the system (encryption, strong passwords/2FA, access rights by role, audit logs, backups); ensure any cloud provider (processor) keeps data in the UK or with adequate protection and has a contract; DPO involved and a data protection impact assessment; staff training; privacy notice telling patients how data is used; process for subject access requests and complaints; report breaches within 72 hours; delete old records according to retention policy. Levels: 5–6 marks: wide range of well-explained points applied to the hospital using correct terms, logically organised; 3–4: several relevant points with some explanation; 1–2: a few basic points, little application.

E Quick quiz

Circle one answer · 8 marks

1 Under data protection law, who is the 'data subject'? [1]

A The organisation that decides how the data is used
B The regulator that enforces the law
C The living person the data is about ✓
D The company that stores the data for an organisation

2 Which body enforces data protection law in the UK? [1]

A The Information Commission ✓
B The police only
C The data subject
D CCEA

3 A gym asks for your blood group even though it has no need for it. Which principle is broken? [1]

A Security
B Accuracy
C Storage limitation
D Data minimisation ✓

4 A request to see the personal data an organisation holds about you is called a... [1]

A data breach
B subject access request ✓

C privacy notice

D licence request

5 Which of these is special category data? [1]

A Home address

B Email address

C Religious beliefs ✓

D Date of birth

6 Within how many hours must a serious personal data breach be reported to the regulator? [1]

A 24

B 72 ✓

C 48

D 168

7 A school emails a list of pupils' addresses to a local takeaway for a marketing campaign. Which principle is broken? [1]

A Purpose limitation ✓

B Accuracy

C Accountability

D Storage limitation

8 Which person advises an organisation on data protection and checks that it follows the law? [1]

A The data subject

B The network manager

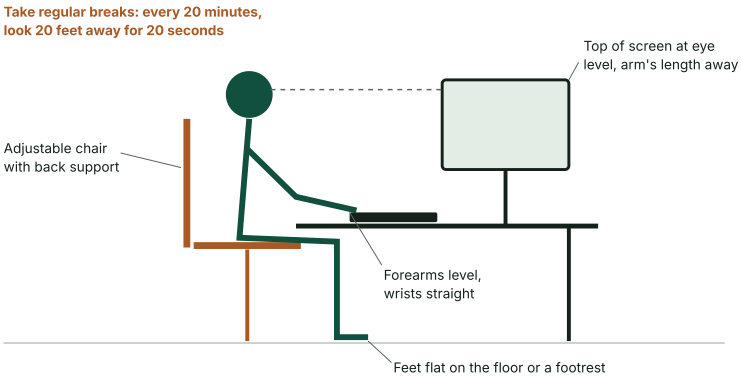
C The Data Protection Officer ✓

D The data processor

Ethics, employment and health and safety

L23 ANSWERS

I can: Describe examples of internet and social media misuse and how personal data is collected without people realising · Weigh up the benefits and concerns of GPS and tracking · Describe four impacts of digital technology on employment, with examples · Describe the causes of RSI, back strain and eye strain and how employees and employers can reduce them



Use this diagram to help you: A correctly set-up computer workstation

A Key terms

Write the letter of the matching definition · 6 marks

D Phishing

A Acceptable use policy

E Trolling

C Cyberbullying

F Fake news

B GPS

A Rules that employees or pupils agree to that set out how they may use an organisation's computers and internet.

B Global Positioning System: uses signals from satellites to work out the exact location of a device in real time.

C Using technology, such as social media or messages, to repeatedly hurt, threaten or embarrass a particular person.

D Using fake emails, texts or adverts that look official to trick people into giving away personal details such as passwords.

E Deliberately posting provocative or offensive messages online to upset people or start arguments; credited separately from cyberbullying.

F Deliberately false or misleading information that is spread online, often made to look like real news.

B Fill the gaps

Use the word bank · 8 marks

teleworking | privacy | upskilling | back strain | displacement | anti-glare | phishing | RSI

Pain in the wrists and hands caused by repeated typing is called **RSI**. When technology replaces jobs that people used to do, this is job **displacement**. Training workers in new skills is called **upskilling**. Using ICT to work from home is **teleworking**. Employers should provide **anti-glare**

screens to reduce eye strain and adjustable chairs to reduce **back strain** . Fake emails that trick people into giving personal details are called **phishing** . Tracking someone's location using GPS without their consent raises **privacy** concerns.

C True or false?

Tick one box · 8 marks

	True	False
Taking regular breaks helps prevent RSI.	☑	☐
Teleworking is an example of a change in work patterns.	☑	☐
Employers have no health and safety responsibilities for staff who use computers. *(Employers must provide a safe workplace; under the Health and Safety at Work (NI) Order 1978 and the Display Screen Equipment Regulations (NI) 1992 they must assess workstations, plan breaks and provide eye tests.)*	☐	☑
Trolling and cyberbullying mean exactly the same thing. *(Cyberbullying repeatedly targets a particular person; trolling is posting provocative comments to upset people or start arguments. Mark schemes credit them as separate points.)*	☐	☑
Loyalty cards allow shops to collect data about customers' buying habits.	☑	☐
A water extinguisher should be used on a fire in a computer room. *(Water conducts electricity; a CO_2 extinguisher should be used for electrical fires.)*	☐	☑
The top of a monitor should be at, or just below, eye level.	☑	☐
Digital technology has only reduced the number of jobs. *(It has replaced some jobs but also created many new ones, such as software engineers and cyber security analysts.)*	☐	☑

D Exam-style questions

36 marks

1 Describe two ethical concerns about the use of GPS tracking. [4]

> Two concerns, each identified (1) and described (1): a person's location can be tracked without their consent (1) which invades their privacy (1); could be used for stalking or controlling someone (1) by following their live location (1); employers may monitor staff constantly (1), even outside working hours (1); apps may sell location data to advertisers (1) without users realising (1); geotagged posts can reveal home or school (1), putting people at risk (1).

2 Ballymena Academy's ICT co-ordinator is writing a talk for Year 10 about how the internet is misused. Describe three different examples of misuse that the talk could include. [6]

> Three ways, each identified (1) and described (1): phishing (1) – fake official-looking emails/adverts trick people into giving personal details (1); theft of usernames and passwords (1) – used to access accounts and make purchases (1); piracy (1) – illegally downloading copyrighted music, films or software (1); plagiarism (1) – copying work and claiming it as your own (1); spreading fake news or hateful/violent content (1) – misleads or harms people (1); using the dark web to access illegal material (1); employees using work internet for personal use (1) – wastes time, leading to acceptable use policies (1).

3 Describe two ways in which a supermarket can collect personal data about its customers without them fully realising it. [4]

> Two ways, each identified (1) and described (1): loyalty cards (1) record what each customer buys and when, used for targeted advertising (1); card payment records (1) show shopping patterns, which may be shared or sold (1); online shopping accounts/cookies (1) track browsing and purchases (1); CCTV (1) records customers in store (1); app/Wi-Fi location data (1) shows when customers visit (1).

4 Describe what is meant by job displacement and give an example. [2]

> Jobs are lost or changed because technology now does the work previously done by people (1); example, e.g. self-service tills replacing checkout staff / robots in car factories / online banking replacing bank clerks (1).

5 Explain why upskilling is needed in a workplace that uses digital technology. [2]

> Technology and software change quickly (1) so workers need training in new skills to do their jobs effectively / keep their jobs / move into new roles (1). Accept: training improves productivity, morale or staff retention (1).

6 Gráinne works in a Belfast insurance office but has been offered the chance to telework three days a week from her home in Banbridge. Explain what teleworking means and describe two advantages it would give her. [6]

> Teleworking means working from home or away from the office, using ICT to stay connected (1) e.g. using broadband, a laptop, cloud apps and video calls to reach the employer's systems (1); two advantages for Gráinne, each identified (1) and described (1): no commute to Belfast (1) saves time and travel costs (1); flexible hours (1) can fit work around family (1); fewer interruptions from colleagues (1) can improve concentration (1); less stress from travel (1) / reduced carbon footprint (1).

7 Describe one change in work patterns caused by digital technology. [2]

> One change identified (1) and described (1): teleworking/hybrid working (1) – staff work from home using broadband and video calls (1); flexible hours (1) – can choose when to work as long as tasks are done (1); shift work (1) – 24-hour call centres/global companies need staff at all hours (1); always contactable (1) – email/messaging blurs home and work (1).

8 Describe one cause of RSI and one measure an employee can take to prevent it. [2]

> Cause: repeating the same movements, e.g. typing or using a mouse for long periods (1); measure: take regular breaks / use a wrist rest / keep wrists straight / vary tasks (1).

9 Describe one measure an employer should take to reduce eye strain. [2]

> One measure identified (1) and described (1): provide anti-glare screens / filters (1) to reduce reflections (1); fit blinds and good lighting (1) to avoid glare on screens (1); provide adjustable/tilting monitors (1) so screens can be positioned away from light (1); pay for regular eye tests (1) as required by the DSE Regulations (NI) 1992 (1); plan work so staff get screen breaks (1).

10 Discuss the ways in which social media can be misused, and the effect this can have on young people in Northern Ireland. [6]

> Indicative content: cyberbullying – repeated targeting causes anxiety, low self-esteem, isolation; trolling – provocative comments cause upset and arguments; hate speech – racist or sectarian posts can increase division and fear; fake profiles and identity theft – victim blamed or reputation damaged, risk of grooming; oversharing – addresses, school or location revealed, risk to safety; sharing images without consent – embarrassment, may be illegal; fake news – young people misled; addiction and comparison with others – affects mental health and sleep. Levels: 5–6 marks: a range of misuses each clearly described with effects on young people, correct terms, well-organised writing; 3–4: several misuses with some effects; 1–2: a few misuses listed with little explanation.

E Quick quiz

Circle one answer · 8 marks

1 Which health problem is most likely to be caused by typing for long periods every day? [1]

A Eye strain
B Hearing loss
C Back strain
D RSI ✓

2 A supermarket installs self-service tills and needs fewer checkout staff. This is an example of... [1]

A upskilling
B job displacement ✓
C teleworking
D phishing

3 Which is an employer measure to reduce back strain? [1]

A Provide adjustable chairs with back support ✓
B Take regular breaks

C Blink more often

D Keep wrists straight

4 Office staff attending a course to learn a new database package is an example of... [1]

A job displacement

B teleworking

C upskilling ✓

D data minimisation

5 Which of these is an ethical concern about GPS tracking? [1]

A Finding a lost phone

B Getting delivery updates

C Better navigation

D Location being tracked without consent ✓

6 An email that looks as if it is from your bank asks you to 'confirm' your password. This is... [1]

A phishing ✓

B trolling

C teleworking

D piracy

7 Which law requires employers to assess computer workstations and offer eye tests to regular screen users? [1]

A Computer Misuse Act 1990

B Health and Safety (Display Screen Equipment) Regulations (NI) 1992 ✓

C Consumer Contracts Regulations 2013

D Copyright, Designs and Patents Act 1988

8 Which of these is an advantage of teleworking for an employee? [1]

A It is easier to switch off from work

B Closer supervision by a manager

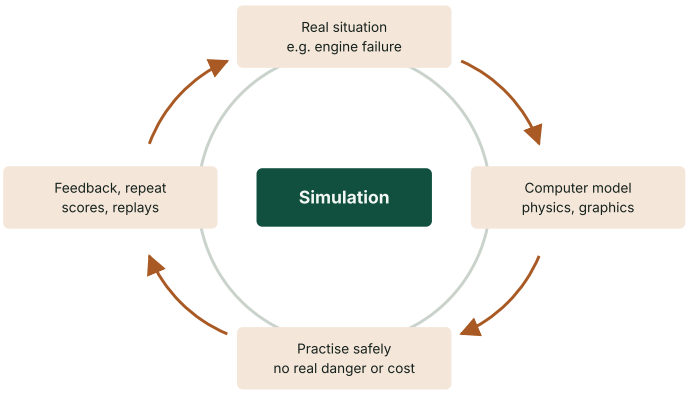
C No time or money spent commuting ✓

D More face-to-face contact with colleagues

Digital applications and extended-answer technique

L24 ANSWERS

I can: Describe the features of gaming applications, simulations and mobile apps · Explain how these applications support education and training, social interaction and work · Weigh up the advantages and disadvantages of online banking, online training and e-commerce for both the user and the organisation · Plan and write a well-organised extended (QWC) answer that reaches a justified conclusion



Use this diagram to help you: How a simulation supports training: model, practise safely, get feedback, repeat

A Key terms

Write the letter of the matching definition · 6 marks

C E-commerce

A Online banking

B Online training

E Gaming application

F Gamification

D Simulation

A Using a bank's secure website or app to manage an account, for example viewing statements, paying bills and transferring money.

B Learning delivered over the internet, often with multimedia and automatic progress tracking, that learners can complete at their own pace.

C The buying and selling of goods and services over the internet.

D A computer model, built from data and mathematical rules, that copies how a real object or situation behaves so it can be tested or practised safely.

E An electronic game played on a console, computer or mobile device, often online with other players.

F Adding game features such as points, badges and leaderboards to a non-game activity to increase motivation.

B Fill the gaps

Use the word bank · 8 marks

phishing | automatic tracking | two-factor authentication | gamification | virtual reality | immediate feedback | simulation | e-commerce

Buying and selling goods over the internet is called **e-commerce**. Adding points, badges and leaderboards to a learning app is known as **gamification**, and games keep learners motivated by giving **immediate feedback**. A **simulation** uses a computer model so trainee pilots can practise an engine

failure safely, while **virtual reality** uses a headset to place the user inside a 3D world. A major risk of online banking is **phishing**, where fake emails or texts try to steal login details, so banks use **two-factor authentication** to check it really is the customer. Online training systems use **automatic tracking** to record the modules each employee has completed.

C True or false?

Tick one box · 9 marks

	True	False
Simulations allow dangerous situations, such as an engine failure, to be practised safely.	☑	☐
Online banking can only be used during branch opening hours. *(Online banking is available 24 hours a day, 7 days a week, from anywhere with an internet connection.)*	☐	☑
A disadvantage of e-commerce for customers is that goods cannot be seen or tried before buying.	☑	☐
Writing 'it is easier' is enough to earn the mark in an extended answer. *(Vague words on their own are not credited; you must explain what is easier and why, using technical terms.)*	☐	☑
In a game, levels usually get harder and must be completed in order.	☑	☐
Online banking saves banks money because fewer branches and staff are needed.	☑	☐
With online training, employers have no way of checking an employee's progress. *(Automatic tracking records the modules completed and assessment scores, so employers can monitor progress.)*	☐	☑
Virtual reality only involves the sense of sight. *(VR can stimulate several senses, such as sight, hearing and touch (through data gloves or controllers), to make the experience feel real.)*	☐	☑
If a question asks for disadvantages to the customer, points about disadvantages to the bank will not earn marks.	☑	☐

D Exam-style questions

44 marks

1 State what is meant by e-commerce. [1]

> Buying and selling goods or services over the internet (1).

2 State two features of gaming applications that help to motivate learners. [2]

> Any two from: leaderboards (1) / points or scores (1) / badges or rewards (1) / levels of increasing difficulty (1) / immediate feedback (1) / multimedia such as animation and sound (1).

3 Describe how gaming applications support social interaction. [2]

> Any one developed point, e.g. online multiplayer lets players compete or co-operate with others anywhere in the world (1) and talk in real time using headsets and microphones or text chat (1). Accept joining online gaming groups/communities and collaborating to solve problems.

4 A flight school at City of Derry Airport is deciding whether to buy a flight simulator for its trainee pilots. Describe two features of a simulator that would make it useful for this training. [4]

> Two features, each identified (1) and described (1), e.g. emergencies such as engine failure or storms can be simulated (1) so the pilot can practise without risk to life (1) / scenarios can be repeated (1) until the pilot is confident (1) / the pilot's actions are recorded (1) so the instructor can give feedback (1) / no fuel used or aircraft taken out of service (1) so training is cheaper (1).

5 The Royal Victoria Hospital in Belfast plans to use virtual reality to train junior doctors. Describe how virtual reality could be used for this training. [3]

> Any three from: doctors wear a VR headset to enter a 3D simulated ward or operating theatre (1) / a virtual patient can be created from body-scan data (1) / they can practise procedures or rehearse an operation before doing it for real (1) / mistakes harm no real patient (1) / scenarios such as rare emergencies can be repeated (1) / data gloves or controllers give touch feedback so it feels realistic (1) / actions are recorded for feedback from a senior doctor (1).

6 Describe how a mobile phone app can support work practices. [2]

> One developed point, e.g. staff can receive announcements or group messages instantly (1) wherever they are, so they can respond quickly (1). Accept: access email/calendar/files away from the office; log jobs or record data on site; video meetings; training modules on the phone.

7 A secondary school in Enniskillen uses a mobile app to communicate with parents and pupils. Describe two ways the app could support education. [4]

> Two ways, each identified (1) and described (1), e.g. teachers can upload resources such as videos, audio and e-books (1) which pupils can use at home or school through the cloud (1) / homework can be set and submitted in the app (1) with notifications reminding pupils of deadlines (1) / parents get messages about events or progress (1) so they can support learning at home (1) / revision quizzes give immediate feedback (1).

8 Margaret, who is 74 and lives on a farm outside Kesh, has been told her local bank branch is closing and she should bank online instead. Describe two disadvantages online banking could have for her. [4]

> Two disadvantages, each identified (1) and described (1), e.g. risk of fraud or phishing (1) where login details are stolen and money taken (1) / needs a device, internet access and digital skills (1) which Margaret may not have, and rural broadband may be poor (1) / no face-to-face help (1) so problems take longer to sort out (1) / system outage or no connection (1) means no access to money (1).

9 A bank with branches across Northern Ireland is moving more of its customers to its app. Explain two benefits this brings to the bank itself. [4]

> Two benefits, each stated (1) and explained (1), e.g. fewer branches needed (1) so lower rent and wage costs (1) / e-statements (1) save printing and postage costs (1) / customers do not need to live near a branch (1) so the bank can attract a larger customer base (1) / transactions are processed automatically (1) reducing staff time (1).

10 A garden centre in Cookstown is thinking of selling its products through a website. Describe two benefits and one drawback of e-commerce for the business. [6]

> Benefits (2 × 2): no need for a large shop or as many checkout staff (1) so costs fall and prices can be lowered (1) / can sell to customers across NI, the UK or worldwide (1) increasing sales (1) / open 24/7 (1) so customers can order when the shop is closed (1) / customer reviews can attract new buyers (1). Drawback (1 × 2): must set up and maintain a secure website and payment system (1) which costs money and needs skills (1) / delivery of bulky or living plants is costly and they may be damaged (1) leading to returns and refunds (1).

11 A supermarket chain with stores across Northern Ireland plans to replace its classroom training days with online training modules. Evaluate the impact of this change on its employees and on the company. (QWC) [6]

> Indicative content. Positives: learn at own pace and convenient time; no travel saves time and money; staff in different locations train together; automatic tracking of modules and scores; engaging multimedia; mobile access; VR simulations give realistic practice. Negatives: limited support (FAQ/email), isolation and delays; needs self-motivation; not all learning styles suited; deadlines hard to manage; employer less in control; reliance on technology (poor connection, browser updates); practical skills harder to learn. Conclusion weighing up, applied to the supermarket's staff and managers. Levels: L3 (5-6) both sides, several well-developed points, justified conclusion, good QWC with specialist terms; L2 (3-4) some developed points, may be one-sided; L1 (1-2) basic list, limited terms.

12 A pupil in Derry~Londonderry buys most of their clothes and games online. Discuss the disadvantages of online shopping for customers. (QWC) [6]

> Indicative content: cannot see/touch/try goods, so they may not match images or descriptions and must be returned; returning goods is inconvenient and refunds may take time; delivery time delays and missed deliveries (especially at Christmas); delivery costs raise total price; fake or scam websites stealing card details; phishing; needs internet and bank card; personal data held by retailers could be breached. Levels: L3 (5-6) at least three well-developed customer disadvantages, organised, accurate specialist terms and good spelling; L2 (3-4) two or three points with some development; L1 (1-2) basic points or list. Points about disadvantages to the business are not credited.

E Quick quiz

Circle one answer · 9 marks

1 Why is a flight simulator useful for training pilots? [1]

A It is more dangerous than flying a real plane
B Emergencies such as engine failure can be practised with no risk to life ✓
C It needs a real aircraft to be taken out of service
D It removes the need for any instructors

2 What is the name for buying and selling goods and services over the internet? [1]

A E-commerce ✓
B E-safety
C Batch processing
D Cloud gaming

3 Which of these is a disadvantage of online banking to the customer? [1]

A 24/7 access to the account
B Instant transfers using Faster Payments
C Risk of phishing attacks stealing login details ✓
D Spending alerts sent to their phone

4 Points, badges and leaderboards added to a revision app are an example of... [1]

A simulation
B encryption
C streaming
D gamification ✓

5 What does an 'evaluate' question require? [1]

A Only the advantages
B A list of key words
C Both sides and a justified conclusion ✓
D A labelled diagram

6 Which is a benefit of online banking for the BANK rather than the customer? [1]

A Fewer branches and staff are needed, cutting costs ✓
B The customer can bank while abroad
C The customer avoids queues
D The customer gets spending alerts

7 How is a 3D virtual patient for surgical training usually created? [1]

A From photographs on social media
B From data collected by body scanners ✓
C By drawing it by hand in a paint program
D From the patient's GP letter

8 Which is a disadvantage of online training for employees? [1]

A They can learn at their own pace
B They do not need to travel to a venue
C Help may be limited to FAQs or email, so they feel unsupported ✓
D Multimedia makes the content more engaging

9 Under the Consumer Contracts Regulations 2013, online shoppers can usually cancel an order for goods within... [1]

A 24 hours
B 7 days
C 30 days
D 14 days of receiving the goods ✓